

mind the caveats about our modeling noted in part (d). How high does the shell go? Now change the launch angle from -1 to 1 . What happens?

Seven Paris Guns were built, but only three were used. They fired a total of 351 shells towards Paris between March 23 and August 9 of 1918. The damage and casualties that they caused were not tactically significant. They were never intended to be so; there was no control over where the shells would fall in Paris, and the amount of explosive carried by each shell was quite small. Instead, they were intended as a form of intimidation, a “scare tactic.” However, military historians agree that they were not effective in that sense either. Their significance turned out to be more scientific than military. The shells that they launched were the first man-made objects to reach the stratosphere, initiating the space age in the science of ballistics.



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CHAPTER 4

The Laplace Transform

4.1 Definition of the Laplace Transform

4.2 Inverse Transforms and Transforms of Derivatives

4.3 Translation Theorems

4.4 Additional Operational Properties

4.5 Dirac Delta Function

4.6 Systems of Linear Differential Equations

Chapter 4 in Review

In Chapter 3 we solved linear mathematical models such as $mx'' + \beta x' + \kappa x = f(t)$ and $Lq'' + Rq' + q/C = E(t)$ where the driving functions $f(t)$ and $E(t)$ were continuous. However, in practice, discontinuous driving functions are not uncommon. Although we solved piecewise linear DEs in Chapter 2, the Laplace transform discussed in this chapter is an especially valuable tool that simplifies the solution of such equations.

4.1 Definition of the Laplace Transform

INTRODUCTION

In elementary calculus you learned that differentiation and integration are *transforms*—this means, roughly speaking, that these operations transform a function into another function. For example, the function $f(x) = x^2$ is transformed, in turn, into a linear function, a family of cubic polynomial functions, and a constant by the operations of differentiation, indefinite integration, and definite integration:

$$\frac{d}{dx}x^2 = 2x, \quad \int x^2 dx = \frac{1}{3}x^3 + c, \quad \int_0^3 x^2 dx = 9.$$

Moreover, these two transforms possess the **linearity property**; this means the transform of a linear combination of functions is a linear combination of the transforms. For α and β constants,

$$\frac{d}{dx}[\alpha f(x) + \beta g(x)] = \alpha f'(x) + \beta g'(x)$$

$$\int [\alpha f(x) + \beta g(x)] dx = \alpha \int f(x) dx + \beta \int g(x) dx$$

and

$$\int_a^b [\alpha f(x) + \beta g(x)] dx = \alpha \int_a^b f(x) dx + \beta \int_a^b g(x) dx$$

provided each derivative and integral exists. In this section we will examine a special type of integral transform called the **Laplace transform**. In addition to possessing the linearity property, the Laplace transform has many other interesting

properties that make it very useful in solving linear initial-value problems.

■ Integral Transform

If $f(x, y)$ is a function of two variables, then a definite integral of f with respect to one of the variables leads to a function of the other variable. For example, by holding y constant we see that $\int_1^2 2xy^2 dx = 3y^2$. Similarly, a definite integral such as e^{-t^2} , $\sqrt{1+t^3}$ transforms a function f of the variable t into a function F of the variable s . We are particularly interested in an **integral transform**, where the interval of integration is the unbounded interval $x_0 = 1$ If $\theta(t)$ is defined for $t \geq 0$, then the improper integral $\int_0^\infty K(s, t)f(t) dt$ is defined as a limit

$$\int_0^\infty K(s, t)f(t) dt = \lim_{b \rightarrow \infty} \int_0^b K(s, t)f(t) dt. \quad (1)$$

If the limit in (1) exists, we say that the integral exists or is **convergent**; if the limit does not exist, the integral does not exist and is **divergent**. The limit in (1) will, in general, exist for only certain values of the variable s .

■ A Definition

The function $y \xrightarrow{-\lambda_2}$ in (1) is said to be the **kernel** of the transform. The choice $K(s, t) = e^{-st}$ as the kernel gives us an especially important integral transform.

DEFINITION 4.1.1 Laplace Transform

Let f be a function defined on $x_0 = 1$ Then the function F defined by

$$F(s) = \int_0^\infty e^{-st}f(t) dt \quad (2)$$

is said to be the **Laplace transform** of f . The domain of $F(s)$ is the set of values of s for which the improper integral (2) converges.

The Laplace transform of a function f is also denoted by $\mathcal{L}\{f(t)\}$. In general discussion and applications, we shall use a lowercase letter to denote the function being transformed and the corresponding capital letter to denote its Laplace transform. For example,

$$\mathcal{L}\{f(t)\} = F(s), \quad \mathcal{L}\{g(t)\} = G(s), \quad \mathcal{L}\{i(t)\} = I(s).$$

The Laplace transform is named in honor of the French mathematician, astronomer, and physicist **Pierre-Simon Marquis de Laplace** (1749–1827), who used the transform in his investigations of probability theory. Most likely the Laplace transform was invented by Leonard Euler. For his services to the Bourbons in restoring the monarchy following the fall of Napoleon Bonaparte in 1814, King Louis XVIII awarded Laplace the title of Marquis.

In a general setting, the variable s could be a complex number. But throughout this chapter we shall assume that s is real. As the next four examples show, the domain of $F(s)$ depends on the function $f(t)$.



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Pierre-Simon Marquis de Laplace

EXAMPLE 1 Using Definition 4.1.1

Find $\mathcal{L}\{1\}$.

SOLUTION

From (2),

$$\begin{aligned}\mathcal{L}\{1\} &= \int_0^{\infty} e^{-st}(1) dt = \lim_{b \rightarrow \infty} \int_0^b e^{-st} dt \\ &= \lim_{b \rightarrow \infty} \left. \frac{-e^{-st}}{s} \right|_0^b = \lim_{b \rightarrow \infty} \frac{-e^{-sb} + 1}{s} = \frac{1}{s}\end{aligned}$$

provided $s > 0$. In other words, when $s > 0$, the exponent $-sb$ is negative and $e^{-sb} \rightarrow 0$ as $b \rightarrow \infty$. The integral $\int_0^{\infty} e^{-st} dt$ diverges for $s < 0$, and so the domain of $F(s) = 1/s$ is defined by $s > 0$. ≡

The use of the limit sign becomes somewhat tedious, so we shall adopt the notation c_1^2 as a shorthand to writing $\lim_{b \rightarrow \infty}(\frac{1}{16})$. For example,

$$\mathcal{L}\{1\} = \int_0^{\infty} e^{-st}(1) dt = \left. \frac{-e^{-st}}{s} \right|_0^{\infty} = \frac{1}{s}, \quad s > 0.$$

At the upper limit, it is understood we mean $e^{-st} \rightarrow 0$ as $t \rightarrow \infty$ for $s > 0$.

EXAMPLE 2 Using Definition 4.1.1

Find $\mathcal{L}\{t\}$.

SOLUTION

From Definition 4.1.1, we have $\mathcal{L}\{t\} = \int_0^{\infty} e^{-st}t dt$. Integrating by parts and using $\lim_{t \rightarrow \infty} te^{-st} = 0$, $s > 0$, along with the result from Example 1, we obtain

$$\mathcal{L}\{t\} = \left. \frac{-te^{-st}}{s} \right|_0^{\infty} + \frac{1}{s} \int_0^{\infty} e^{-st} dt = \frac{1}{s} \mathcal{L}\{1\} = \frac{1}{s} \left(\frac{1}{s} \right) = \frac{1}{s^2}.$$

≡

EXAMPLE 3 Using Definition 4.1.1

Find

- (a) $\mathcal{L}\{e^{-3t}\}$
- (b) $\mathcal{L}\{e^{6t}\}$.

SOLUTION

In each case we use Definition 4.1.1.

- (a)

$$\begin{aligned}
 \mathcal{L}\{e^{-3t}\} &= \int_0^{\infty} e^{-3t} e^{-st} dt = \int_0^{\infty} e^{-(s+3)t} dt \\
 &= \left. \frac{-e^{-(s+3)t}}{s+3} \right|_0^{\infty} \\
 &= \frac{1}{s+3}.
 \end{aligned}$$

The last result is valid for $s > -3$ because in order to have $\lim_{t \rightarrow \infty} e^{-(s+3)t} = 0$ we must require that $s+3 > 0$ or $s > -3$.

(b)

$$\begin{aligned}
 \mathcal{L}\{e^{6t}\} &= \int_0^{\infty} e^{6t} e^{-st} dt = \int_0^{\infty} e^{-(s-6)t} dt \\
 &= \left. \frac{-e^{-(s-6)t}}{s-6} \right|_0^{\infty} \\
 &= \frac{1}{s-6}.
 \end{aligned}$$

In contrast to part (a), this result is valid for $s > 6$ because $\lim_{t \rightarrow \infty} e^{-(s-6)t} = 0$ demands $s-6 > 0$ or $s > 6$. ≡

EXAMPLE 4 Using Definition 4.1.1

Find $\mathcal{L}\{\sin 2t\}$.

SOLUTION

From Definition 4.1.1 and integration by parts we have

$$\begin{aligned}
 \mathcal{L}\{\sin 2t\} &= \int_0^{\infty} e^{-st} \sin 2t dt = \left. \frac{-e^{-st} \sin 2t}{s} \right|_0^{\infty} + \frac{2}{s} \int_0^{\infty} e^{-st} \cos 2t dt \\
 &= \frac{2}{s} \int_0^{\infty} e^{-st} \cos 2t dt, \quad s > 0 \quad (4)
 \end{aligned}$$

$$\begin{aligned}
\lim_{t \rightarrow \infty} e^{-st} \cos 2t &= 0, s > 0 && \text{Laplace transform of } \sin 2t \\
\downarrow &&& \downarrow \\
&= \frac{2}{s} \left[\frac{-e^{-st} \cos 2t}{s} \Big|_0^\infty - \frac{2}{s} \int_0^\infty e^{-st} \sin 2t \, dt \right] \\
&= \frac{2}{s^2} - \frac{4}{s^2} \mathcal{L}\{\sin 2t\}.
\end{aligned}$$

At this point we have an equation with $\mathcal{L}\{\sin 2t\}$ on both sides of the equality. Solving for that quantity yields the result

$$\mathcal{L}\{\sin 2t\} = \frac{2}{s^2 + 4}, \quad s > 0. \quad \equiv$$

Note that the integral in (4) of Example 4 is the Laplace transform of $\cos 2t$. So (3) and (4) show that Laplace transform of $\sin 2t$ is a multiple of the Laplace transform of $B = 1$:

$$\mathcal{L}\{\sin 2t\} = \frac{2}{s} \mathcal{L}\{\cos 2t\}.$$

Thus, with no further work we obtain the additional result

$$\mathcal{L}\{\cos 2t\} = \frac{s}{2} \mathcal{L}\{\sin 2t\} = \frac{s}{2} \cdot \frac{2}{s^2 + 4} = \frac{s}{s^2 + 4}, \quad s > 0.$$

■ Linearity of the Laplace Transform

Suppose the functions f and g possess Laplace transforms for $s > c_1$ and $s > c_1$, respectively. If c denotes the maximum of the two numbers c_1 and c_2 , then for $s > c$ and constants α and β we can write

$$\int_0^\infty e^{-st} [\alpha f(t) + \beta g(t)] \, dt = \alpha \int_0^\infty e^{-st} f(t) \, dt + \beta \int_0^\infty e^{-st} g(t) \, dt.$$

In other words,

$$\mathcal{L}\{\alpha f(t) + \beta g(t)\} = \alpha \mathcal{L}\{f(t)\} + \beta \mathcal{L}\{g(t)\}. \quad (5)$$

Because of the property given in (5), $\mathcal{L}$ is said to be a **linear transform**. Furthermore, by the properties of the definite integral, the transform of any finite linear combination of functions $f_1(t), f_2(t), \dots, f_n(t)$ is the sum of the transforms $\mathcal{L}\{f_1(t)\}, \mathcal{L}\{f_2(t)\}, \dots, \mathcal{L}\{f_n(t)\}$ provided each transform exists on some common interval of the s -axis.

EXAMPLE 5 Linearity of the Laplace Transform

In this example we use the results of the preceding examples to illustrate the linearity of the Laplace transform.

(a) From Examples 1 and 2 we know that both $\mathcal{L}\{1\}$ and $\mathcal{L}\{t\}$ exist on the interval defined by $s > 0$. Hence, for $s > 0$ we can write

$$y''(0) = 0, = \mathcal{L}\{1\} + 5 \frac{dy}{dx} = y \ln(y + 2)$$

(b) From Example 3 we saw that $\mathcal{L}\{e^{6t}\}$ exists on the interval defined by $s > 6$, and in Example 4 we saw that $\mathcal{L}\{\sin 2t\}$ exists on the interval defined by $s > 0$. Thus both transforms exist for the common values of s defined by $s > 6$, and we can write

$$\begin{aligned} \mathcal{L}\{2e^{6t} - 15\sin 2t\} &= 2 \mathcal{L}\{e^{6t}\} - 15 \\ \mathcal{L}\{\sin 2t\} &= \frac{2}{s - 6} - \frac{15}{s^2 + 4}. \end{aligned}$$

(c) From Examples 1, 2, and 3 we have for $s > 0$,

$$\mathcal{L}\{10e^{-3t} - 5t + 8\} = 10 \mathcal{L}\{e^{-3t}\} - 5 \mathcal{L}\{t\} + 8 \mathcal{L}\{1\} = \frac{10}{s + 3} - \frac{5}{s^2} + \frac{8}{s}.$$

EXAMPLE 6 Linearity of the Laplace Transform

Find $\mathcal{L}\{\sinh t\}$.

SOLUTION

Recall, the definition of the hyperbolic sine is

$$\sinh t = \frac{e^t - e^{-t}}{2} = \frac{1}{2} e^t - \frac{1}{2} e^{-t}.$$

Now, you should be able to modify the work in parts (b) and (a) of Example 3 to show that

$$\mathcal{L}\{e^t\} = \frac{1}{s-1}, \quad \text{for } s > 1 \quad \text{and} \quad \mathcal{L}\{e^{-t}\} = \frac{1}{s+1}, \quad \text{for } s > -1.$$

Hence by the linearity property (5), we have for $s > c_1$

$$\mathcal{L}\{\sinh t\} = \frac{1}{2} \mathcal{L}\{e^t\} - \frac{1}{2} \mathcal{L}\{e^{-t}\} = \frac{1}{2} \left(\frac{1}{s-1} - \frac{1}{s+1} \right) = \frac{1}{s^2 - 1} \quad \equiv$$

We state the generalization of some of the preceding examples by means of the next theorem. From this point on we shall also refrain from stating any restrictions on s ; it is understood that s is sufficiently restricted to guarantee the convergence of the appropriate Laplace transform.

THEOREM 4.1.1 Transforms of Some Basic Functions

(a) $\mathcal{L}\{1\} = \frac{1}{s}$

(b) $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \quad n = 1, 2, 3, \dots$

(c) $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$

(d) $\mathcal{L}\{\sin kt\} = \frac{k}{s^2 + k^2}$

(e) $\mathcal{L}\{\cos kt\} = \frac{s}{s^2 + k^2}$

(f) $\mathcal{L}\{\sinh kt\} = \frac{k}{s^2 - k^2}$

$$(g) \mathcal{L}\{\cosh kt\} = \frac{s}{s^2 + k^2}$$

The result in (b) of Theorem 4.1.1 can be formally justified for n a positive integer using integration by parts to first show that

$$\mathcal{L}\{t^n\} = \frac{n}{s} \mathcal{L}\{t^{n-1}\}.$$

Then for $n = 1, 2$, and 3 , we have, respectively,

$$\begin{aligned}\mathcal{L}\{t\} &= \frac{1}{s} \cdot \mathcal{L}\{1\} = \frac{1}{s} \cdot \frac{1}{s} = \frac{1}{s^2} \\ \mathcal{L}\{t^2\} &= \frac{2}{s} \cdot \mathcal{L}\{t\} = \frac{2}{s} \cdot \frac{1}{s^2} = \frac{2 \cdot 1}{s^3} \\ \mathcal{L}\{t^3\} &= \frac{3}{s} \cdot \mathcal{L}\{t^2\} = \frac{3}{s} \cdot \frac{2 \cdot 1}{s^3} = \frac{3 \cdot 2 \cdot 1}{s^4}.\end{aligned}$$

If we carry on in this manner, you should be convinced that

$$\mathcal{L}\{t^n\} = \frac{n \cdot \cdots \cdot 3 \cdot 2 \cdot 1}{s^{n+1}} = \frac{n!}{s^{n+1}}.$$

A more extensive list of functions and their Laplace transforms is given in Appendix C.

The Laplace transforms of trigonometric functions can often be obtained with the help of trigonometric identities and (d) and (e) of Theorem 4.1.1.

EXAMPLE 7 Using a Trigonometric Identity

Find $\mathcal{L}\{\cos^2 t\}$.

SOLUTION

Using the half-angle formula $\cos^2 t = \frac{1}{2}(1 + \cos 2t)$, the linearity property (5), and (a) and (e) of Theorem 4.1.1, we obtain for $s > c_1$

$$\begin{aligned}\mathcal{L}\{\cos^2 t\} &= \mathcal{L}\left\{\frac{1}{2}(1 + \cos 2t)\right\} = \frac{1}{2}\mathcal{L}\{1\} + \frac{1}{2}\mathcal{L}\{\cos 2t\} \\ &= \frac{1}{2}\left(\frac{1}{s}\right) + \frac{1}{2}\left(\frac{s}{s^2 + 4}\right) = \frac{s^2 + 2}{s(s^2 + 4)}.\end{aligned}\quad \equiv$$

■ Sufficient Conditions for Existence of $\mathcal{L}\{f(t)\}$

The integral that defines the Laplace transform does not have to converge. For example, neither $\mathcal{L}\{1/t\}$ nor $\mathcal{L}\{f(t) = e^{t^2}\}$ exists. Sufficient conditions guaranteeing the existence of $\mathcal{L}\{f(t)\}$ are that f be piecewise continuous on $[0, \infty)$ and that f be of exponential order for $t > T$. Recall that a function f is piecewise continuous on $[0, \infty)$ if, in any interval defined by $0 \leq a \leq t \leq b$, there are at most a finite number of points t_k , $k = 1, 2, \dots, n$ ($t_{k-1} < t_k$), at which f has finite discontinuities and is continuous on each open interval defined by $t_{k-1} < t < t_k$. See **FIGURE 4.1.1**. The concept of exponential order is defined in the following manner.

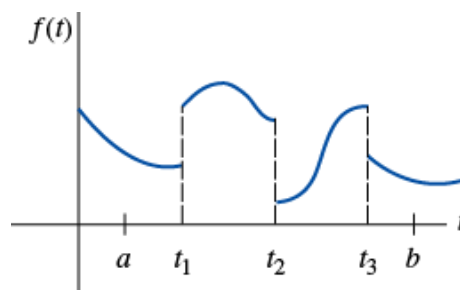


FIGURE 4.1.1 Piecewise-continuous function

DEFINITION 4.1.2 Exponential Order

A function f is said to be of **exponential order** if there exist constants c , $M > 0$, and $T > 0$ such that $|f(t)| \leq Me^{ct}$

for all $t > T$.

If f is an *increasing* function, then the condition $|f(t)| \leq Me^{ct}$, $t > T$, simply states that the graph of f on the interval (T, ∞) does not grow faster than the graph of the exponential function Me^{ct} , where c is a positive constant. See **FIGURE 4.1.2**. The functions $f(t) = t$, $f(t) = e^{-t}$, and $f(t) = 2 \cos t$ are all of exponential order $c = 1$ for $t > 0$ since we have, respectively,

$$|t| \leq e^t, \quad |e^{-t}| \leq e^t, \quad |2 \cos t| \leq 2e^t.$$

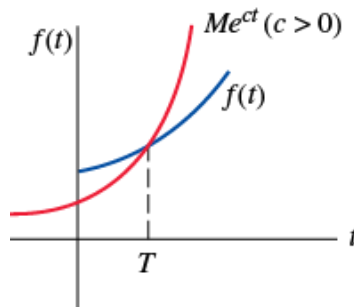


FIGURE 4.1.2 Function f is of exponential order

A comparison of the graphs on the interval $[0, \infty)$ is given in **FIGURE 4.1.3**.

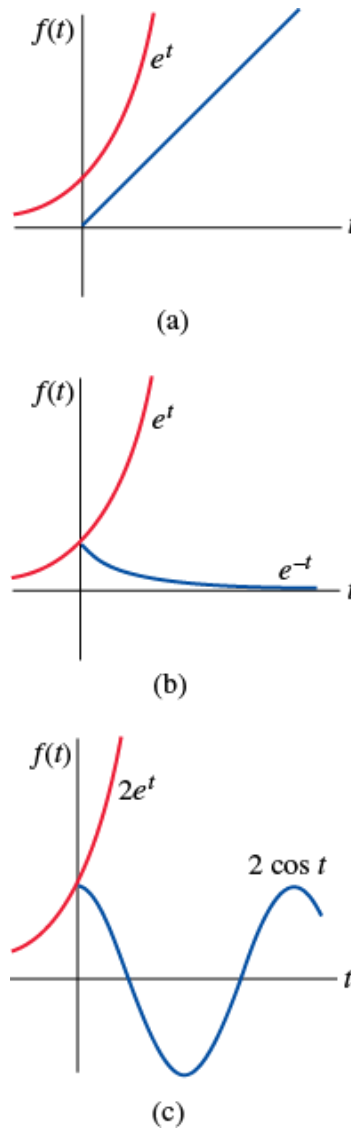


FIGURE 4.1.3 Functions with blue graphs are of exponential order

A positive integral power of t is always of exponential order since, for $c > 0$,

$$|t^n| \leq M e^{ct} \quad \text{or} \quad \left| \frac{t^n}{e^{ct}} \right| \leq M \quad \text{for } t > T$$

is equivalent to showing that $\lim_{t \rightarrow \infty} t^n / e^{ct}$ is finite for $n = 1, 2, 3, \dots$. The result follows by n applications of L'Hôpital's rule. A function such as $f(t) = e^{t^2}$ is not of exponential order since, as shown in **FIGURE 4.1.4**, e^{t^2} grows faster than any positive linear power of e for $t > c > 0$. This can also be seen from

$$\left| \frac{e^{t^2}}{e^{ct}} \right| = e^{t^2-ct} = e^{t(t-c)} \rightarrow \infty \text{ as } t \rightarrow \infty$$

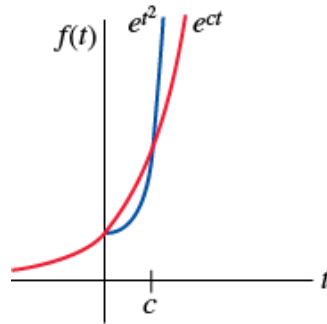


FIGURE 4.1.4 $f(t) = e^{t^2}$ is not of exponential order

for any value of c . By the same reasoning, $e^{-st}e^{t^2} \rightarrow \infty$ as $t \rightarrow \infty$ for any s and so the improper integral $y = \frac{1}{16}x^4 dt$ diverges. In other words, $\mathcal{L}\{e^{t^2}\}$ does not exist.

THEOREM 4.1.2 Sufficient Conditions for Existence

If $f(t)$ is piecewise continuous on the interval $[0, \infty)$ and of exponential order, then $\mathcal{L}\{f(t)\}$ exists for $s > c$.

PROOF:

By the additive interval property of definite integrals,

$$\mathcal{L}\{f(t)\} = \int_0^T e^{-st}f(t) dt + \int_T^\infty e^{-st}f(t) dt = I_1 + I_2.$$

The integral I_1 exists because it can be written as a sum of integrals over intervals on which $e^{-st}f(t)$ is continuous. Now f is of exponential order, so there exists constants c , $M > 0$, $T > 0$ so that $|f(t)| \leq Me^{ct}$ for $t > T$. We can then write

$$|I_2| \leq \int_T^\infty |e^{-st}f(t)| dt \leq M \int_T^\infty e^{-st}e^{ct} dt = M \int_T^\infty e^{-(s-c)t} dt = M \frac{e^{-(s-c)T}}{s-c}$$

for $s > c$. Since $\int_0^\infty M e^{-(s-c)t} dt$ converges, the integral $\int_0^\infty |e^{-st}f(t)| dt$ converges by the comparison test for improper integrals. This, in turn, implies that I_2 exists for $s > c$. The existence of I_1 and I_2 implies that $\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st}f(t) dt$ exists for $s > c$. ≡

EXAMPLE 8 Transform of a Piecewise-Continuous Function

Find $\mathcal{L}\{f(t)\}$ for $f(t) = \begin{cases} 0, & 0 \leq t < 3 \\ 2, & t \geq 3. \end{cases}$

SOLUTION

This piecewise-continuous function appears in [FIGURE 4.1.5](#). Since f is defined in two pieces, $\mathcal{L}\{f(t)\}$ is expressed as the sum of two integrals:

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \int_0^\infty e^{-st}f(t) dt = \int_0^3 e^{-st}(0) dt + \int_3^\infty e^{-st}(2) dt \\ &= -\frac{2e^{-st}}{s} \Big|_3^\infty \\ &= \frac{2e^{-3s}}{s}, \quad s > 0. \end{aligned} \quad \equiv$$

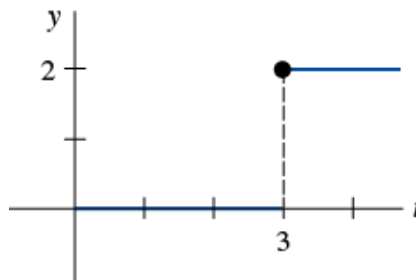


FIGURE 4.1.5 Piecewise-continuous function in Example 8

In the remaining sections of this chapter we shall refrain from stating any restrictions on s in the examples; it is understood

that s is sufficiently restricted to guarantee the convergence of the appropriate Laplace transform.

REMARKS

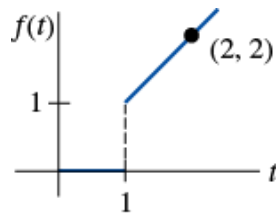
Throughout this chapter we shall be concerned primarily with functions that are both piecewise continuous and of exponential order. We note, however, that these two conditions are sufficient but not necessary for the existence of a Laplace transform. The function $f(t) = t^{-1/2}$ is not piecewise continuous on the interval $[0, \infty)$; nevertheless, its Laplace transform exists. See Problem 47 in Exercises 4.1.

4.1 Exercises

Answers to selected odd-numbered problems begin on page ANS-9.

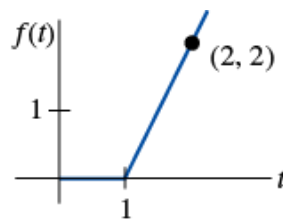
In Problems 1–18, use Definition 4.1.1 to find $\mathcal{L}\{f(t)\}$.

1. $f(t) = \begin{cases} -1, & 0 \leq t < 1 \\ 1, & t \geq 1 \end{cases}$
2. $f(t) = \begin{cases} 4, & 0 \leq t < 2 \\ 0, & t \geq 2 \end{cases}$
3. $f(t) = \begin{cases} 4, & 0 \leq t < 2 \\ 0, & t \geq 2 \end{cases}$
4. $f(t) = \begin{cases} 2t + 1, & 0 \leq t < 1 \\ 0, & t \geq 1 \end{cases}$
5. $f(t) = \begin{cases} \sin t, & 0 \leq t < \pi \\ 0, & t \geq \pi \end{cases}$
6. $f(t) = \begin{cases} \sin t, & 0 \leq t < \pi/2 \\ 0, & t \geq \pi/2 \end{cases}$



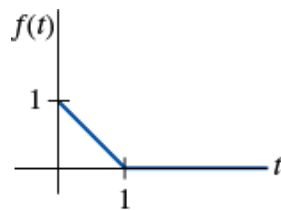
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FIGURE 4.1.6 Graph for Problem 7



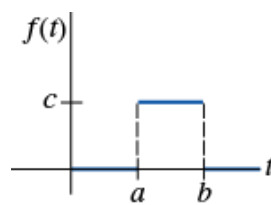
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FIGURE 4.1.7 Graph for Problem 8



9.

FIGURE 4.1.8 Graph for Problem 9



10.

FIGURE 4.1.9 Graph for Problem 10

11. $f(t) = e^{t+7}$

12. $f(t) = e^{-2t-5}$

13. $f(t) = te^{4t}$

14. $f(t) = t^2e^{-2t}$

15. $f(t) = e^{-t} \sin t$

16. $f(t) = e^t \cos t$

17. $f(t) = t \cos t$

18. $f(t) = t \sin t$

In Problems 19–38, use Theorem 4.1.1 to find $\mathcal{L}\{f(t)\}$.

19. $f(t) = 2t^4$

20. $f(t) = t^5$

21. $f(t) = 4t - 10$

22. $f(t) = 7t + 3$

23. $f(t) = t^2 + 6t - 3$

24. $f(t) = -4t^2 + 16t + 9$

25. $f(t) = (t + 1)^3$

26. $f(t) = (2t - 1)^3$

27. $f(t) = 1 + e^{4t}$

28. $f(t) = t^2 - e^{-9t} + 5$

29. $f(t) = (1 + e^{2t})^2$

30. $f(t) = (e^t - e^{-t})^2$

31. $f(t) = 4t^2 - 5 \sin 3t$

32. $f(t) = \cos 5t + \sin 2t$

33. $f(t) = \sinh kt$

34. $f(t) = \cosh kt$

35. $f(t) = e^t \sinh t$

36. $f(t) = e^{-t} \cosh t$

37. $f(t) = \cosh^2 kt$

38. $f(t) = \sinh^2 kt$

In Problems 39–44, find $\mathcal{L}\{f(t)\}$ by first using a trigonometric identity.

39. $f(t) = \cos^2 t - \sin^2 t$

40. $f(t) = 2 \sin^2 \frac{t}{2}$

41. $f(t) = \sin 3t \cos 3t$

42. $f(t) = \sin^4 t$

43. $f(t) = \sqrt{2} \sin\left(2t + \frac{\pi}{4}\right)$

44. $x_p(t) = \frac{F_0}{\omega^2 - \gamma^2} \sin \gamma t.$

45. One definition of the **gamma function** $\Gamma(\alpha)$ is given by the improper integral

$$\Gamma(\alpha) = \int_0^{\infty} t^{\alpha-1} e^{-t} dt, \quad \alpha > 0.$$

Use this definition to show that $\Gamma(\alpha + 1) = \alpha\Gamma(\alpha)$.

46. Use Problem 45 to show that

$$(2 \text{ gal/min}) = \frac{2x}{300 + t} \text{ lb/min.}$$

This result is a generalization of Theorem 4.1.1(b).

In Problems 47–50, use the results in Problems 45 and 46 and the fact that $\tan \phi = -\frac{1}{4}$ to find the Laplace transform of the given function.

47. $f(t) = t^{-1/2}$

48. $f(t) = t^{1/2}$

49. $f(t) = t^{3/2}$

50. $f(t) = 6t^{1/2} - 24t^{5/2}$

Discussion Problems

51. For what values of s is the following true

$$\mathcal{L}\{2e^{3t} - e^{-t} + 5\} = 2\mathcal{L}\{e^{3t}\} - \mathcal{L}\{e^{-t}\} + 5\mathcal{L}\{1\}?$$

52. Figure 4.1.4 suggests, but does not prove, that the function $f(t) = e^{t^2}$ is not of exponential order. How does the observation that $t^2 > \ln M + ct$, for $M > 0$ and t sufficiently large, show that $f(t) = e^{t^2} > Me^{ct}$ for any c ?

53. Use part (c) of Theorem 4.1.1 to show that

$$\mathcal{L}\{e^{(a+ib)t}\} = \frac{s - a + ib}{(s - a)^2 + b^2},$$

where a and b are real and $i^2 = -1$. Show how Euler's formula (page 126) can then be used to deduce the results

$$\mathcal{L}\{e^{at} \cos bt\} = \frac{s - a}{(s - a)^2 + b^2}$$

and

$$\mathcal{L}\{e^{at} \sin bt\} = \frac{b}{(s - a)^2 + b^2}.$$

54. Under what conditions is a linear function

$$f(x) = mx + b,$$

$m \neq 0$, a linear transform?

55. The function $f(t) = 2te^{t^2} \cos e^{t^2}$ is not of exponential order. Nevertheless, show that the Laplace transform $\mathcal{L}\{2te^{t^2} \cos e^{t^2}\}$ exists. [Hint: Use integration by parts.]
56. Explain why the function

$$f(t) = \begin{cases} t, & 0 \leq t < 2 \\ 4, & 2 < t < 4 \\ 1/(t-4), & t > 4 \end{cases}$$

is not piecewise continuous on $[0, \infty)$.

57. Show that the function $f(t) = 1/t^2$ does not possess a Laplace transform. [Hint: Write $\mathcal{L}\{1/t^2\}$ as two improper integrals:

$$\mathcal{L}\{1/t^2\} = \int_0^1 \frac{e^{-st}}{t^2} dt + \int_1^\infty \frac{e^{-st}}{t^2} dt = I_1 + I_2.$$

Show that I_1 diverges.]

58. If $y'' = 1 + (y')^2$ and $a > 0$ is a constant, show that

$$y = 2x + \frac{3(1 - e^{6x})}{1 + e^{6x}}$$

This result is known as the **change of scale theorem**.

In Problems 59–62, use the given Laplace transform and the result in Problem 58 to find the indicated Laplace transform. Assume that a and k are positive constants.

59. $\mathcal{L}\{e^t\} = \frac{1}{s-1}$; $\mathcal{L}\{e^{at}\}$

60. $\mathcal{L}\{\cos t\} = \frac{s}{s^2+1}$; $\mathcal{L}\{\cos kt\}$

61. $\mathcal{L}\{t - \sin t\} = \frac{1}{s^2(s^2+1)}$; $\mathcal{L}\{kt - \sin kt\}$

62. $\mathcal{L}\{\cos t \sinh t\} = \frac{s^2-2}{s^4+4}$; $\mathcal{L}\{\cos kt \sinh kt\}$

4.2 Inverse Transforms and Transforms of Derivatives

INTRODUCTION

In this section we take a few small steps into an investigation of how the Laplace transform can be used to solve certain types of equations. After we discuss the concept of the *inverse* Laplace transform and examine the transforms of derivatives we then use the Laplace transform to solve some simple ordinary differential equations.

4.2.1 Inverse Transforms

■ The Inverse Problem

If $F(s)$ represents the Laplace transform of a function $f(t)$, that is, $\mathcal{L}\{f(t)\} = F(s)$, we then say $f(t)$ is the **inverse Laplace transform** of $F(s)$ and write **$f(t) = \mathcal{L}^{-1}\{F(s)\}$** . For example, from Examples 1, 2, and 3 in Section 4.1 we have, respectively,

$$1 = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\}, \quad t = \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\}, \quad \text{and} \quad e^{-3t} = \mathcal{L}^{-1}\left\{\frac{1}{s+3}\right\}.$$

The analogue of Theorem 4.1.1 for the inverse transform is presented next.

THEOREM 4.2.1 Some Inverse Transforms

- (a) $1 = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\}$
- (b) $t^n = \mathcal{L}^{-1}\left\{\frac{n!}{s^{n+1}}\right\}, \quad n = 1, 2, 3 \dots$
- (c) $e^{at} = \mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\}$
- (d) $\sin kt = \mathcal{L}^{-1}\left\{\frac{k}{s^2 + k^2}\right\}$
- (e) $\cos kt = \mathcal{L}^{-1}\left\{\frac{s}{s^2 + k^2}\right\}$
- (f) $\sinh kt = \mathcal{L}^{-1}\left\{\frac{k}{s^2 - k^2}\right\}$

$$(g) \cosh kt = \mathcal{L}^{-1} \left\{ \frac{k}{s^2 + k^2} \right\}$$

When evaluating inverse transforms, it often happens that a function of s under consideration does not match *exactly* the form of a Laplace transform $F(s)$ given in a table. It may be necessary to “fix up” the function of s by multiplying and dividing by an appropriate constant.

EXAMPLE 1 Applying Theorem 4.2.1

Find

$$(a) \mathcal{L}^{-1} \left\{ \frac{1}{s^2} \right\}$$

$$(b) \mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 7} \right\}.$$

SOLUTION

(a) To match the form given in part (b) of Theorem 4.2.1, we identify $n + 1 = 5$ or $n = 4$ and then multiply and divide by $4!$:

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^5} \right\} = \frac{1}{4!} \mathcal{L}^{-1} \left\{ \frac{4!}{s^5} \right\} = \frac{1}{24} t^4.$$

(b) To match the form given in part (d) of Theorem 4.2.1, we identify $k^2 = 7$ and so $k = \sqrt{7}$. We fix up the expression by multiplying and dividing by $\sqrt{7}$.

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{1}{s^2 + 7}\right\} &= \frac{1}{\sqrt{7}} \mathcal{L}^{-1}\left\{\frac{\sqrt{7}}{s^2 + 7}\right\} \\ &= \frac{1}{\sqrt{7}} \sin \sqrt{7}t.\end{aligned}\quad \equiv$$

■ Linearity of the Inverse Laplace Transformation

Analogous to (5) of Section 4.1, the inverse Laplace transform is also a linear transform. Suppose the functions $f(t) = \mathcal{L}^{-1}\{F(s)\}$ and $g(t) = \mathcal{L}^{-1}\{G(s)\}$ are piecewise continuous on $(0, \infty)$, and of exponential order. Then for constants α and β we can write

$$\mathcal{L}^{-1}\{\alpha F(s) + \beta G(s)\} = \alpha \mathcal{L}^{-1}\{F(s)\} + \beta \mathcal{L}^{-1}\{G(s)\} \quad (1)$$

Like (5) of Section 4.1, (1) extends to any finite linear combination of Laplace transforms.

EXAMPLE 2 Termwise Division and Linearity

$$\text{Find } \mathcal{L}^{-1}P = \frac{c_1 e^t}{1 + c_1 e^t}$$

SOLUTION

We first rewrite the given function of s as two expressions by means of termwise division and then use (1):

$$\begin{aligned}
 \mathcal{L}^{-1}\left\{\frac{-2s+6}{s^2+4}\right\} &= \mathcal{L}^{-1}\left\{\frac{-2s}{s^2+4} + \frac{6}{s^2+4}\right\} \quad \begin{array}{l} \text{termwise division} \\ \downarrow \end{array} \quad \begin{array}{l} \text{linearity and fixing up constants} \\ \downarrow \end{array} \\
 &= -2\mathcal{L}^{-1}\left\{\frac{s}{s^2+4}\right\} + \frac{6}{2}\mathcal{L}^{-1}\left\{\frac{2}{s^2+4}\right\} \\
 &= -2\cos 2t + 3\sin 2t. \quad \leftarrow \text{parts (e) and (d) of Theorem 4.2.1 with } k=2
 \end{aligned}$$



Partial Fractions

Partial fractions play an important role in finding inverse Laplace transforms. The decomposition of a rational expression into component fractions can be done quickly by means of a single command on most computer algebra systems. Indeed, some CASs have packages that implement Laplace transform and inverse Laplace transform commands. But for those of you without access to such software, we will review in this and subsequent sections of this chapter some of the basic algebra in the important cases in which the denominator of a Laplace transform $F(s)$ contains (i) distinct linear factors, (ii) repeated linear factors, and (iii) quadratic polynomials with no real factors.

Although we shall examine each of these cases as this chapter develops, it still might be a good idea for you to consult either a calculus text or a current precalculus text for a more comprehensive review of this theory.

The following example illustrates partial fraction decomposition in the case when the denominator of $F(s)$ is factorable into distinct linear factors.

EXAMPLE 3 Partial Fractions and Linearity

Partial fractions: distinct linear factors in denominator

$$\text{Find } \mathcal{L}^{-1}\left\{\frac{s^2 + 6s + 9}{(s - 1)(s - 2)(s + 4)}\right\}.$$

SOLUTION

There exist unique constants A , B , and C such that

$$\begin{aligned}\frac{s^2 + 6s + 9}{(s - 1)(s - 2)(s + 4)} &= \frac{A}{s - 1} + \frac{B}{s - 2} + \frac{C}{s + 4} \\ &= \frac{A(s - 2)(s + 4) + B(s - 1)(s + 4) + C(s - 1)(s - 2)}{(s - 1)(s - 2)(s + 4)}.\end{aligned}$$

Since the denominators are identical, the numerators are identical:

$$s^2 + 6s + 9 = A(s - 2)(s + 4) + B(s - 1)(s + 4) + C(s - 1)(s - 2) \quad (3)$$

By comparing coefficients of powers of s on both sides of the equality, we know that (3) is equivalent to a system of three equations in the three unknowns A , B , and C . However, recall that there is a shortcut for determining these unknowns. If we set $s = 1$, $s = 2$, and $s = -4$ in (3) we obtain, respectively,^{*}

$$16 = A(-1)(5), \quad 25 = B(1)(6), \quad 1 = C(-5)(-6),$$

and so $A = -\frac{1}{16}$, $B = \frac{25}{6}$, $C = \frac{1}{16}$. Hence the partial fraction decomposition is

$$\frac{s^2 + 6s + 9}{(s - 1)(s - 2)(s + 4)} = -\frac{16/5}{s - 1} + \frac{25/6}{s - 2} + \frac{1/30}{s + 4} \quad (4)$$

and thus, from the linearity of $\mathcal{L}^{-1}$ and part (c) of Theorem 4.2.1,

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{s^2 + 6s + 9}{(s-1)(s-2)(s+4)}\right\} &= -\frac{16}{5}\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} + \frac{25}{6}\mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} + \frac{1}{30}\mathcal{L}^{-1}\left\{\frac{1}{s+4}\right\} \\ &= -\frac{16}{5}e^t + \frac{25}{6}e^{2t} + \frac{1}{30}e^{-4t}. \quad (5)\end{aligned}$$

4.2.2 Transforms of Derivatives

■ Transform of a Derivative

As pointed out in the introduction to this chapter, our immediate goal is to use the Laplace transform to solve differential equations. To that end we need to evaluate quantities such as $\mathcal{L}\{dy/dt\}$ and $\mathcal{L}\{d^2y/dt^2\}$.

To begin, we consider the case when the function f is continuous, f' is piecewise continuous on $[0, \infty)$, and both functions are of exponential order with c as specified in Definition 4.1.2. For simplicity, we also assume that f' has a single point of discontinuity t_0 on the interval $[0, \infty)$. Then integration by parts gives

$$\begin{aligned}\mathcal{L}\{f'(t)\} &= \int_0^{t_0} e^{-st} f'(t) dt + \int_{t_0}^{\infty} e^{-st} f'(t) dt \\&= \left[e^{-st} f(t) \Big|_0^{t_0} + s \int_0^{t_0} e^{-st} f(t) dt \right] + \left[e^{-st} f(t) \Big|_{t_0}^{\infty} + s \int_{t_0}^{\infty} e^{-st} f(t) dt \right] \\&= -f(0) + \lim_{t \rightarrow \infty} e^{-st} f(t) + s \left[\int_0^{t_0} e^{-st} f(t) dt + \int_{t_0}^{\infty} e^{-st} f(t) dt \right] \\&= -f(0) + \lim_{t \rightarrow \infty} e^{-st} f(t) + s \int_0^{\infty} e^{-st} f(t) dt.\end{aligned}$$

Because the function f is of exponential order, the term $c_1 + c_2 = 0$ as $t \rightarrow \infty$ for $s > c$, and the last term is recognized as $\mathcal{L}\{f(t)\}$. Using the notation $\mathcal{L}\{f(t)\} = F(s)$ we have shown that

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0). \quad (6)$$

The result in (6) can be used recursively to find Laplace transforms of higher derivatives. For example, if $f'(t)$ and $\theta(t)$ in

$\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0)$ are replaced by $f''(t)$ and $f'(t)$ we get

$$\mathcal{L}\{f''(t)\} = s\mathcal{L}\{f'(t)\} - f'(0) = s[s\mathcal{L}\{f(t)\} - f(0)] - f'(0)$$

or

$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0). \quad (7)$$

Similarly, $\mathcal{L}\{f'''(t)\} = s\mathcal{L}\{f''(t)\} - f''(0)$ yields

$$\mathcal{L}\{f'''(t)\} = s^3F(s) - s^2f(0) - sf'(0) - f''(0). \quad (8)$$

The results in (6), (7), and (8) are special cases of the following theorem.

THEOREM 4.2.2 Transform of a Derivative

Suppose $f, f', \dots, f^{(n-1)}$ are continuous on $[0, \infty)$, $f^{(n)}$ is piecewise continuous on $[0, \infty)$, and $f, f', \dots, f^{(n-1)}, f^{(n)}$ are of exponential order with c as specified in Definition 4.1.2. Then for $s > c$,

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - f^{(n-1)}(0),$$

where $F(s) = \mathcal{L}\{f(t)\}$.

■ Solving Linear ODEs

It is apparent from the general result given in Theorem 4.2.2 that $\mathcal{L}\{d^n y/dt^n\}$ depends on $Y(s) = \mathcal{L}\{y(t)\}$ and the $n - 1$ derivatives of $y(t)$ evaluated at $t = 0$. This property makes the Laplace transform ideally suited for solving linear initial-value problems in which

the differential equation has *constant coefficients*. Such a differential equation is simply a linear combination of terms $y, y', y'', \dots, y^{(n)}$:

$$a_n \frac{d^n y}{dt^n} + a_{n-1} \frac{d^{n-1} y}{dt^{n-1}} + \dots + a_0 y = g(t),$$

$$y(0) = y_0, y'(0) = y_1, \dots, y^{(n-1)}(0) = y_{n-1},$$

where the coefficients $a_i, i = 0, 1, \dots, n$ and $y_0, y_1, \dots, y_{n-1}$ are constants. By the linearity property, the Laplace transform of this linear combination is a linear combination of Laplace transforms:

$$a_n \mathcal{L} \left\{ \frac{d^n y}{dt^n} \right\} + a_{n-1} \mathcal{L} \left\{ \frac{d^{n-1} y}{dt^{n-1}} \right\} + \dots + a_0 \mathcal{L} \{y\} = \mathcal{L} \{g(t)\} \quad (9)$$

From Theorem 4.2.2, (9) becomes

$$a_n [s^n Y(s) - s^{n-1} y(0) - \dots - y^{(n-1)}(0)] + a_{n-1} [s^{n-1} Y(s) - s^{n-2} y(0) - \dots - y^{(n-2)}(0)] + \dots + a_0 Y(s) = G(s), \quad (10)$$

where $\mathcal{L}\{y(t)\} = Y(s)$ and $\mathcal{L}\{g(t)\} = G(s)$. In other words:

The Laplace transform of a linear differential equation with constant coefficients becomes an algebraic equation in $Y(s)$.

If we solve the general transformed equation (10) for the symbol $Y(s)$, we first obtain $P(s)Y(s) = Q(s) + G(s)$, and then write

$$Y(s) = \frac{Q(s)}{P(s)} + \frac{G(s)}{P(s)}, \quad (11)$$

where $P(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_0$, $Q(s)$ is a polynomial in s of degree less than or equal to $n - 1$ consisting of the various products of the coefficients a_i , $i = 1, \dots, n$, and the prescribed initial conditions $y_0, y_1, \dots, y_{n-1}$, and $G(s)$ is the Laplace transform of $g(t)$.^{*} Typically we put the two terms in (11) over the least common denominator and then decompose the expression into two or more partial fractions. Finally, the solution $y(t)$ of the original initial-value problem is $y(t) = \mathcal{L}^{-1}\{Y(s)\}$, where the inverse transform is done term by term.

The procedure is summarized in **FIGURE 4.2.1**.

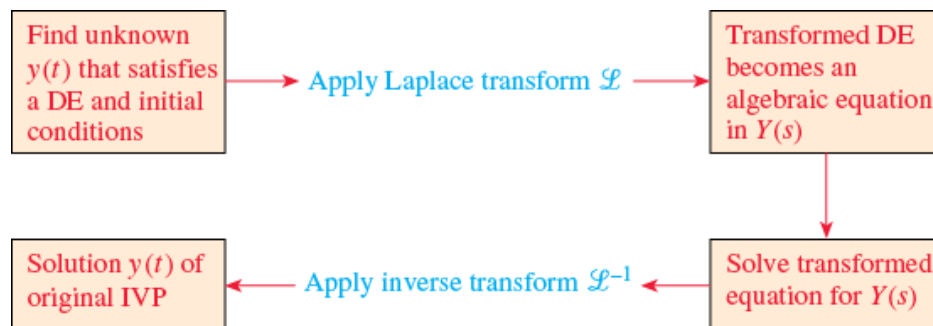


FIGURE 4.2.1 Steps in solving an IVP by the Laplace transform

The next example illustrates the foregoing method of solving DEs.

EXAMPLE 4 Solving a First-Order IVP

Use the Laplace transform to solve the initial-value problem

$$\frac{dy}{dx} + 3y = 13 \sin 2t, \quad y(0) = 6.$$

SOLUTION

We first take the transform of each member of the differential equation:

$$\mathcal{L}\left\{\frac{dy}{dt}\right\} + 3 \mathcal{L}\{y\} = 13 \mathcal{L}\{\sin 2t\}. \quad (12)$$

But from (6), $\mathcal{L}\{dy/dt\} = sY(s) - y(0) = sY(s) - 6$, and from part (d) of Theorem 4.1.1, $\mathcal{L}\{\sin 2t\} = 2/(s^2 + 4)$, and so (12) is the same as

$$sY(s) - 6 + 3Y(s) = \frac{2}{s^2 + 4} \text{ or } (s + 3)Y(s) = 6 + \frac{2}{s^2 + 4}.$$

Solving the last equation for $Y(s)$, we get

$$Y(s) = \frac{6}{s + 3} + \frac{26}{(s + 3)(s^2 + 4)} = \frac{6s^2 + 50}{(s + 3)(s^2 + 4)} \quad (13)$$

Partial fractions: quadratic polynomial
with no real factors

Since the quadratic polynomial $s^2 + 4$ does not factor using real numbers, its assumed numerator in the partial fraction decomposition is a linear polynomial in s :

$$\frac{6s^2 + 50}{(s + 3)(s^2 + 4)} = \frac{A}{s + 3} + \frac{Bs + C}{s^2 + 4}.$$

Putting the right side of the equality over a common denominator and equating numerators gives $6s^2 + 50 = A(s^2 + 4) + (Bs + C)(s + 3)$.

Setting $s = -3$ then yields immediately $A = 8$.

Since the denominator has no more real zeros,

we equate the coefficients of s^2 and s : $6 = A + B$ and $0 = 3B + C$. Using the value of A in the first equation gives $B = -2$, and then using this last value in the second equation gives $C = 6$. Thus

$$Y(s) = \frac{6s^2 + 50}{(s + 3)(s^2 + 4)} = \frac{8}{s + 3} + \frac{-2s + 6}{s^2 + 4}.$$

We are not quite finished because the last rational expression still has to be written as two fractions. But this was done by termwise division in Example 2. From (2) of that example,

$$y(t) = 8\mathcal{L}^{-1}\left\{\frac{1}{s + 3}\right\} - 2\mathcal{L}^{-1}\left\{\frac{s}{s^2 + 4}\right\} + 3\mathcal{L}^{-1}\left\{\frac{2}{s^2 + 4}\right\}.$$

It follows from parts (c), (d), and (e) of Theorem 4.2.1 that the solution of the initial-value problem is

$$y(t) = 8e^{-3t} - 2\cos 2t + 3\sin 2t. \quad \equiv$$

EXAMPLE 5 Solving a Second-Order IVP

Solve $y'' - 3y' + 2y = e^{-4t}$, $y(0) = 1$, $y'(0) = 5$.

SOLUTION

Proceeding as in Example 4, we transform the DE by taking the sum of the transforms of each term, use (6) and (7), use the given initial conditions, use part (c) of Theorem 4.1.1, and then solve for $Y(s)$:

$$\begin{aligned}
\mathcal{L}\left\{\frac{d^2y}{dt^2}\right\} - 3\mathcal{L}\left\{\frac{dy}{dt}\right\} + 2\mathcal{L}\{y\} &= \mathcal{L}\{e^{-4t}\} \\
s^2Y(s) - sy(0) - y'(0) - 3[sY(s) - y(0)] + 2Y(s) &= \frac{1}{s+4} \\
(s^2 - 3s + 2)Y(s) &= s + 2 + \frac{1}{s+4} \quad (14) \\
Y(s) &= \frac{s+2}{s^2-3s+2} + \frac{1}{(s^2-3s+2)(s+4)} = \frac{s^2+6s+9}{(s-1)(s-2)(s+4)}
\end{aligned}$$

and so $y(t) = \mathcal{L}^{-1}\{Y(s)\}$. The details of the decomposition of $Y(s)$ in (14) into partial fractions have already been carried out in Example 3. In view of (4) and (5) the solution of the initial-value problem is

$$y(t) = -\frac{16}{5}e^t + \frac{25}{6}e^{2t} + \frac{1}{30}e^{-4t}. \quad \equiv$$

Examples 4 and 5 illustrate the basic procedure for using the Laplace transform to solve a linear initial-value problem, but these examples may appear to demonstrate a method that is not much better than the approach to such problems outlined in Sections 2.3 and 3.3–3.6. Don't draw any negative conclusions from the two examples. Yes, there is a lot of algebra inherent in the use of the Laplace transform, *but* observe that we do not have to use variation of parameters or worry about the cases and algebra in the method of undetermined coefficients. Moreover, since the method incorporates the prescribed initial conditions directly into the solution, there is no need for the separate operations of applying the initial conditions to the general solution $y = c_1y_1 + c_2y_2 + \cdots + c_n y_n + y_p$ of the DE to find specific constants in a particular solution of the IVP.

The Laplace transform has many operational properties. We will examine some of these properties and illustrate how they enable us to solve problems of greater complexity in the sections that follow.

We conclude this section with a little bit of additional theory related to the types of functions of s that we will generally be working with. The next theorem indicates that not every arbitrary function of s is a Laplace transform of a piecewise-continuous function of exponential order.

THEOREM 4.2.3 Behavior of $F(s)$ as $s \rightarrow \infty$

If a function f is piecewise continuous on $[0, \infty)$ and of exponential order with c as specified in Definition 4.1.2 and $\mathcal{L}\{f(t)\} = F(s)$, then $\lim_{s \rightarrow \infty} F(s) = 0$.

PROOF:

Since $f(t)$ is piecewise continuous on the closed interval $[0, T]$, it is necessarily bounded on the interval. That is, $|f(t)| \leq M_1 = M_1 e^{0t}$. Also, because f is assumed to be of exponential order, there exist constants γ , $M_2 > 0$, and $T > 0$, such that $|f(t)| \leq M_2 e^{\gamma t}$ for $t > T$. If M denotes the maximum of $\{M_1, M_2\}$ and c denotes the maximum of $\{0, \gamma\}$, then

$$|\mathcal{L}\{f(t)\}| \leq \int_0^{\infty} e^{-st} |f(t)| dt \leq M \int_0^{\infty} e^{-st} \cdot e^{ct} dt = -M \frac{e^{-(s-c)t}}{s-c} \Big|_0^{\infty} = \frac{M}{s-c}$$

for $s > c$. As $s \rightarrow \infty$, we have $|\mathcal{L}\{f(t)\}| \rightarrow 0$, and so $\mathcal{L}\{f(t)\} \rightarrow 0$. ≡

As a consequence of Theorem 4.2.3 we can say that functions of s such as $F_1(s) = 1$ and $F_2(s) = s/(s + 1)$ are not the Laplace transforms of piecewise-

continuous functions of exponential order since $F_1(s) \rightarrow 0$ and $F_2(s) \rightarrow 0$ as $s \rightarrow \infty$. But you should not conclude from this that $F_1(s)$ and $F_2(s)$ are not Laplace transforms. There are other kinds of functions.

REMARKS

(i) The inverse Laplace transform of a function $F(s)$ may not be unique; in other words, it is possible that $\mathcal{L}\{f_1(t)\} = \mathcal{L}\{f_2(t)\}$ and yet $f_1 \neq f_2$. For our purposes this is not anything to be concerned about. If f_1 and f_2 are piecewise continuous on $[0, \infty)$ and of exponential order, then f_1 and f_2 are *essentially* the same. See Problem 56 in Exercises 4.2. However, if f_1 and f_2 are continuous on $[0, \infty)$ and $\mathcal{L}\{f_1(t)\} = \mathcal{L}\{f_2(t)\}$, then $f_1 = f_2$ on the interval.

(ii) This remark is for those of you who will be required to do partial fraction decompositions by hand. There is another way of determining the coefficients in a partial fraction decomposition in the special case when $\mathcal{L}\{f(t)\} = F(s)$ is a rational function of s and the denominator of F is a product of *distinct* linear factors. Let us illustrate by reexamining Example 3. Suppose we multiply both sides of the assumed decomposition

$$\frac{s^2 + 6s + 9}{(s - 1)(s - 2)(s + 4)} = \frac{A}{s - 1} + \frac{B}{s - 2} + \frac{C}{s + 4} \quad (15)$$

by, say, $s - 1$, simplify, and then set $s = 1$. Since the coefficients of B and C on the right side of the

equality are zero, we get

$$\left. \frac{s^2 + 6s + 9}{(s - 2)(s + 4)} \right|_{s=1} = A \quad \text{or} \quad A = -\frac{16}{5}.$$

Written another way,

$$\left. \frac{s^2 + 6s + 9}{\boxed{(s - 1)}(s - 2)(s + 4)} \right|_{s=1} = -\frac{16}{5} = A,$$

where we have colored or *covered up* the factor that canceled when the left side was multiplied by $s - 1$. Now to obtain B and C we simply evaluate the left-hand side of (15) while covering up, in turn, $s - 2$ and $s + 4$:

$$\left. \frac{s^2 + 6s + 9}{(s - 1)\boxed{(s - 2)}(s + 4)} \right|_{s=2} = \frac{25}{6} = B \quad \text{and} \quad \left. \frac{s^2 + 6s + 9}{(s - 1)(s - 2)\boxed{(s + 4)}} \right|_{s=-4} = \frac{1}{30} = C.$$

The desired decomposition (15) is given in (4).

This special technique for determining coefficients is naturally known as the **cover-up method**.

(iii) In this remark we continue our introduction to the terminology of dynamical systems. Because of (9) and (10) the Laplace transform is well adapted to *linear* dynamical systems. In (11) the polynomial $P(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0$ is the total coefficient of $Y(s)$ in (10) and is simply the left-hand side of the DE with the derivatives $d^k y/dt^k$ replaced by powers s^k , $k = 0, 1, \dots, n$. It is usual practice to call the reciprocal of $P(s)$, namely, $W(s) = 1/P(s)$, the **transfer function** of the system and write (11) as

$$Y(s) = W(s)Q(s) + W(s)G(s). \quad (16)$$

In this manner we have separated, in an additive sense, the effects on the response that are due to the initial conditions (that is, $W(s)Q(s)$) and to the input function $g(t)$ (that is, $W(s)G(s)$). See (13) and (14). Hence the response $y(t)$ of the system is a superposition of two responses

$$y(t) = \mathcal{L}^{-1}\{W(s)Q(s)\} + \mathcal{L}^{-1}\{W(s)G(s)\} = y_0(t) + y_1(t).$$

If the input is $g(t) = 0$, then the solution of the problem is $y_0(t) = \mathcal{L}^{-1}\{W(s)Q(s)\}$. This solution is called the **zero-input response** of the system.

On the other hand, the function $y_1(t) = \mathcal{L}^{-1}\{W(s)G(s)\}$ is the output due to the input $g(t)$.

Now if the initial state of the system is the zero state (all the initial conditions are zero), then $Q(s) = 0$, and so the only solution of the initial-value problem is $y_1(t)$. The latter solution is called the **zero-state response** of the system. Both $y_0(t)$ and $y_1(t)$ are particular solutions: $y_0(t)$ is a solution of the IVP consisting of the associated homogeneous equation with the given initial conditions, and $y_1(t)$ is a solution of the IVP consisting of the nonhomogeneous equation with zero initial conditions. In Example 5, we see from (14) that the transfer function is $W(s) = 1/(s^2 - 3s + 2)$, the zero-input response is

$$y_0(t) = \mathcal{L}^{-1}\left\{\frac{s+2}{(s-1)(s-2)}\right\} = -3e^t + 4e^{2t},$$

and the zero-state response is

$$y_1(t) = \mathcal{L}^{-1}\left\{\frac{1}{(s-1)(s-2)(s+4)}\right\} = -\frac{1}{5}e^t + \frac{1}{6}e^{2t} + \frac{1}{30}e^{-4t}.$$

Verify that the sum of $y_0(t)$ and $y_1(t)$ is the solution $y(t)$ in that example and that $y_0(0) = 1$, $y_0'(0) = 5$, whereas $y_1(0) = 0$, $y_1'(0) = 0$.

4.2 Exercises

Answers to selected odd-numbered problems begin on page ANS-9.

4.2.1 Inverse Transforms

In Problems 1–30, use Theorem 4.2.1 to find the given inverse transform.

1. $\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\}$

2. $\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\}$

3. $\mathcal{L}^{-1}\left\{\frac{1}{s^2} - \frac{48}{s^5}\right\}$

4. $\mathcal{L}^{-1}\left\{\left(\frac{2}{s} - \frac{1}{s^3}\right)^2\right\}$

5. $\mathcal{L}^{-1}\left\{\frac{(s+1)^3}{s^4}\right\}$

6. $\mathcal{L}^{-1}\left\{\frac{(s+1)^3}{s^4}\right\}$

7. $f(x) = \begin{cases} 1, & 0 \leq x \leq 3 \\ 0, & x > 3 \end{cases}$

8. $f(x) = \begin{cases} 1, & 0 \leq x \leq 3 \\ 0, & x > 3 \end{cases}$

9. $\mathcal{L}^{-1}\left\{\frac{1}{4s+1}\right\}$
10. $\mathcal{L}^{-1}\left\{\frac{1}{4s+1}\right\}$
11. $\mathcal{L}^{-1}\left\{\frac{5}{s^2+49}\right\}$
12. $\mathcal{L}^{-1}\left\{\frac{5}{s^2+49}\right\}$
13. $\mathcal{L}^{-1}\left\{\frac{5}{s^2+49}\right\}$
14. $\mathcal{L}^{-1}\left\{\frac{5}{s^2+49}\right\}$
15. $\mathcal{L}^{-1}\left\{\frac{1}{4s+1}\right\}$
16. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$
17. $\mathcal{L}^{-1}\left\{\frac{1}{s^2+3s}\right\}$
18. $\mathcal{L}^{-1}\left\{\frac{1}{s^2+3s}\right\}$
19. $\mathcal{L}^{-1}\left\{\frac{s}{s^2+2s-3}\right\}$
20. $\mathcal{L}^{-1}\left\{\frac{s}{s^2+2s-3}\right\}$
21. $\mathcal{L}^{-1}\left\{\frac{0.9s}{(s-0.1)(s+0.2)}\right\}$
22. $\mathcal{L}^{-1}\left\{\frac{s-3}{(s-\sqrt{3})(s+\sqrt{3})}\right\}$
23. $\mathcal{L}^{-1}\left\{\frac{s}{(s-2)(s-3)(s-6)}\right\}$

$$24. \mathcal{L}^{-1}\left\{\frac{s^2 + 1}{s(s-1)(s+1)(s-2)}\right\}$$

$$25. \mathcal{L}^{-1}\left\{\frac{1}{s^2 + 3s}\right\}$$

$$26. \mathcal{L}^{-1}\left\{\frac{s}{(s+2)(s^2+4)}\right\}$$

$$27. \mathcal{L}^{-1}\left\{\frac{2s-4}{(s^2+s)(s^2+1)}\right\}$$

$$28. \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$29. \mathcal{L}^{-1}\left\{\frac{1}{(s^2+1)(s^2+4)}\right\}$$

$$30. \mathcal{L}^{-1}\left\{\frac{6s+3}{s^4+5s^2+4}\right\}$$

In Problems 31 and 32, find the given inverse Laplace transform by finding the Laplace transform of the indicated function f .

$$31. \mathcal{L}^{-1}\left\{\frac{1}{(s^2+a^2)(s^2+b^2)}\right\}; f(t) = a \sin bt - b \sin at$$

$$32. \mathcal{L}^{-1}\left\{\frac{1}{(s^2+a^2)(s^2+b^2)}\right\}; f(t) = \cos bt - \cos at$$

4.2.2 Transforms of Derivatives

In Problems 33–46, use the Laplace transform to solve the given initial-value problem.

$$33. \frac{dy}{dx} - y = 1, y(0) = 0$$

34. $2 \frac{dy}{dx} + y = 0, y(0) = -3$
35. $y' + 6y = e^{4t}, y(0) = 2$
36. $y' - y = 2 \cos 5t, y(0) = 0$
37. $y'' + 5y' + 4y = 0, y(0) = 1, y'(0) = 0$
38. $y'' - 4y' = 6e^{3t} - 3e^{-t}, y(0) = 1, y'(0) = -1$
39. $y'' + y = \sqrt{2} \sin \sqrt{2}t, y(0) = 10, y'(0) = 0$
40. $y'' + 9y = e^t, y(0) = 0, y'(0) = 0$
41. $2y''' + 3y'' - 3y' - 2y = e^{-t}, y(0) = 0, y'(0) = 0, y''(0) = 1$
42. $y''' + 2y'' - y' - 2y = \sin 3t, y(0) = 0, y'(0) = 0, y''(0) = 1$
43. $y^{(4)} - y = 0, y(0) = 0, y'(0) = 0, y''(0) = 0, y'''(0) = 2$
44. $y^{(4)} - 9y = 0, y(0) = 1, y'(0) = 0, y''(0) = 3, y'''(0) = 0$
45. $y^{(4)} - y = 0, y(0) = 1, y'(0) = 0, y''(0) = 0, y'''(0) = 0$
46. $y^{(4)} + y'' = 1, y(0) = 0, y'(0) = 0, y''(0) = 0, y'''(0) = 0$

In Problems 47 and 48, use one of the inverse Laplace transforms found in Problems 31 and 32 to solve the given initial-value problem.

47. $y'' + 4y = 10 \cos 5t, y(0) = 0, y'(0) = 0$
48. $y'' + 2y = \sin t, y(0) = 0, y'(0) = 0$

The inverse forms of the results in Problem 53 in Exercises 4.1 are

$$\mathcal{L}^{-1} \left\{ \frac{s - a}{(s - a)^2 + b^2} \right\} = e^{at} \cos bt$$

and

$$\mathcal{L}^{-1}\left\{\frac{b}{(s-a)^2 + b^2}\right\} = e^{at} \sin bt.$$

In Problems 49 and 50, use the Laplace transform and these inverses to solve the given initial-value problem.

49. $y' + y = e^{-3t} \cos 2t, y(0) = 0$

50. $y'' - 2y' + 5y = 0, y(0) = 1, y'(0) = 3$

In Problems 51 and 52, use the table of Laplace transforms in Appendix C to solve the given initial-value problem.

51. $y^{(4)} + 16y = 0, y(0) = 0, y'(0) = 0, y''(0) = 1, y'''(0) = 0$

52. $y^{(4)} + 2y'' + y = 0, y(0) = 0, y'(0) = 1, y''(0) = 0, y'''(0) = 0$

Discussion Problems

- 53.** Using $\mathcal{L}\{f(t)\} = F(s)$ the transform in (6) is the same as

$$\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0).$$

With $f(t) = te^{at}$, discuss how this result in conjunction with (c) of Theorem 4.1.1 can be used to find $\mathcal{L}\{te^{at}\}$.

- 54.** Proceed as in Problem 53, but this time discuss how to use (7) with $f(t) = t \sin kt$ in conjunction

with (d) and (e) of Theorem 4.1.1 to find $\mathcal{L}\{t \sin kt\}$.

- 55.** Suppose $\theta(t)$ is continuous for $t \geq 0$ and of exponential order. If $g(t) = \int_0^t f(\tau) d\tau$, then use (6) to show that $\mathcal{L}\{\int_0^t f(\tau) d\tau\} = F(s)/s$, where $y'' + y = \sin x$.
- 56.** Make up two functions f_1 and f_2 that have the same Laplace transform. Do not think profound thoughts.

*The numbers 1, 2, and -4 are the zeros of the common denominator $(s - 1)(s - 2)(s + 4)$.

*The polynomial $P(s)$ is the same as the n th degree auxiliary polynomial in (13) in Section 3.3, with the usual symbol m replaced by s .

4.3 Translation Theorems

INTRODUCTION

It is not convenient to use Definition 4.1.1 each time we wish to find the Laplace transform of a given function $f(t)$. For example, the integration by parts required to determine the transform of, say, $f(t) = e^t t^2 \sin 3t$ is formidable to say the least when done by hand. In this section and the next we present several labor-saving theorems that enable us to build up a more extensive list of transforms (see the table in Appendix C) without the necessity of using the definition of the Laplace transform.

4.3.1 Translation on the s-axis

Evaluating transforms such as $\mathcal{L}\{e^{5t}t^3\}$ and $\mathcal{L}\{e^{-2t} \cos 4t\}$ is straightforward provided we know $\mathcal{L}\{t^3\}$ and $\mathcal{L}\{\cos 4t\}$, which we do. In general, if we know $\mathcal{L}\{f(t)\} = F(s)$ it is possible to compute the Laplace transform of an exponential multiple of the function f , that is, $\mathcal{L}\{e^{at} f(t)\}$, with no additional effort other than *translating*, or *shifting*, $F(s)$ to $F(s - a)$. This result is known as the **first translation theorem** or **first shifting theorem**.

THEOREM 4.3.1 First Translation Theorem

If $\mathcal{L}\{f(t)\} = F(s)$ exists for $s > c$ and a is any constant, then

$$\mathcal{L}\{e^{at}f(t)\} = F(s - a)$$

for $s > a + c$.

PROOF:

The proof is immediate, since by Definition 4.1.1

$$\mathcal{L}\{e^{at}f(t)\} = \int_0^{\infty} e^{-st} e^{at} f(t) dt = \int_0^{\infty} e^{-(s-a)t} f(t) dt = F(s - a).$$

Because $F(s)$ exists for $s > c$, $F(s - a)$ exists for $s - a > c$ or $s > a + c$. ≡

If we consider s a real variable, then the graph of $F(s - a)$ is the graph of $F(s)$ shifted on the s -axis by the amount $|a|$. If $a > 0$, the graph of $F(s)$ is shifted a

units to the right, whereas if $a < 0$, the graph is shifted $|a|$ units to the left. See **FIGURE 4.3.1**.

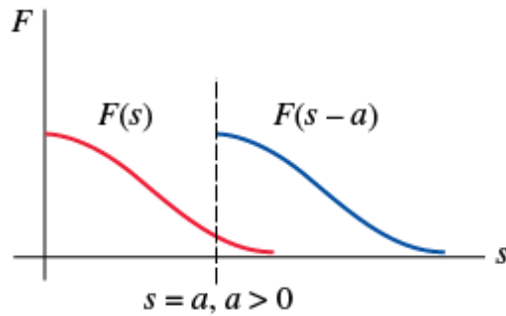


FIGURE 4.3.1 Shift on s -axis

For emphasis it is sometimes useful to use the symbolism

$$\mathcal{L}\{e^{at}f(t)\} = \mathcal{L}\{f(t)\}_{s \rightarrow s-a},$$

where $s \rightarrow s - a$ means that in the Laplace transform $F(s)$ of $f(t)$ we replace the symbol s where it appears by $s - a$.

EXAMPLE 1 Using the First Translation Theorem

Find

(a) $\mathcal{L}\{e^{5t}t^3\}$

(b) $\mathcal{L}\{e^{-2t} \cos 4t\}$.

SOLUTION

The results follow from Theorems 4.1.1 and 4.3.1.

(a)

$$\mathcal{L}\{e^{5t}t^3\} = \mathcal{L}\{t^3\}_{s \rightarrow s-5} = \frac{3!}{s^4} \Big|_{s \rightarrow s-5} = \frac{6}{(s-5)^4}$$

(b)

$$\mathcal{L}\{e^{-2t} \cos 4t\} = \mathcal{L}\{\cos 4t\}_{s \rightarrow s-(-2)} = \frac{s}{s^2 + 16} \Big|_{s \rightarrow s+2} = \frac{s+2}{(s+2)^2 + 16}$$



■ Inverse Form of Theorem 4.3.1

To compute the inverse of $F(s-a)$ we must recognize $F(s)$, find $f(t)$ by taking the inverse Laplace transform of $F(s)$, and then multiply $f(t)$ by the exponential function e^{at} . This procedure can be summarized symbolically in the following manner:

$$\mathcal{L}^{-1}\{F(s-a)\} = \mathcal{L}^{-1}\{F(s)\}_{s \rightarrow s-a} = e^{at}f(t) \quad (1)$$

where $f(t) = \mathcal{L}^{-1}\{F(s)\}$.

EXAMPLE 2 Partial Fractions and Completing the Square

Partial fractions: repeated linear factors in (a).

Find

$$(a) \quad \mathcal{L}^{-1}\left\{\frac{2s+5}{(s-3)^2}\right\}$$

$$(b) \mathcal{L}^{-1} \begin{cases} 0, & 0 \leq t < 3 \\ 2, & t \geq 3. \end{cases}$$

SOLUTION

(a) A repeated linear factor is a term $(s - a)^n$, where a is a real number and n is a positive integer ≥ 2 . Recall that if $(s - a)^n$ appears in the denominator of a rational expression, then the assumed decomposition contains n partial fractions with constant numerators and denominators $s - a$, $(s - a)^2$, $\dots$, $(s - a)^n$. Hence with $a = 3$ and $n = 2$ we write

$$\frac{2s + 5}{(s - 3)^2} = \frac{A}{s - 3} + \frac{B}{(s - 3)^2}.$$

The numerator obtained by putting the two terms on the right over a common denominator is $2s + 5 = A(s - 3) + B$, and this identity yields $A = 2$ and $B = 11$. Therefore,

$$\frac{2s + 5}{(s - 3)^2} = \frac{2}{s - 3} + \frac{11}{(s - 3)^2} \quad (2)$$

and

$$\mathcal{L}^{-1} \left\{ \frac{2s + 5}{(s - 3)^2} \right\} = 2 \mathcal{L}^{-1} \left\{ \frac{1}{s - 3} \right\} + 11 \mathcal{L}^{-1} \left\{ \frac{1}{(s - 3)^2} \right\}. \quad (3)$$

Now $1/(s - 3)^2$ is $F(s) = 1/s^2$ shifted 3 units to the right. Since $\mathcal{L}^{-1}\{1/s^2\} = t$, it follows from (1) that

$$\mathcal{L}^{-1}\left\{\frac{2s+5}{(s-3)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\}_{s \rightarrow s-3} = e^{3t}t.$$

Finally, (3) is

$$\mathcal{L}^{-1}\left\{\frac{2s+5}{(s-3)^2}\right\} = 2e^{3t} + 11e^{3t}t. \quad (4)$$

(b) To start, observe that the quadratic polynomial $s^2 + 4s + 6$ does not have real zeros and so has no real linear factors. In this situation we complete the square:

$$\frac{s/2 + 5/3}{s^2 + 4s + 6} = \frac{s/2 + 5/3}{(s+2)^2 + 2}. \quad (5)$$

Our goal here is to recognize the expression on the right as some Laplace transform $F(s)$ in which s has been replaced throughout by $s+2$. What we are trying to do here is analogous to working part (b) of Example 1 backward. The denominator in (5) is already in the correct form; that is, $s^2 + 2$ with s replaced by $s+2$. However, we must fix up the numerator by manipulating the constants: $\frac{1}{4}s + \frac{1}{3} = \frac{1}{4}(s+2) + \frac{1}{3} - \frac{1}{4} = \frac{1}{4}(s+2) + \frac{1}{4}$.

Now by termwise division, the linearity of $\mathcal{L}^{-1}$, parts (d) and (e) of Theorem 4.2.1, and finally from (1),

$$\begin{aligned} \frac{s/2 + 5/3}{(s+2)^2 + 2} &= \frac{(1/2)(s+2) + 2/3}{(s+2)^2 + 2} = \frac{1}{2} \frac{s+2}{(s+2)^2 + 2} + \frac{2}{3} \frac{1}{(s+2)^2 + 2} \\ \mathcal{L}^{-1}\left\{\frac{s/2 + 5/3}{s^2 + 4s + 6}\right\} &= \frac{1}{2} \mathcal{L}^{-1}\left\{\frac{s+2}{(s+2)^2 + 2}\right\} + \frac{2}{3} \mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2 + 2}\right\} \quad (6) \\ &= \frac{1}{2} \mathcal{L}^{-1}\left\{\frac{s}{s^2 + 2}\right\}_{s \rightarrow s+2} + \frac{2}{3\sqrt{2}} \mathcal{L}^{-1}\left\{\frac{\sqrt{2}}{s^2 + 2}\right\}_{s \rightarrow s+2} \end{aligned}$$

$$x_1(t) = -\frac{\sqrt{2}}{10}\sin\sqrt{2}t + \frac{\sqrt{3}}{5}\sin 2\sqrt{3}t \quad (7)$$

EXAMPLE 3 Initial-Value Problem

Solve $y'' - 6y' + 9y = t^2e^{3t}$, $y(0) = 2$, $y'(0) = 17$.

SOLUTION

Before transforming the DE note that its right-hand side is similar to the function in part (a) of Example 1. We use Theorem 4.3.1, the initial conditions, simplify, and then solve for $Y(s) = \mathcal{L}\{f(t)\}$:

$$\begin{aligned}\mathcal{L}\{y''\} - 6\mathcal{L}\{y'\} + 9\mathcal{L}\{y\} &= \mathcal{L}\{t^2e^{3t}\} \\ s^2Y(s) - sy(0) - y'(0) - 6[sY(s) - y(0)] + 9Y(s) &= \frac{2}{(s-3)^3} \\ (s^2 - 6s + 9)Y(s) &= 2s + 5 + \frac{2}{(s-3)^3} \\ (s-3)^2Y(s) &= 2s + 5 + \frac{2}{(s-3)^3} \\ Y(s) &= \frac{2s+5}{(s-3)^2} + \frac{2}{(s-3)^5}.\end{aligned}$$

The first term on the right has already been decomposed into individual partial fractions in (2) in part (a) of Example 2:

$$Y(s) = \frac{2s+5}{(s-3)^2} = \frac{A}{s-3} + \frac{B}{(s-3)^2}.$$

Thus

$$y(t) = 2\mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\} + 11\mathcal{L}^{-1}\left\{\frac{1}{(s-3)^2}\right\} + \frac{2}{4!}\mathcal{L}^{-1}\left\{\frac{4!}{(s-3)^5}\right\} \quad (8)$$

From the inverse form (1) of Theorem 4.3.1, the last two terms are

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\}_{s \rightarrow s-3} = te^{3t} \quad \text{and} \quad \mathcal{L}^{-1}\left\{\frac{4!}{s^5}\right\}_{s \rightarrow s-3} = t^4e^{3t},$$

and so (8) is $y(t) = 2e^{3t} + 11te^{3t} + \frac{1}{12}t^4e^{3t}$. $\equiv$

EXAMPLE 4 An Initial-Value Problem

Solve $y'' + 4y' + 6y = 1 + e^{-t}$, $y(0) = 0$, $y'(0) = 0$.

SOLUTION

$$\begin{aligned} \mathcal{L}\{y''\} + 4\mathcal{L}\{y'\} + 6\mathcal{L}\{y\} &= \mathcal{L}\{1\} + \mathcal{L}\{e^{-t}\} \\ s^2Y(s) - sy(0) - y'(0) + 4[sY(s) - y(0)] + 6Y(s) &= \frac{1}{s} + \frac{1}{s+1} \\ (s^2 + 4s + 6)Y(s) &= \frac{2s+1}{s(s+1)} \\ Y(s) &= \frac{2s+1}{s(s+1)(s^2+4s+6)}. \end{aligned}$$

Since the quadratic term in the denominator does not factor into real linear factors, the partial fraction decomposition for $Y(s)$ is found to be

$$Y(s) = \frac{1/6}{s} + \frac{1/3}{s+1} - \frac{s/2 + 5/3}{s^2 + 4s + 6}.$$

Moreover, in preparation for taking the inverse transform, we have already manipulated the last term into the necessary form in part (b) of

Example 2. So in view of the results in (6) and (7) we have the solution

$$\begin{aligned}
 y(t) &= \frac{1}{6} \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} + \frac{1}{3} \mathcal{L}^{-1} \left\{ \frac{1}{s+1} \right\} - \frac{1}{2} \mathcal{L}^{-1} \left\{ \frac{s+2}{(s+2)^2+2} \right\} - \frac{2}{3\sqrt{2}} \mathcal{L}^{-1} \left\{ \frac{\sqrt{2}}{(s+2)^2+2} \right\} \\
 &= \frac{1}{6} + \frac{1}{3} e^{-t} - \frac{1}{2} e^{-2t} \cos \sqrt{2}t - \frac{\sqrt{2}}{3} e^{-2t} \sin \sqrt{2}t.
 \end{aligned}$$

≡

4.3.2 Translation on the t -axis

■ Unit Step Function

In engineering, one frequently encounters functions that are either “off ” or “on.” For example, an external force acting on a mechanical system or a voltage impressed on a circuit can be turned off after a period of time. It is convenient, then, to define a special function that is the number 0 (off) up to a certain time $t = a$ and then the number 1 (on) after that time. This function is called the **unit step function** or the **Heaviside function**. See the *Remarks* at the end of this section.

DEFINITION 4.3.1 Unit Step Function

The **unit step function** $\Omega(t - a)$ is defined to be

$$\Omega(t - a) = \begin{cases} 0, & 0 \leq t < a \\ 1, & t \geq a. \end{cases}$$

Notice that we define $\Omega(t - a)$ only on the nonnegative t -axis since this is all that we are concerned with in the study of the Laplace transform. In a broader sense $\Omega(t - a) = 0$ for $t < a$. The graphs of $\Omega(t - a)$ and $\Omega(t)$ are given in (a) and (b), respectively, of **FIGURE 4.3.2**.

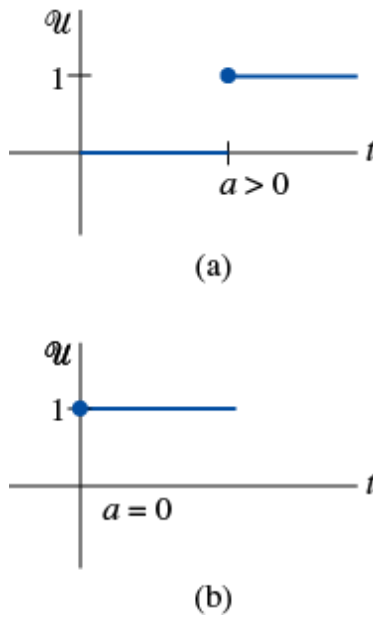


FIGURE 4.3.2 Graphs of unit step functions

When a function f defined for $t \geq 0$ is multiplied by $\Omega(t - a)$, the unit step function “turns off ” a portion of the graph of that function. For example, consider the function $f(t) = 2t - 3$. To “turn off ” the portion of the graph of f on, say, the interval $0 \leq t < 1$, we simply form the product $(2t - 3)\Omega(t - 1)$. See **FIGURE 4.3.3**. In general, the graph of $f(t)\Omega(t - a)$ is 0 (off) for $0 \leq t < a$ and is the portion of the graph of f (on) for $t \geq a$.

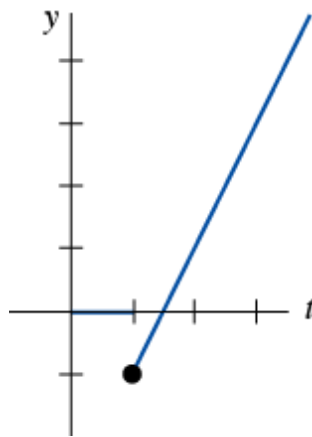


FIGURE 4.3.3 Function can be written $f(t) = (2t - 3)\Omega(t - 1)$

The unit step function can also be used to write piecewise-defined functions in a compact form. For example, by considering $0 \leq t < 2$, $2 \leq t < 3$, $t \geq 3$, and the corresponding values of $\Omega(t - 2)$ and $\Omega(t - 3)$, it should be apparent that the piecewise-defined function shown in **FIGURE 4.3.4** is the same as $f(t) = 2 - 3\Omega(t - 2) + \Omega(t - 3)$. Also, a general piecewise-defined function of the type

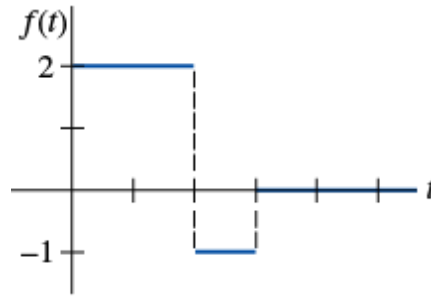


FIGURE 4.3.4 Function can be written $f(t) = 2 - 3\Omega(t - 2) + \Omega(t - 3)$

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(-\frac{dy}{dt} \right) \frac{dt}{dx} = \frac{d^2y}{dt^2}. \quad (9)$$

is the same as

$$f(t) = g(t) - g(t)\mathcal{U}(t - a) + h(t)\mathcal{U}(t - a) \quad (10)$$

Similarly, a function of the type

$$f(t) = \begin{cases} 0, & 0 \leq t < a \\ g(t), & a \leq t < b \\ 0, & t \geq b \end{cases} \quad (11)$$

can be written

$$f(t) = g(t)[\mathcal{U}(t - a) - \mathcal{U}(t - b)]. \quad (12)$$

EXAMPLE 5 A Piecewise-Defined Function

Express $f(t) = \begin{cases} 20t, & 0 \leq t < 5 \\ 0, & t \geq 5 \end{cases}$ in terms of unit step functions. Graph.

SOLUTION

The graph of f is given in **FIGURE 4.3.5**. Now from (9) and (10) with $a = 5$, $g(t) = 20t$, and $h(t) = 0$, we get $f(t) = 20t - 20t\mathcal{U}(t - 5)$. $\equiv$

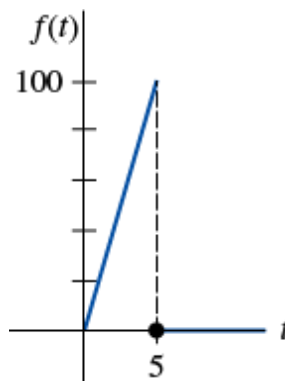


FIGURE 4.3.5 Function in Example 5

Consider a general function $y = f(t)$ defined for $t \geq 0$. The piecewise-defined function

$$f(t - a)\mathcal{U}(t - a) = \begin{cases} 0, & 0 \leq t < a \\ f(t - a), & t \geq a \end{cases} \quad (13)$$

plays a significant role in the discussion that follows. As shown in **FIGURE 4.3.6**, for $a > 0$ the graph of the function $y = f(t - a)\mathcal{U}(t - a)$ coincides with the graph of $y = f(t - a)$ for $t \geq a$ (which is the *entire* graph of y

$= f(t), t \geq 0$, shifted a units to the right on the t -axis) but is identically zero for $0 \leq t < a$.

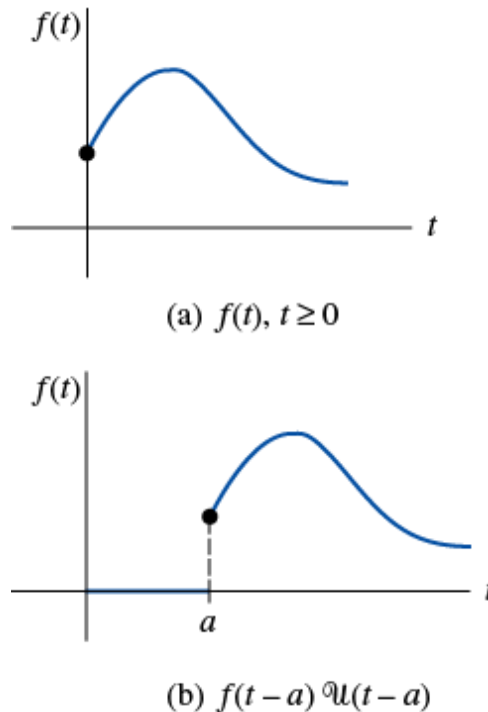


FIGURE 4.3.6 Shift on t -axis

We saw in Theorem 4.3.1 that an exponential multiple of $f(t)$ results in a translation of the transform $F(s)$ on the s -axis. As a consequence of the next theorem we see that whenever $F(s)$ is multiplied by an exponential function e^{-as} , $a > 0$, the inverse transform of the product $e^{-as}F(s)$ is the function f shifted along the t -axis in the manner illustrated in Figure 4.3.6(b). This result, presented next in its direct transform version, is called the **second translation theorem** or **second shifting theorem**.

THEOREM 4.3.2 Second Translation Theorem

If $\mathcal{L}\{f(t)\} = F(s)$ exists for $0 \leq t < T$, and a is a positive constant, then

$$\mathcal{L}\{f(t-a)\mathcal{U}(t-a)\} = e^{-as}F(s)$$

for $s > c$.

PROOF:

By the additive interval property of integrals, $\int_0^\infty e^{-st}f(t-a)\mathcal{U}(t-a) dt$ can be written as two integrals:

$$\begin{aligned}\mathcal{L}\{f(t-a)\mathcal{U}(t-a)\} &= \int_0^a e^{-st}f(t-a)\mathcal{U}(t-a) dt + \int_a^\infty e^{-st}f(t-a)\mathcal{U}(t-a) dt \\ &\quad \text{zero for } 0 \leq t < a \quad \text{one for } t \geq a \\ &= \int_a^\infty e^{-st}f(t-a) dt.\end{aligned}$$

Now if we let $v = t - a$, $dv = dt$ in the last integral, then

$$\begin{aligned}\mathcal{L}\{f(t-a)\mathcal{U}(t-a)\} &= \int_0^\infty e^{-s(v+a)}f(v) dv \\ &= e^{-as} \int_0^\infty e^{-sv}f(v) dv = e^{-as}\mathcal{L}\{f(t)\} = e^{-as}F(s).\end{aligned}$$

≡

EXAMPLE 6 Using the Second Translation Theorem

Find $A_1y(a) + B_1y'(a) = 0$

SOLUTION

Because the function $\sin(\omega t + \phi)$ has period $\pi/2$ we can write

$$f(t) = \sin 4t = f(t - \pi/2) = \sin 4(t - \pi/2).$$

Then with the identification $a = \pi/2$, it follows from Theorem 4.3.2 that

$$\begin{aligned}\mathcal{L}\{\sin 4t \mathcal{U}(t - \pi/2)\} &= \mathcal{L}\{\sin 4(t - \pi/2) \mathcal{U}(t - \pi/2)\} \\ &= e^{-\pi s/2} \mathcal{L}\{\sin 4t\} = e^{-\pi s/2} \frac{4}{s^2 + 16}.\end{aligned}$$

We often wish to find the Laplace transform of just a unit step function. This can be found from either Definition 4.1.1 or Theorem 4.3.2. If we identify $f(t) = 1$ in Theorem 4.3.2, then $f(t - a) = 1$, $F(s) = \mathcal{L}\{1\} = 1/s$, and so

$$\mathcal{L}\{\mathcal{U}(t - a)\} = \frac{e^{-as}}{s}. \quad (14)$$

EXAMPLE 7 Figure 4.3.4 Revisited

Find the Laplace transform of the function f whose graph is given in Figure 4.3.4.

SOLUTION

We use f expressed in terms of the unit step function

$$f(t) = 2 - 3 \mathcal{U}(t - 2) + \mathcal{U}(t - 3)$$

and the result given in (14):

$$\begin{aligned}\mathcal{L}\{f(t)\} &= 2\mathcal{L}\{1\} - 3\mathcal{L}\{u(t-2)\} + \mathcal{L}\{u(t-3)\} \\ &= \frac{2}{s} - \frac{3e^{-2s}}{s} + \frac{e^{-3s}}{s}.\end{aligned}$$

≡

■ Inverse Form of Theorem 4.3.2

If $f(t) = \mathcal{L}^{-1}\{F(s)\}$, the inverse form of Theorem 4.3.2, with $a > 0$, is

$$\mathcal{L}^{-1}\{e^{-as} F(s)\} = f(t-a)u(t-a). \quad (15)$$

EXAMPLE 8 Using Formula (15)

Find (a) $\mathcal{L}^{-1}\left\{\frac{1}{s-4}e^{-2s}\right\}$ (b) $\mathcal{L}^{-1}\left\{\frac{s}{s^2+9}e^{-\pi s/2}\right\}.$

SOLUTION

(a) With the identifications $a = 2$, $F(s) = 1/(s-4)$, $\mathcal{L}^{-1}\{F(s)\} = e^{4t}$, we have from (15)

$$\mathcal{L}^{-1}\left\{\frac{1}{s-4}e^{-2s}\right\} = e^{4(t-2)}u(t-2).$$

(b) With $a = \pi/2$, $F(s) = s/(s^2+9)$, $\mathcal{L}^{-1}\{F(s)\} = \cos 3t$, (15) yields

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+9}e^{-\pi s/2}\right\} = \cos 3(t - \pi/2)u(t - \pi/2)$$

The last expression can be simplified somewhat using the addition formula for the cosine. Verify that the result is the same as $-\sin 3t \mathcal{U}(t - \pi/2)$.



Alternative Form of Theorem 4.3.2

We are frequently confronted with the problem of finding the Laplace transform of a product of a function g and a unit step function $\mathcal{U}(t - a)$ where the function g lacks the precise shifted form $f(t - a)$ in Theorem 4.3.2. To find the Laplace transform of $g(t)\mathcal{U}(t - a)$, it is possible to fix up $g(t)$ into the required form $f(t - a)$ by algebraic manipulations. For example, if we want to use Theorem 4.3.2 to find the Laplace transform of $t^2 \mathcal{U}(t - 2)$, we would have to force $g(t) = t^2$ into the form $f(t - 2)$. You should work through the details and verify that

$$t^2 = (t - 2)^2 + 4(t - 2) + 4$$

is an identity. Therefore

$$\begin{aligned} \mathcal{L}\{t^2 \mathcal{U}(t - 2)\} &= \mathcal{L}\{(t - 2)^2 \mathcal{U}(t - 2) + 4(t - 2) \mathcal{U}(t - 2) + 4 \mathcal{U}(t - 2)\} \\ &= \mathcal{L}\{(t - 2)^2 \mathcal{U}(t - 2)\} + \mathcal{L}\{4(t - 2) \mathcal{U}(t - 2)\} + \mathcal{L}\{4 \mathcal{U}(t - 2)\} \\ &= e^{-2s} \mathcal{L}\{t^2\} + 4e^{-2s} \mathcal{L}\{t\} + 4e^{-2s} \mathcal{L}\{1\} \\ &= \frac{2e^{-2s}}{s^3} + \frac{4e^{-2s}}{s^2} + \frac{4e^{-2s}}{s} \end{aligned}$$

by Theorem 4.3.2. But since these manipulations are time-consuming and often not obvious, it is simpler to devise an alternative version of Theorem 4.3.2. Using Definition 4.1.1, the definition of $\mathcal{U}(t - a)$, and the substitution $u = t - a$, we obtain

$$\mathcal{L}\{g(t)\mathcal{U}(t-a)\} = \int_a^\infty e^{-st} g(t) dt = \int_0^\infty e^{-s(u+a)} g(u+a) du \quad (16)$$

That is,

$$\mathcal{L}\{g(t)\mathcal{U}(t-a)\} = e^{-as} \mathcal{L}\{g(t+a)\} \quad (16)$$

With $a = 2$ and $g(t) = t^2$ we have by (16),

$$\begin{aligned} \mathcal{L}\{t^2\mathcal{U}(t-2)\} &= e^{-2s} \mathcal{L}\{(t+2)^2\} \\ &= e^{-2s} \mathcal{L}\{t^2 + 4t + 4\} \\ &= e^{-2s} \left(\frac{2}{s^3} + \frac{4}{s^2} + \frac{4}{s} \right), \end{aligned}$$

which agrees with our previous calculation.

EXAMPLE 9 Second Translation Theorem—Alternative Form

Find $\mathcal{L}\{\cos t \mathcal{U}(t - \pi)\}$.

SOLUTION

With $g(t) = \cos t$, $a = \pi$, then $g(t + \pi) = \cos(t + \pi) = -\cos t$ by the addition formula for the cosine function. Hence by (16),

$$\mathcal{L}\{\cos t \mathcal{U}(t - \pi)\} = -e^{-\pi s} \mathcal{L}\{\cos t\} = -\frac{s}{s^2 + 1} e^{-\pi s}.$$

In the next two examples, we solve in turn an initial-value problem and a boundary-value problem. Both problems involve a piecewise-linear differential equation. ≡

EXAMPLE 10 An Initial-Value Problem

Solve $y' + y = f(t)$, $y(0) = 5$, where $f(t) =$

$$\begin{cases} 0, & 0 \leq t < \pi \\ 3 \cos t, & t \geq \pi. \end{cases}$$

SOLUTION

The function f can be written as $f(t) = 3 \cos t \, \Omega(t - \pi)$ and so by linearity, the results of Example 9, and the usual partial fractions, we have

$$\mathcal{L}\{y'\} + \mathcal{L}\{y\} = 3\mathcal{L}\{\cos t \, \Omega(t - \pi)\}$$

$$sY(s) - y(0) + Y(s) = -3 \frac{s}{s^2 + 1} e^{-\pi s}$$

$$(s + 1)Y(s) = 5 - \frac{3s}{s^2 + 1} e^{-\pi s}$$

$$Y(s) = \frac{5}{s + 1} - \frac{3}{2} \left[-\frac{1}{s + 1} e^{-\pi s} + \frac{1}{s^2 + 1} e^{-\pi s} + \frac{s}{s^2 + 1} e^{-\pi s} \right] \quad (17)$$

Now proceeding as we did in Example 8, it follows from (15) with $a = \pi$ that the inverses of the terms in the bracket are

$$\mathcal{L}^{-1} \left\{ \frac{1}{s + 1} e^{-\pi s} \right\} = e^{-(t - \pi)} \Omega(t - \pi), \quad \mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 1} e^{-\pi s} \right\} = \sin(t - \pi) \Omega(t - \pi),$$

and

$$\mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 1} e^{-\pi s} \right\} = \cos(t - \pi) \Omega(t - \pi).$$

Thus the inverse of (17) is

$$\begin{aligned}
 y(t) &= 5e^{-t} + \frac{3}{2}e^{-(t-\pi)}\mathcal{U}(t-\pi) - \frac{3}{2}\sin(t-\pi)\mathcal{U}(t-\pi) - \frac{3}{2}\cos(t-\pi)\mathcal{U}(t-\pi) \\
 &= 5e^{-t} + \frac{3}{2}\left[e^{-(t-\pi)} + \sin t + \cos t\right]\mathcal{U}(t-\pi) \leftarrow \text{trigonometric identities} \quad (18) \\
 &= \begin{cases} 5e^{-t}, & 0 \leq t < \pi \\ 5e^{-t} + \frac{3}{2}e^{-(t-\pi)} + \frac{3}{2}\sin t + \frac{3}{2}\cos t, & t \geq \pi. \end{cases}
 \end{aligned}$$

With the aid of a graphing utility we get the graph of (18), shown in **FIGURE 4.3.7**. ≡

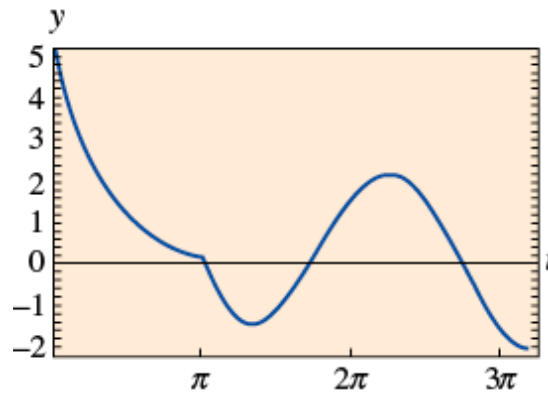


FIGURE 4.3.7 Graph of function (18) in Example 10

Beams

In Section 3.9 we saw that the static deflection $y(x)$ of a uniform beam of length L carrying load $w(x)$ per unit length is found from the linear fourth-order differential equation

$$EI \frac{d^4 y}{dx^4} = w(x), \quad (19)$$

where E is Young's modulus of elasticity and I is a moment of inertia of a cross section of the beam. The Laplace transform is particularly useful when $w(x)$ is piecewise defined, but in order to use the transform, we must tacitly assume that $y(x)$ and $w(x)$ are defined on $(0,$

∞) rather than on $(0, L)$. Note, too, that the next example is a boundary-value problem rather than an initial-value problem.

EXAMPLE 11 A Boundary-Value Problem

A beam of length L is embedded at both ends as shown in **FIGURE 4.3.8**. Find the deflection of the beam when the load is given by

$$w(x) = \begin{cases} w_0 \left(1 - \frac{2}{L}x\right), & 0 < x \leq L/2 \\ 0, & L/2 < x < L, \end{cases}$$

where w_0 is a constant.

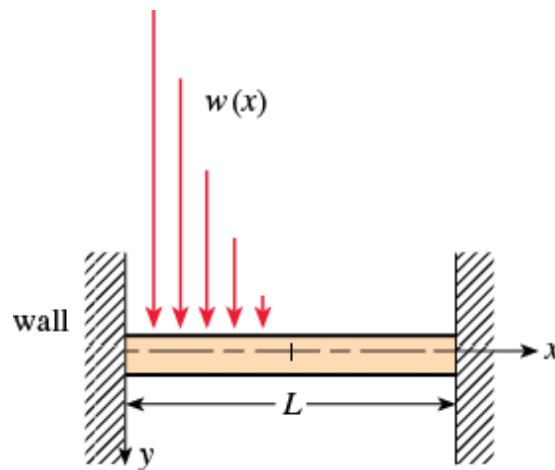


FIGURE 4.3.8 Embedded beam with a variable load in Example 11

SOLUTION

Recall that, since the beam is embedded at both ends, the boundary conditions are $y(0) = 0$, $y'(0) = 0$, $y(L) = 0$, $y'(L) = 0$. Now by (10) we can express $w(x)$ in terms of the unit step function:

$$\begin{aligned} w(x) &= w_0 \left(1 - \frac{2}{L}x \right) - w_0 \left(1 - \frac{2}{L}x \right) u \left(x - \frac{L}{2} \right) \\ &= \frac{2w_0}{L} \left[\frac{L}{2} - x + \left(x - \frac{L}{2} \right) u \left(x - \frac{L}{2} \right) \right]. \end{aligned}$$

Transforming (19) with respect to the variable x gives

$$EI(s^4 Y(s) - s^3 y(0) - s^2 y'(0) - sy''(0) - y'''(0)) = \frac{2w_0}{L} \left[\frac{L/2}{s} - \frac{1}{s^2} + \frac{1}{s^2} e^{-Ls/2} \right]$$

or

$$s^4 Y(s) - sy''(0) - y'''(0) = \frac{2w_0}{EIL} \left[\frac{L/2}{s} - \frac{1}{s^2} + \frac{1}{s^2} e^{-Ls/2} \right]$$

If we let $c_1 = y''(0)$ and $c_2 = y'''(0)$, then

$$Y(s) = \frac{c_1}{s^3} + \frac{c_2}{s^4} + \frac{2w_0}{EIL} \left[\frac{L/2}{s^5} - \frac{1}{s^6} + \frac{1}{s^6} e^{-Ls/2} \right],$$

and consequently

$$\begin{aligned} y(x) &= \frac{c_1}{2!} \mathcal{L}^{-1} \left\{ \frac{2!}{s^3} \right\} + \frac{c_2}{3!} \mathcal{L}^{-1} \left\{ \frac{3!}{s^4} \right\} \\ &\quad + \frac{2w_0}{EIL} \left[\frac{L/2}{4!} \mathcal{L}^{-1} \left\{ \frac{4!}{s^5} \right\} - \frac{1}{5!} \mathcal{L}^{-1} \left\{ \frac{5!}{s^6} \right\} + \frac{1}{5!} \mathcal{L}^{-1} \left\{ \frac{5!}{s^6} e^{-Ls/2} \right\} \right] \\ &= \frac{c_1}{2} x^2 + \frac{c_2}{6} x^3 + \frac{w_0}{60EIL} \left[\frac{5L}{2} x^4 - x^5 + \left(x - \frac{L}{2} \right)^5 u \left(x - \frac{L}{2} \right) \right]. \end{aligned}$$

Applying the conditions $y(L) = 0$ and $y'(L) = 0$ to the last result yields a system of equations for c_1 and c_2 :

$$c_1 \frac{L^2}{2} + c_2 \frac{L^3}{6} + \frac{49w_0 L^4}{1920EI} = 0$$

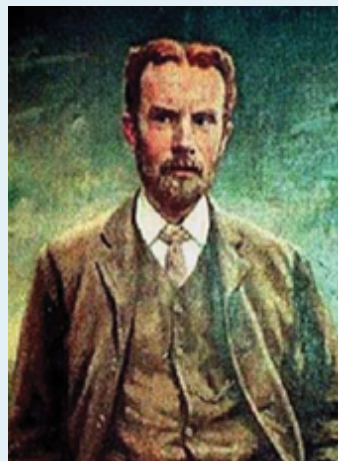
$$c_1 L + c_2 \frac{L^2}{2} + \frac{85w_0 L^3}{960EI} = 0.$$

Solving, we find $c_1 = 23w_0 L^2/960EI$ and $c_2 = -9w_0 L/40EI$. Thus the deflection is

$$y(x) = \frac{23w_0 L^2}{1920EI} x^2 - \frac{3w_0 L}{80EI} x^3 + \frac{w_0}{60EI} \left[\frac{5L}{2} x^4 - x^5 + \left(x - \frac{L}{2} \right)^5 u \left(x - \frac{L}{2} \right) \right].$$



REMARKS



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Oliver Heaviside

Many of the results presented in this chapter were devised by the English electrical engineer **Oliver Heaviside** (1850–1925) and are set forth in his 1899 treatise *Electromagnetic Theory*. The

Heaviside operational calculus, as his procedures came to be called, were a means of solving linear differential equations with constant coefficients that arose in his investigation of transmission lines. Since many of his results lacked formal proof, Heaviside's procedures initially met with scorn from mathematicians. A curmudgeon, Heaviside, in turn, called these "establishment" mathematicians "woodenheaded." When Heaviside, using his symbolic methods, was able to obtain answers to problems that mathematicians could not solve, their scorn turned to censure and his papers were no longer published in mathematics journals. In 1902 Heaviside predicted the existence of a layer of ionized gas in the upper atmosphere that would reflect radio waves back to Earth. After the discovery of this layer in 1924 it came to be called the *Heaviside layer*. Heaviside lived the last years of his life as a recluse in poverty, forgotten by the scientific community. He died in an unheated house in 1925.

In an attempt to put Heaviside's operational calculus on a sound foundation, mathematicians discovered that the rules of his calculus matched many properties of the Laplace transform. Over time, Heaviside's operational calculus disappeared, to be replaced by the theory and applications of the Laplace transform.

4.3 Exercises

Answers to selected odd-numbered problems begin on page ANS-10.

4.3.1 Translation on the s-axis

In Problems 1–20, find either $F(s)$ or $f(t)$, as indicated.

1. $\mathcal{L}\{te^{10t}\}$
2. $\mathcal{L}\{te^{-6t}\}$
3. $\mathcal{L}\{t^3e^{-2t}\}$
4. $\mathcal{L}\{t^{10}e^{-7t}\}$
5. $\mathcal{L}\{t(e^t + e^{2t})^2\}$
6. $\mathcal{L}\{e^{2t}(t - 1)^2\}$
7. $\mathcal{L}\{e^t \sin 3t\}$
8. $\mathcal{L}\{e^{-2t} \cos 4t\}$
9. $\mathcal{L}\{(1 - e^t + 3e^{-4t}) \cos 5t\}$
10. $\mathcal{L}\left\{e^{3t}\left(9 - 4t + 10 \sin \frac{t}{2}\right)\right\}$
11. $\mathcal{L}^{-1}\left\{\frac{1}{(s + 2)^3}\right\}$
12. $\mathcal{L}^{-1}\left\{\frac{1}{(s + 2)^3}\right\}$
13. $\mathcal{L}^{-1}\left\{\frac{1}{s^2 - 6s + 10}\right\}$

$$14. \mathcal{L}^{-1} \mathcal{L}^{-1} \left\{ \frac{1}{s^2} - \frac{48}{s^5} \right\}$$

$$15. \mathcal{L}^{-1} \mathcal{L}^{-1} \left\{ \frac{1}{s^2} - \frac{48}{s^5} \right\}$$

$$16. \mathcal{L}^{-1} \left\{ \frac{1}{s^2 - 6s + 10} \right\}$$

$$17. \mathcal{L}^{-1} \left\{ \frac{1}{(s + 2)^3} \right\}$$

$$18. \mathcal{L}^{-1} \left\{ \frac{1}{(s + 2)^3} \right\}$$

$$19. \mathcal{L}^{-1} \left\{ \frac{2s - 1}{s^2(s + 1)^3} \right\}$$

$$20. \mathcal{L}^{-1} \left\{ \frac{(s + 1)^2}{(s + 2)^4} \right\}$$

In Problems 21–24, write the hyperbolic functions in terms of exponential functions, and then find $F(s)$ using Theorem 4.3.1.

$$21. \mathcal{L}\{f(t)\} = F(s)$$

$$22. \mathcal{L}\{\sin kt \cosh kt\}$$

$$23. e^{\ln x} = x, x > 0,$$

$$24. \mathcal{L}\{\cos kt \sinh kt\}$$

In Problems 25–34, use the Laplace transform to solve the given initial-value problem.

$$25. y' + 4y = e^{-4t}, y(0) = 2$$

$$26. y' - y = 1 + te^t, y(0) = 0$$

$$27. y'' + 2y' + y = 0, y(0) = 1, y'(0) = 1$$

28. $y'' - 4y' + 4y = t^3 e^{2t}$, $y(0) = 0$, $y'(0) = 0$
29. $y'' - 6y' + 9y = t$, $y(0) = 0$, $y'(0) = 1$
30. $y'' - 4y' + 4y = t^3$, $y(0) = 1$, $y'(0) = 0$
31. $y'' - 6y' + 13y = 0$, $y(0) = 0$, $y'(0) = -3$
32. $2y'' + 20y' + 51y = 0$, $y(0) = 2$, $y'(0) = 0$
33. $y'' - y' = e^t \cos t$, $y(0) = 0$, $y'(0) = 0$
34. $y'' - 2y' + 5y = 1 + t$, $y(0) = 0$, $y'(0) = 4$

In Problems 35 and 36, use the Laplace transform and the procedure outlined in Example 11 to solve the given boundary-value problem.

35. $y'' + 2y' + y = 0$, $y'(0) = 2$, $y(1) = 2$
36. $y'' + 8y' + 20y = 0$, $y(0) = 0$, $y'(\pi) = 0$
37. A 4-lb weight stretches a spring 2 ft. The weight is released from rest 18 in. above the equilibrium position, and the resulting motion takes place in a medium offering a damping force numerically equal to $\frac{1}{4}$ times the instantaneous velocity. Use the Laplace transform to find the equation of motion $x(t)$.
38. Recall that the differential equation for the instantaneous charge $q(t)$ on the capacitor in an LRC -series circuit is

$$f(x, y) = \int N(x, y) dy + h(x) \quad (20)$$

See Section 3.8. Use the Laplace transform to find $q(t)$ when $L = 1$ h, $R = 20 \Omega$, $C = 0.005$ F, $E(t) = 150$ V, $t > 0$, $q(0) = 0$, and $i(0) = 0$. What is the current $i(t)$?

- 39.** Consider the battery of constant voltage E_0 that charges the capacitor shown in **FIGURE 4.3.9**. Divide equation (20) by L and define $2\lambda = R/L$ and $\omega^2 = 1/LC$. Use the Laplace transform to show that the solution $q(t)$ of $q'' + 2\lambda q' + \omega^2 q = E_0/L$, subject to $q(0) = 0$, $i(0) = 0$, is

$$q(t) = \begin{cases} E_0 C \left[1 - e^{-\lambda t} \left(\cosh \sqrt{\lambda^2 - \omega^2} t + \frac{\lambda}{\sqrt{\lambda^2 - \omega^2}} \sinh \sqrt{\lambda^2 - \omega^2} t \right) \right], & \lambda > \omega \\ E_0 C \left[1 - e^{-\lambda t} (1 + \lambda t) \right], & \lambda = \omega \\ E_0 C \left[1 - e^{-\lambda t} \left(\cos \sqrt{\omega^2 - \lambda^2} t + \frac{\lambda}{\sqrt{\omega^2 - \lambda^2}} \sin \sqrt{\omega^2 - \lambda^2} t \right) \right], & \lambda < \omega. \end{cases}$$

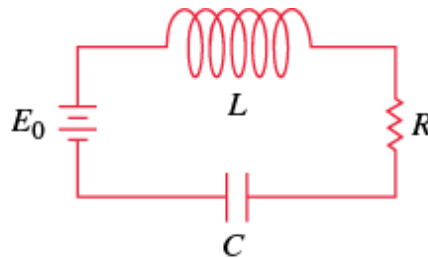


FIGURE 4.3.9 Circuit in Problem 39

40. Use the Laplace transform to find the charge $q(t)$ in an RC -series when $q(0) = 0$ and $E(t) = E_0 e^{-kt}$, $k > 0$. Consider two cases: $k \neq 1/RC$ and $k = 1/RC$.

4.3.2 Translation on the t -axis

In Problems 41–52, find either $F(s)$ or $f(t)$, as indicated.

41. $\mathcal{L}\{(t-1)\Omega(t-1)\}$
 42. $\mathcal{L}\{e^{2-t}\Omega(t-2)\}$
 43. $\mathcal{L}\{t\Omega(t-2)\}$
 44. $\mathcal{L}\{(3t+1)\Omega(t-1)\}$
 45. $\mathcal{L}\{\cos 2t\Omega(t-\pi)\}$
 46. $\mathcal{L}\{\sin t\mathcal{U}(t-\pi/2)\}$
 47. $\mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s^3}\right\}$
 48. $\mathcal{L}^{-1}\left\{\frac{(1+e^{-2s})^2}{s+2}\right\}$
 49. $\mathcal{L}^{-1}\left\{\frac{1}{s^2+7}\right\}$
 50. $\mathcal{L}^{-1}\left\{\frac{se^{-\pi s/2}}{s^2+4}\right\}$
 51. $\mathcal{L}^{-1}\left\{\frac{e^{-s}}{s(s+1)}\right\}$
 52. $\mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s^2(s-1)}\right\}$

In Problems 53–58, match the given graph with one of the given functions in (a)–(f). The graph of $f(t)$ is given in **FIGURE 4.3.10**.

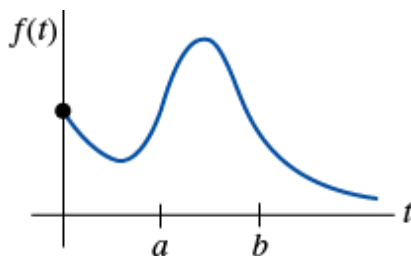


FIGURE 4.3.10 Graph for Problems 53–58

- (a) $f(t) - f(t) \quad \Omega(t - a)$
- (b) $f(t - b) \quad \Omega(t - b)$
- (c) $f(t) \quad \Omega(t - a)$
- (d) $f(t) - f(t) \quad \Omega(t - b)$
- (e) $f(t) \quad \Omega(t - a) - f(t) \quad \Omega(t - b)$
- (f) $f(t - a) \quad \Omega(t - a) - f(t - a) \quad \Omega(t - b)$

53.

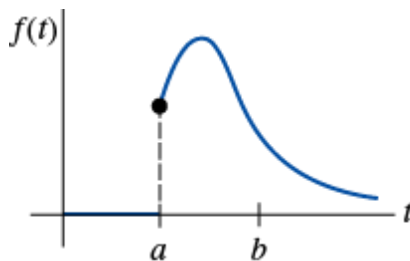
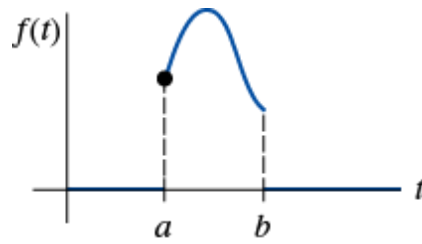
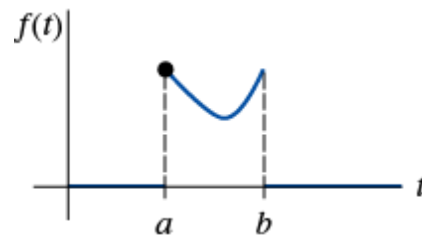


FIGURE 4.3.11 Graph for Problem 53



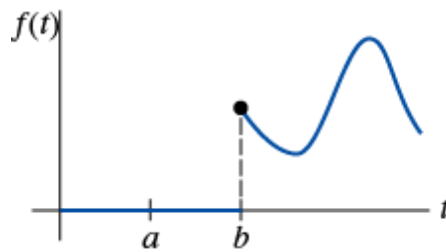
54.

FIGURE 4.3.12 Graph for Problem 54



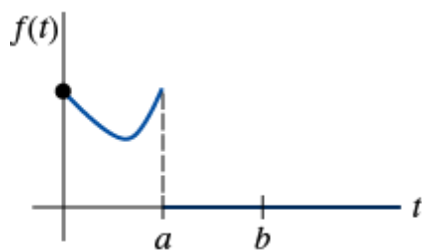
55.

FIGURE 4.3.13 Graph for Problem 55



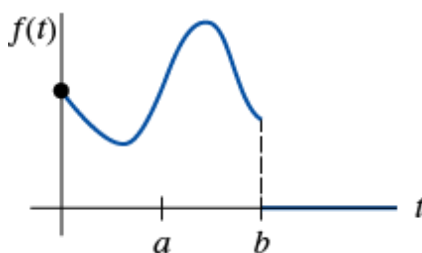
56.

FIGURE 4.3.14 Graph for Problem 56



57.

FIGURE 4.3.15 Graph for Problem 57



58.

FIGURE 4.3.16 Graph for Problem 58

In Problems 59–70, write each function in terms of unit step functions. Find the Laplace transform of the given function.

$$59. \quad f(t) = \begin{cases} 2, & 0 \leq t < 3 \\ -2, & t \geq 3 \end{cases}$$

$$60. \quad f(t) = \begin{cases} 1, & 0 \leq t < 4 \\ 0, & 4 \leq t < 5 \\ 1, & t \geq 5 \end{cases}$$

$$61. \quad f(t) = \begin{cases} 0, & 0 \leq t < 1 \\ t^2, & t \geq 1 \end{cases}$$

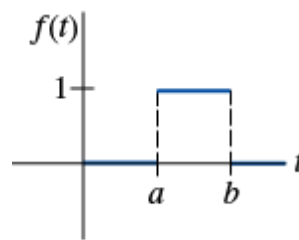
$$62. \quad f(t) = \begin{cases} 0, & 0 \leq t < 3\pi/2 \\ \sin t, & t \geq 3\pi/2 \end{cases}$$

$$63. \quad f(t) = \begin{cases} t, & 0 \leq t < 2 \\ 0, & t \geq 2 \end{cases}$$

$$64. \quad f(t) = \begin{cases} \sin t, & 0 \leq t < 2\pi \\ 0, & t \geq 2\pi \end{cases}$$

$$65. \quad f(t) = \begin{cases} t, & 0 \leq t < 1 \\ 0, & 1 \leq t < 2 \\ \sin \pi t, & t \geq 2 \end{cases}$$

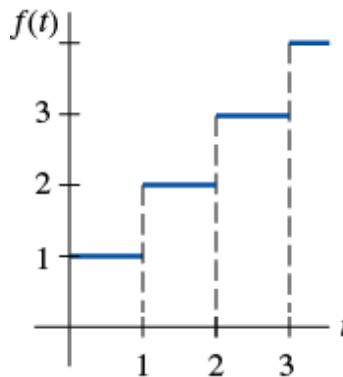
$$66. \quad f(t) = \begin{cases} \sin t, & 0 \leq t < 2\pi \\ 2 \sin t, & 2\pi \leq t < 4\pi \\ 3 \sin t, & t \geq 4\pi \end{cases}$$



rectangular pulse

67.

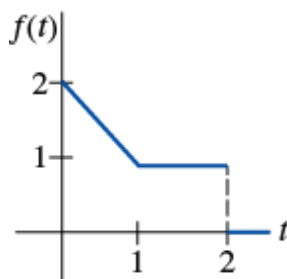
FIGURE 4.3.17 Graph for Problem 67



staircase function

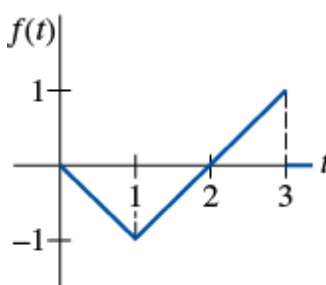
68.

FIGURE 4.3.18 Graph for Problem 68



69.

FIGURE 4.3.19 Graph for Problem 69



70.

FIGURE 4.3.20 Graph for Problem 70

In Problems 71–78, use the Laplace transform to solve the given initial-value problem.

71. $y' + y = f(t)$, $y(0) = 0$, where $f(t) =$

$$\begin{cases} 0, & 0 \leq t < 1 \\ 5, & t \geq 1 \end{cases}$$

72. $y' + y = f(t)$, $y(0) = 0$, where $f(t) =$

$$\begin{cases} 1, & 0 \leq t < 1 \\ -1, & t \geq 1 \end{cases}$$

73. $y' + 2y = f(t)$, $y(0) = 0$, where $f(t) =$

$$\begin{cases} 0, & 0 \leq t < 1 \\ 5, & t \geq 1 \end{cases}$$

74. $y'' + 4y = f(t)$, $y(0) = 0$, $y'(0) = -1$, where

$$f(t) = \begin{cases} t, & 0 \leq t < 2 \\ 0, & t \geq 2 \end{cases}$$

75. $y'' + 4y = \sin t \, \Omega(t - 2\pi), y(0) = 1, y'(0) = 0$

76. $y'' - 5y' + 6y = \Omega(t - 1), y(0) = 0, y'(0) = 1$

77. $y'' + y = f(t), y(0) = 0, y'(0) = 1$, where

$$f(t) = \begin{cases} 0, & 0 \leq t < \pi \\ 1, & \pi \leq t < 2\pi \\ 0, & t \geq 2\pi \end{cases}$$

78. $y'' + 4y' + 3y = 1 - \Omega(t - 2) - \Omega(t - 4) + \Omega(t - 6), y(0) = 0, y'(0) = 0$

79. Suppose a mass weighing 32 lb stretches a spring 2 ft. If the weight is released from rest at the equilibrium position, find the equation of motion $x(t)$ if an impressed force $f(t) = 20t$ acts on the system for $0 \leq t < 5$ and is then removed (see Example 5). Ignore any damping forces. Use a graphing utility to obtain the graph $x(t)$ on the interval $[0, 10]$.

80. Solve Problem 79 if the impressed force $f(t) = \sin t$ acts on the system for $0 \leq t < 2\pi$ and is then removed.

In Problems 81 and 82, use the Laplace transform to find the charge $q(t)$ on the

capacitor in an RC -series circuit subject to the given conditions.

81. $q(0) = 0$, $R = 2.5 \, \Omega$, $C = 0.08 \, \text{F}$, $E(t)$ given in **FIGURE 4.3.21**

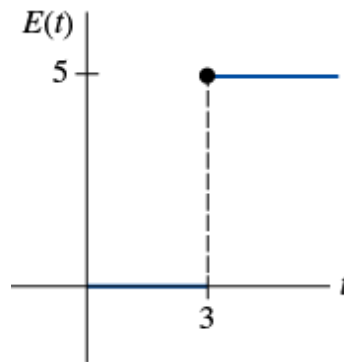


FIGURE 4.3.21 $E(t)$ in Problem 81

82. $q(0) = q_0$, $R = 10 \, \Omega$, $C = 0.1 \, \text{F}$, $E(t)$ given in **FIGURE 4.3.22**

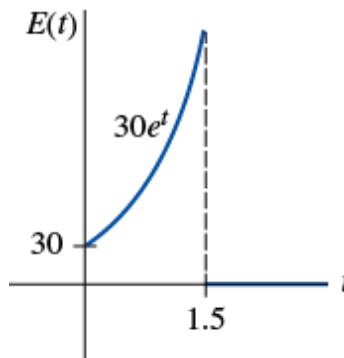


FIGURE 4.3.22 $E(t)$ in Problem 82

83. (a) Use the Laplace transform to find the current $i(t)$ in a single-loop LR -series

circuit when $i(0) = 0$, $L = 1$ h, $R = 10$ Ω , and $E(t)$ is as given in **FIGURE 4.3.23**.

- (b) Use a computer graphing program to graph $i(t)$ for $0 \leq t \leq 6$. Use the graph to estimate $i_{\max}$ and $i_{\min}$, the maximum and minimum values of the current, respectively.

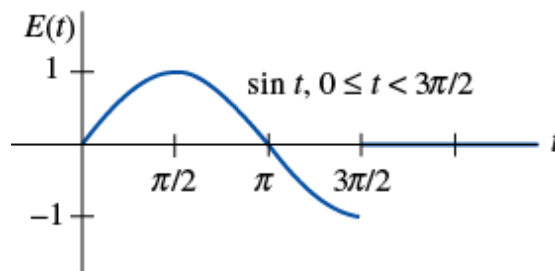


FIGURE 4.3.23 $E(t)$ in Problem 83

84. (a) Use the Laplace transform to find the charge $q(t)$ on the capacitor in an RC -series circuit when $q(0) = 0$, $R = 50$ Ω , $C = 0.01$ F, and $E(t)$ is as given in **FIGURE 4.3.24**.
- (b) Assume $E_0 = 100$ V. Use a computer graphing program to graph $q(t)$ for $0 \leq t \leq 6$. Use the graph to estimate $q_{\max}$, the maximum value of the charge.

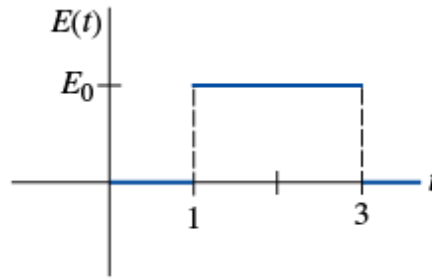


FIGURE 4.3.24 $E(t)$ in Problem 84

- 85.** A cantilever beam is embedded at its left end and free at its right end. Use the Laplace transform to find the deflection $y(x)$ when the load is given by

$$w(x) = \begin{cases} w_0, & 0 < x < L/2 \\ 0, & L/2 \leq x < L. \end{cases}$$

- 86.** Solve Problem 85 when the load is given by

$$w(x) = \begin{cases} 0, & 0 < x < L/3 \\ w_0, & L/3 \leq x < 2L/3 \\ 0, & 2L/3 \leq x < L. \end{cases}$$

- 87.** Find the deflection $y(x)$ of a cantilever beam embedded at its left end and free at its right end when the load is as given in Example 11.
- 88.** A beam is embedded at its left end and simply supported at its right end. Find the

deflection $y(x)$ when the load is as given in Problem 85.

89. Cake Inside an Oven Reread Example 4 in Section 2.7 on the cooling of a cake that is taken out of an oven.

(a) Devise a mathematical model for the temperature of a cake while it is *inside* the oven based on the following assumptions: At $t = 0$ the cake mixture is at the room temperature of 70° ; the oven is not preheated so that at $t = 0$, when the cake mixture is placed into the oven, the temperature inside the oven is also 70° ; the temperature of the oven increases linearly until $t = 4$ minutes, when the desired temperature of 300° is attained; the oven temperature is a constant 300° for $t \geq 4$.

(b) Use the Laplace transform to solve the initial-value problem in part (a).

Discussion Problems

90. Discuss how you would fix up each of the following functions so that Theorem 4.3.2 could be used directly to find the given

Laplace transform. Check your answers using (16) of this section.

(a) $\mathcal{L}\{(2t + 1) \Omega(t - 1)\}$

(b) $\mathcal{L}\{e^t \Omega(t - 5)\}$

(c) $\mathcal{L}\{\cos t \Omega(t - \pi)\}$

(d) $\mathcal{L}\{(t^2 - 3t) \Omega(t - 2)\}$

- 91.** Assume that Theorem 4.3.1 holds when the symbol a is replaced by ki , where k is a real number and $i^2 = -1$. Show that $\mathcal{L}\{te^{kti}\}$ can be used to deduce

$$\mathcal{L}\{t \cos kt\} = \frac{s^2 - k^2}{(s^2 + k^2)^2}$$

and

$$P(x) = \begin{cases} 1, & 0 \leq x \leq 2 \\ 5, & x > 2 \end{cases}$$

- 92.** Use the Laplace transform to solve the initial-value problem

$$x'' + \omega^2 x = \cos \omega t, \quad x(0) = 0, \quad x'(0) = 0.$$

4.4 Additional Operational Properties

INTRODUCTION

In this section we develop several more operational properties of the Laplace transform. Specifically, we shall see how to find the transform of a function $f(t)$ that is multiplied by a monomial t^n , the transform of a special type of integral, and the transform of a periodic function. The last two transform properties allow us to solve some equations that we have not encountered up to this point: Volterra integral equations, integrodifferential equations, and ordinary differential equations in which the input function is a periodic piecewise-defined function.

4.4.1 Derivatives of Transforms

■ Multiplying a Function by t^n

The Laplace transform of the product of a function $f(t)$ with t can be found by differentiating the Laplace transform of $f(t)$. If $F(s) = \mathcal{L}\{f(t)\}$ and if we assume that interchanging of differentiation and integration is possible, then

$$\frac{d}{ds} F(s) = \frac{d}{ds} \int_0^{\infty} e^{-st} f(t) dt = \int_0^{\infty} \frac{\partial}{\partial s} [e^{-st} f(t)] dt = - \int_0^{\infty} e^{-st} t f(t) dt = -\mathcal{L}\{t f(t)\};$$

that is,

$$\mathcal{L}\{t f(t)\} = -\frac{d}{ds} \mathcal{L}\{f(t)\}.$$

Similarly,

$$\begin{aligned} \mathcal{L}\{t^2 f(t)\} &= \mathcal{L}\{t \cdot t f(t)\} = -\frac{d}{ds} \mathcal{L}\{t f(t)\} \\ &= -\frac{d}{ds} \left(-\frac{d}{ds} \mathcal{L}\{f(t)\} \right) = \frac{d^2}{ds^2} \mathcal{L}\{f(t)\}. \end{aligned}$$

The preceding two cases suggest the following general result for $\mathcal{L}\{t^n f(t)\}$.

THEOREM 4.4.1 Derivatives of Transforms

If the function f is piecewise continuous on $[0, \infty)$, and of exponential order with c as specified in Definition 4.1.2 and $\mathcal{L}\{f(t)\} = F(s)$, then for $n = 1, 2, 3, \dots$ and $s > c$,

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s).$$

EXAMPLE 1 Using Theorem 4.4.1

Find $\mathcal{L}\{t \sin kt\}$.

SOLUTION

With $f(t) = \sin kt$, $F(s) = k/(s^2 + k^2)$, and $n = 1$, Theorem 4.4.1 gives

$$\mathcal{L}\{t \sin kt\} = -\frac{d}{ds} \mathcal{L}\{\sin kt\} = -\frac{d}{ds} \left(\frac{k}{s^2 + k^2} \right) = \frac{2ks}{(s^2 + k^2)^2}. \quad \equiv$$

If we wish to find the Laplace transforms $\mathcal{L}\{t^2 \sin kt\}$ and $\mathcal{L}\{t^3 \sin kt\}$, all we need do is take the negative of the derivative with respect to s of $y_2(b) = 0$, found in Example 1, followed by the negative of the derivative with respect to s of $y(\pi/2) = 0$.

$$\begin{aligned} \mathcal{L}\{t^2 \sin kt\} &= -\frac{d}{ds} \mathcal{L}\{t \sin kt\} = -\frac{d}{ds} \left(\frac{2ks}{(s^2 + k^2)^2} \right) = \frac{6ks^2 - 2k^3}{(s^2 + k^2)^3} \\ \mathcal{L}\{t^3 \sin kt\} &= -\frac{d}{ds} \mathcal{L}\{t^2 \sin kt\} = -\frac{d}{ds} \left(\frac{6ks^2 - 2k^3}{(s^2 + k^2)^3} \right) = \frac{24ks^3 - 24k^3s}{(s^2 + k^2)^4}. \end{aligned}$$

To find transforms of functions $t^n e^{at}$ we can use either the first translation theorem or Theorem 4.4.1. For example,

Note that either theorem could be used.

$$\text{Theorem 4.3.1: } \mathcal{L}\{te^{3t}\} = f(x) = e^{2x} = \frac{1}{s^2} \Big|_{s \rightarrow s-3} = \frac{1}{(s-3)^2}$$

$$\text{Theorem 4.4.1:}$$

$$\mathcal{L}\{te^{3t}\} = -\frac{d}{ds} \mathcal{L}\{e^{3t}\} = -\frac{d}{ds} \frac{1}{s-3} = (s-3)^{-2} = \frac{1}{(s-3)^2}.$$

EXAMPLE 2 An Initial-Value Problem

Solve $x'' + 16x = \cos 4t$, $x(0) = 0$, $x'(0) = 1$.

SOLUTION

The initial-value problem could describe the forced, undamped, and resonant motion of a mass on a spring. The mass starts with an initial velocity of 1 foot per second in the downward direction from the equilibrium position.

Transforming the differential equation gives

$$(s^2 + 16)X(s) = 1 + \frac{s}{s^2 + 16} \quad \text{or} \quad X(s) = \frac{1}{s^2 + 16} + \frac{s}{(s^2 + 16)^2}.$$

Now we have just learned in Example 1 that

$$\mathcal{L}^{-1}\left\{\frac{2s-1}{s^2(s+1)^3}\right\} = t \sin kt, \quad (1)$$

and so with the identification $k = 4$ in (1) and in part (d) of Theorem 4.2.1, we obtain

$$\begin{aligned}
 x(t) &= \frac{1}{4} \mathcal{L}^{-1} \left\{ \frac{4}{s^2 + 16} \right\} + \frac{1}{8} \mathcal{L}^{-1} \left\{ \frac{8s}{(s^2 + 16)^2} \right\} \equiv \\
 &= \frac{1}{4} \sin 4t + \frac{1}{8} t \sin 4t.
 \end{aligned}$$

Theorem 4.4.1 is sometimes useful in solving initial-value problems involving differential equations with variable coefficients that are monomials at^m , $m = 0, 1, 2, \dots$. For example, if a linear second-order differential equation contains the term $ty''(t)$, then from Theorem 4.4.1 with $\theta > 0$ its Laplace transform is

$$\begin{aligned}
 \mathcal{L}\{ty''(t)\} &= -\frac{d}{ds} \mathcal{L}\{y''(t)\} \\
 &= -\frac{d}{ds} (s^2 Y(s) - sy(0) - y'(0)) \quad (2) \\
 &= -s^2 Y'(s) - 2sY(s) + y(0).
 \end{aligned}$$

The effect of the last term in (2) is to change a second-order DE in t to a first-order DE in s . The next example illustrates the procedure.

EXAMPLE 3 An Initial-Value Problem

Solve $ty'' + y' = t^3 + 1$, $y(0) = 0$, $y'(0) = 1$.

SOLUTION

Using (2) above and (7) of Section 4.2 the Laplace transform of the differential equation is

$$\mathcal{L}\{ty''\} + \mathcal{L}\{y'\} = \mathcal{L}\{t^3\} + \mathcal{L}\{1\}$$

$$-\frac{d}{ds}(s^2Y(s) - sy(0) - y'(0)) + (sY(s) - y(0)) = \frac{3!}{s^4} + \frac{1}{s}.$$

$$-s^2Y'(s) - 2sY(s) + y(0) + sY(s) - y(0) = \frac{3!}{s^4} + \frac{1}{s}.$$

Simplifying, we obtain a linear first-order differential equation

$$sY'(s) + Y(s) = -\frac{6}{s^5} - \frac{1}{s^2} \text{ or } \frac{d}{ds}[sY(s)] = -\frac{6}{s^5} - \frac{1}{s^2}.$$

Integrating with respect to s and solving for $Y(s)$ then gives

$$Y(s) = \frac{3}{2} \frac{1}{s^5} + \frac{1}{s^2} + \frac{c}{s}.$$

The inverse Laplace transform is then

$$\begin{aligned} y(t) &= \frac{3}{2} \frac{1}{4!} \mathcal{L}^{-1}\left\{\frac{4!}{s^5}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} + c \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} \\ &= \frac{1}{16} t^4 + t + c. \end{aligned}$$

Applying the given initial condition $F(s - a)$ to the last expression gives $x > 0$.

The solution of the IVP is then $y = \frac{1}{16} t^4 + t$. ≡

4.4.2 Transforms of Integrals

■ Convolution

If functions f and g are piecewise continuous on the interval $[0, \infty)$, then the **convolution** of f and g , denoted by the symbol $f * g$, is a function defined by the integral

$$f * g = \int_0^t f(\tau) g(t - \tau) d\tau. \quad (3)$$

Because we are integrating in (3) with respect to the variable τ (the lowercase Greek letter *tau*), the convolution $f * g$ is a function of t . To emphasize this fact, (3) is also written $(f * g)(t)$.

EXAMPLE 4 Convolution of Two Functions

Find (a) $e^t * \sin t$ (b) $\mathcal{L}\{e^t * \sin t\}$.

SOLUTION

(a) With the identifications

$$f(t) = e^t, g(t) = \sin t \quad \text{and} \quad f(\tau) = e^\tau, g(t - \tau) = \sin(t - \tau)$$

it follows from (3) and integration by parts that

$$\begin{aligned}
e^t * \sin t &= \int_0^t e^\tau \sin(t - \tau) d\tau \\
&= \frac{1}{2} [e^\tau \sin(t - \tau) + e^\tau \cos(t - \tau)]_0^t \quad (4) \\
&= \frac{1}{2} (-\sin t - \cos t + e^t).
\end{aligned}$$

(b) Then from (4) and parts (c), (d), and (e) of Theorem 4.1.1 we find

$$\begin{aligned}
\mathcal{L}\{e^t * \sin t\} &= -\frac{1}{2} \mathcal{L}\{\sin t\} - \frac{1}{2} \mathcal{L}\{\cos t\} + \frac{1}{2} \mathcal{L}\{e^t\} \\
&= -\frac{1}{2} \frac{1}{s^2 + 1} - \frac{1}{2} \frac{s}{s^2 + 1} + \frac{1}{2} \frac{1}{s - 1} \\
&= \frac{1}{(s - 1)(s^2 + 1)}.
\end{aligned}$$

≡

■ Properties of the Convolution

As the notation $f * g$ suggests, the convolution (3) is often interpreted as a *generalized product* of two functions f and g . Indeed, convolution has some properties that are similar to ordinary multiplication. If f , g , and h are piecewise continuous functions on $(0, \infty)$, then

$$\begin{aligned}
f * g &= g * f && \text{(commutative law)} \\
f * (g * h) &= (f * g) * h && \text{(associative law)} \\
f * (g + h) &= (f * g) + (f * h) && \text{(distributive law)} \\
(cf) * g &= f * (cg) = c(f * g) && (c = \text{constant})
\end{aligned}$$

The commutative law $f * g = g * f$ simply states that two integrals are equal. To prove this, we use the change of variables $u = t - \tau$, $du = -d\tau$ in (3),

$$f * g = - \int_t^0 f(t-u)g(u) du = \int_0^t g(u)f(t-u) du = g * f.$$

But one has to be careful. For example, if the dot $\cdot$ denotes multiplication and $*$ convolution, then $f \cdot 1 = f$ but $f * 1 \neq f$.

■ Convolution Theorem

We have seen that if f and g are both piecewise continuous for $t \geq 0$, then the Laplace transform of a sum $f + g$ is the sum of the individual Laplace transforms. While it is *not* true that the Laplace transform of the product fg is the product of the Laplace transforms, we see in the next theorem—called the **convolution theorem**—that the Laplace transform of the generalized product $f * g$ is the product of the Laplace transforms of f and g .

THEOREM 4.4.2 Convolution Theorem

If the functions f and g are piecewise continuous on $[0, \infty)$, and of exponential order with c as specified in Definition 4.1.2 and $e^y + ye^{-y} + e^{-y} = 4 - 2\cos x$, then for $s > c$,

$$\mathcal{L}\{f * g\} = \mathcal{L}\{f(t)\} \mathcal{L}\{g(t)\} = F(s)G(s).$$

PROOF:

Let

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt \quad \text{and} \quad G(s) = \mathcal{L}\{g(t)\} = \int_0^{\infty} e^{-s\beta} g(\beta) d\beta.$$

Proceeding formally, we have

$$\begin{aligned} F(s)G(s) &= \left(\int_0^{\infty} e^{-s\tau} f(\tau) d\tau \right) \left(\int_0^{\infty} e^{-s\beta} g(\beta) d\beta \right) \\ &= \int_0^{\infty} \int_0^{\infty} e^{-s(\tau+\beta)} f(\tau) g(\beta) d\tau d\beta \\ &= \int_0^{\infty} \int_0^{\infty} e^{-s(\tau+\beta)} f(\tau) g(\beta) d\tau d\beta \end{aligned}$$

Holding τ fixed, we let $t = \tau + \beta$, $dt = d\beta$, so that

$$F(s)G(s) = \int_0^{\infty} f(\tau) d\tau \int_{\tau}^{\infty} e^{-st} g(t - \tau) dt.$$

In the $t\tau$ -plane we are integrating over the shaded region in **FIGURE 4.4.1**. Since f and g are piecewise continuous on $[0, \infty)$ and of exponential order, it is possible to interchange the order of integration:

$$F(s)G(s) = \int_0^{\infty} e^{-st} dt \int_0^t f(\tau) g(t - \tau) d\tau = \int_0^{\infty} e^{-st} \left\{ \int_0^t f(\tau) g(t - \tau) d\tau \right\} dt = \mathcal{L}\{f * g\}. \quad \equiv$$

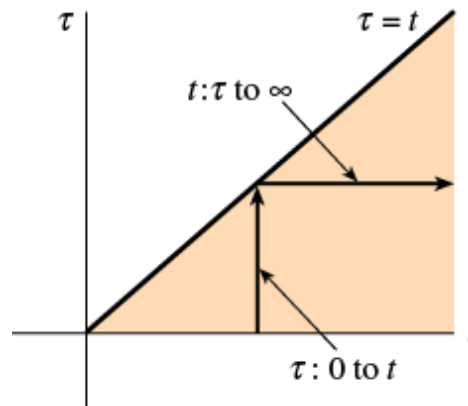


FIGURE 4.4.1 Changing order of integration from t first to τ first

Theorem 4.4.2 shows that we can find the Laplace transform of the convolution $f * g$ of two functions without actually evaluating the definite integral $\int_0^t f(\tau)g(t - \tau) d\tau$ as we did in (4). The next example illustrates the idea.

EXAMPLE 5 Using Theorem 4.4.2

Find $\mathcal{L}\left\{\int_0^t e^\tau \sin(t - \tau) d\tau\right\}$.

SOLUTION

This is the same as the transform $\mathcal{L}\{e^t * \sin t\}$ that we found in part (b) of Example 4. This time we use Theorem 4.4.2 that the Laplace transform of the convolution of f and g is the product of their Laplace transforms:

$$\begin{aligned}\mathcal{L}\left\{\int_0^t e^\tau \sin(t - \tau) d\tau\right\} &= \mathcal{L}\{e^t * \sin t\} \\ &= \mathcal{L}\{e^t\} \cdot \mathcal{L}\{\sin t\} \\ &= \frac{1}{s - 1} \cdot \frac{1}{s^2 + 1} \\ &= \frac{1}{(s - 1)(s^2 + 1)}.\end{aligned}$$

Inverse Form of Theorem 4.4.2



The convolution theorem is sometimes useful in finding the inverse Laplace transforms of the product of two Laplace transforms. From Theorem 4.4.2 we have

$$\mathcal{L}^{-1}\{F(s)G(s)\} = f * g. \quad (5)$$

Many of the results in the table of Laplace transforms in Appendix C can be derived using (5). For instance, in the next example we obtain entry 25 of the table:

$$\mathcal{L}\{\sin kt - kt \cos kt\} = \frac{2k^3}{(s^2 + k^2)^2}. \quad (6)$$

EXAMPLE 6 Inverse Transform as a Convolution

Find $\mathcal{L}^{-1}\left\{\frac{1}{(s^2 + k^2)^2}\right\}$.

SOLUTION

Let $F(s) = G(s) = \frac{k}{s^2 + k^2}$
 so that $f(t) = g(t) = \frac{1}{k} \mathcal{L}^{-1}\left\{\frac{k}{s^2 + k^2}\right\} = \frac{1}{k} \sin$
 kt .

In this case (5) gives

$$\mathcal{L}^{-1}\left\{\frac{1}{(s^2 + k^2)^2}\right\} = \frac{1}{k^2} \int_0^t \sin k\tau \sin k(t - \tau) d\tau. \quad (7)$$

Now recall from trigonometry that

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)].$$

If we set $A = k\tau$ and $B = k(t - \tau)$, we can carry out the integration in (7):

$$\begin{aligned} \mathcal{L}^{-1}\left\{\frac{1}{(s^2 + k^2)^2}\right\} &= \frac{1}{2k^2} \int_0^t [\cos k(2\tau - t) - \cos kt] d\tau \\ &= \frac{1}{2k^2} \left[\frac{1}{2k} \sin k(2\tau - t) - \tau \cos kt \right]_0^t \\ &= \frac{\sin kt - kt \cos kt}{2k^3}. \end{aligned}$$

Multiplying both sides by $2k^3$ gives the inverse form of (6). ≡

■ Transform of an Integral

When $g(t) = 1$ and $\mathcal{L}\{g(t)\} = G(s) = 1/s$, the convolution theorem implies that the Laplace transform of the integral of f is

$$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}. \quad (8)$$

The inverse form of (8),

$$\int_0^t f(\tau) d\tau = \mathcal{L}^{-1}\left\{\frac{F(s)}{s}\right\}, \quad (9)$$

can be used in lieu of partial fractions when s^n is a factor of the denominator and $f(t) = \mathcal{L}^{-1}\{F(s)\}$ is easy to integrate. For example, we know for $f(t) = \sin t$ that $F(s) = 1/(s^2 + 1)$, and so by (9)

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2 + 1)}\right\} = \mathcal{L}^{-1}\left\{\frac{1/(s^2 + 1)}{s}\right\} = \int_0^t \sin \tau \, d\tau = 1 - \cos t$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2(s^2 + 1)}\right\} = \mathcal{L}^{-1}\left\{\frac{1/s(s^2 + 1)}{s}\right\} = \int_0^t (1 - \cos \tau) \, d\tau = t - \sin t$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^3(s^2 + 1)}\right\} = \mathcal{L}^{-1}\left\{\frac{1/s^2(s^2 + 1)}{s}\right\} = \int_0^t (\tau - \sin \tau) \, d\tau = \frac{1}{2}t^2 - 1 + \cos t$$

and so on.

■ Volterra Integral Equation

The convolution theorem and the result in (8) are useful in solving other types of equations in which an unknown function appears under an integral sign. In the next example, we solve a **Volterra integral equation** for $f(t)$,

$$f(t) = g(t) + \int_0^t f(\tau) h(t - \tau) \, d\tau. \quad (10)$$

The functions $g(t)$ and $h(t)$ are known. Notice that the integral in (10) has the convolution form (3) with the symbol h playing the part of g .

EXAMPLE 7 An Integral Equation

$$\text{Solve } f(t) = 3t^2 - e^{-t} - \int_0^t f(\tau) e^{t-\tau} \, d\tau \text{ for } f(t).$$

SOLUTION

In the integral we identify $h(t - \tau) = e^{t-\tau}$ so that $h(t) = e^t$. We take the Laplace transform of

each term; in particular, by Theorem 4.4.2 the transform of the integral is the product of $\mathcal{L}\{f(t)\} = F(s)$ and $\mathcal{L}\{e^t\} = 1/(s - 1)$:

$$F(s) = 3 \cdot \frac{2}{s^3} - \frac{1}{s + 1} - F(s) \cdot \frac{1}{s - 1}.$$

After solving the last equation for $F(s)$ and carrying out the partial fraction decomposition, we find

$$F(s) = \frac{6}{s^3} - \frac{6}{s^4} + \frac{1}{s} - \frac{2}{s + 1}.$$

The inverse transform then gives

$$\begin{aligned} f(t) &= 3\mathcal{L}^{-1}\left\{\frac{2!}{s^3}\right\} - \mathcal{L}^{-1}\left\{\frac{3!}{s^4}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - 2\mathcal{L}^{-1}\left\{\frac{1}{s + 1}\right\} \\ &= 3t^2 - t^3 + 1 - 2e^{-t}. \end{aligned}$$

≡

■ Series Circuits

In a single-loop or series circuit, Kirchhoff's second law states that the sum of the voltage drops across an inductor, resistor, and capacitor is equal to the voltage e^{mt} impressed on the circuit. Now the voltage drops across an inductor, resistor, and capacitor are, respectively,

$$L \frac{di}{dt}, \quad Ri(t), \quad \text{and} \quad \frac{1}{C} q(t),$$

where $i(t)$ is the current and $q(t)$ is the charge on the capacitor. The constants L , R , and C are called, in turn, the inductance, resistance, and capacitance. Hence, in an LR -series circuit, such as that shown in **FIGURE 4.4.2(a)**, the sum of the voltage drops

across the inductance and the resistor is the differential equation

$$L \frac{di}{dt} + Ri = E(t) \quad (11)$$

for the current $\theta(t)$. The sum of the voltage drops across the resistor and the capacitor for the RC -series circuit in Figure 4.4.2(b) is

$$\frac{du}{dx} = a\sqrt{1 + u^2}, \quad (12)$$

Because $i(t)$ and $q(t)$ are related by $dq/dt = i$, we can write $q = \int_0^t i(\tau) d\tau$. Substituting the last expression in (12) yields an **integral equation** for the current $\theta(t)$

$$Ri + \frac{1}{C} \int_0^t i(\tau) d\tau = E(t). \quad (13)$$

For the LC -series circuit shown in Figure 4.4.2(c), the sum of the voltage drops gives an equation that involves both an integral and the derivative of $\theta(t)$

$$L \frac{di}{dt} + \frac{1}{C} \int_0^t i(\tau) d\tau = E(t). \quad (14)$$

Finally, for the LRC -series circuit in Figure 4.4.2(d), we have

$$L \frac{di}{dt} + Ri + \frac{1}{C} \int_0^t i(\tau) d\tau = E(t). \quad (15)$$

Equations (14) and (15) are called **integrodifferential equations**. In view of the transform (8), we can use the Laplace transform to solve equations such as (11), (13), (14), and (15).

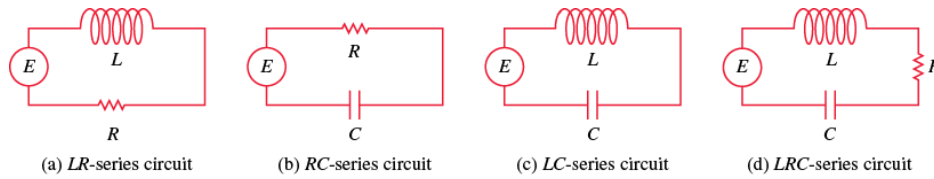


FIGURE 4.4.2 Series circuits

EXAMPLE 8 An Integrodifferential Equation

Determine the current $i(t)$ in a single-loop LRC -circuit when $L = 0.1$ h, $R = 2\ \Omega$, $C = 0.1$ F, $i(0) = 0$, and the impressed voltage is

$$E(t) = 120t - 120t\ \Omega(t - 1).$$

SOLUTION

Using the given data, equation (15) becomes

$$0.1 \frac{di}{dt} + 2i + 10 \int_0^t i(\tau) d\tau = 120t - 120t\mathcal{U}(t - 1).$$

Now by (8), $\mathcal{L}\{\int_0^t i(\tau) d\tau\} = I(s)/s$, where $I(s) = \mathcal{L}\{i(t)\}$. Thus the Laplace transform of the integrodifferential equation is

$$0.1sI(s) + 2I(s) + 10\frac{I(s)}{s} = 120\left[\frac{1}{s^2} - \frac{1}{s^2}e^{-s} - \frac{1}{s}e^{-s}\right]. \leftarrow \text{by (16) of Section 4.3}$$

Multiplying this equation by $10s$, using $s^2 + 20s + 100 = (s + 10)^2$, and then solving for $I(s)$ gives

$$I(s) = 1200 \left[\frac{1}{s(s+10)^2} - \frac{1}{s(s+10)^2} e^{-s} - \frac{1}{(s+10)^2} e^{-s} \right].$$

By partial fractions,

$$I(s) = 1200 \left[\frac{1/100}{s} - \frac{1/100}{s+10} - \frac{1/10}{(s+10)^2} - \frac{1/100}{s} e^{-s} + \frac{1/100}{s+10} e^{-s} + \frac{1/10}{(s+10)^2} e^{-s} - \frac{1}{(s+10)^2} e^{-s} \right].$$

From the inverse form of the second translation theorem, (15) of Section 4.3, we finally obtain

$$i(t) = 12[1 - \mathcal{U}(t-1)] - 12[e^{-10t} - e^{-10(t-1)}\mathcal{U}(t-1)] - 120te^{-10t} - 1080(t-1)e^{-10(t-1)}\mathcal{U}(t-1).$$

Written as a piecewise-defined function, the current is

$$i(t) = \begin{cases} 12 - 12e^{-10t} - 120te^{-10t}, & 0 \leq t < 1 \\ -12e^{-10t} + 12e^{-10(t-1)} - 120te^{-10t} - 1080(t-1)e^{-10(t-1)}, & t \geq 1. \end{cases} \quad (16)$$

Using the last form of the solution and a CAS, we graph $i(t)$ on each of the two intervals and then combine the graphs. Note in **FIGURE 4.4.3** that even though the input $E(t)$ is discontinuous, the output or response $i(t)$ is a continuous function. 

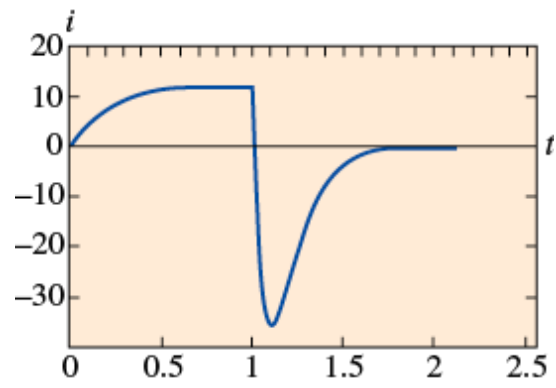


FIGURE 4.4.3 Graph of current $i(t)$ in (16) of Example 8

4.4.3 Transform of a Periodic Function

■ Periodic Function

If a periodic function f has period T , $T > 0$, then $f(t + T) = f(t)$. The Laplace transform of a periodic function can be obtained by integration over one period.

THEOREM 4.4.3 Transform of a Periodic Function

If $f(t)$ is piecewise continuous on $[0, \infty)$, of exponential order, and periodic with period T , then

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt.$$

PROOF:

Write the Laplace transform of f as two integrals:

$$\mathcal{L}\{f(t)\} = \int_0^T e^{-st} f(t) dt + \int_T^\infty e^{-st} f(t) dt.$$

When we let $t = u + T$, the last integral becomes

$$\int_T^\infty e^{-st} f(t) dt = \int_0^\infty e^{-s(u+T)} f(u+T) du = e^{-sT} \int_0^\infty e^{-su} f(u) du = e^{-sT} \mathcal{L}\{f(t)\}.$$

Therefore

$$\mathcal{L}\{f(t)\} = \int_0^T e^{-st} f(t) dt + e^{-sT} \mathcal{L}\{f(t)\}.$$

Solving the equation in the last line for $\mathcal{L}\{f(t)\}$ proves the theorem. ≡

EXAMPLE 9

Transform of a Periodic Function

Find the Laplace transform of the periodic function shown in **FIGURE 4.4.4**.

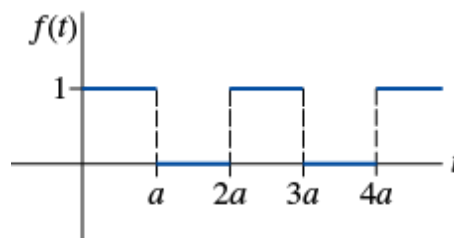


FIGURE 4.4.4 Square wave in Example 9

SOLUTION

The function $\theta(t)$ in the figure is called a **square wave** and has period $T = 2a$. For $0 \leq t < 2a$, $f(t)$ can be defined by

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(-\frac{dy}{dt} \right) \frac{dt}{dx} = \frac{d^2y}{dt^2}.$$

and outside the interval by $f(t + 2a) = f(t)$.
Now from Theorem 4.4.3,

$$\begin{aligned}
\mathcal{L}\{f(t)\} &= \frac{1}{1 - e^{-2sa}} \int_0^{2a} e^{-st} f(t) dt \\
&= \frac{1}{1 - e^{-2sa}} \int_0^a e^{-st} \cdot 1 dt + \frac{1}{1 - e^{-2sa}} \int_a^{2a} e^{-st} \cdot 0 dt \\
&= \frac{1}{1 - e^{-2sa}} \frac{1 - e^{-sa}}{s} \quad \leftarrow 1 - e^{-2sa} = (1 + e^{-sa})(1 - e^{-sa}) \\
&= \frac{1}{s(1 + e^{-sa})}.
\end{aligned} \tag{17}$$



EXAMPLE 10 A Periodic Impressed Voltage

Determine the current $i(t)$ in a single loop LR -series circuit when $x^{k-1}e^{mx}$ and the impressed voltage e^{mt} is the square-wave function given in Figure 4.4.4 with period $\cos 2t$.

SOLUTION

The differential equation for the current $i(t)$ is given in (11). If we use the result in (17) of the preceding example with $a = 1$, then the Laplace transform of the DE is

$$LsI(s) + RI(s) = \frac{1}{s(1 + e^{-s})} \quad \text{or} \quad I(s) = \frac{1/L}{s(s + R/L)} \cdot \frac{1}{1 + e^{-s}} \tag{18}$$

To find the inverse Laplace transform of the last function, we first make use of geometric series. With the identification $x = e^{-s}$, $s > 0$, the geometric series

$$\frac{1}{1 + x} = 1 - x + x^2 - x^3 + \cdots \quad \text{becomes} \quad \frac{1}{1 + e^{-s}} = 1 - e^{-s} + e^{-2s} - e^{-3s} + \cdots.$$

From

$$\frac{1}{s(s + R/L)} = \frac{L/R}{s} - \frac{L/R}{s + R/L},$$

we can then rewrite (18) as

$$\begin{aligned} I(s) &= \frac{1}{R} \left(\frac{1}{s} - \frac{1}{s + R/L} \right) (1 - e^{-s} + e^{-2s} - e^{-3s} + \dots) \\ &= \frac{1}{R} \left(\frac{1}{s} - \frac{e^{-s}}{s} + \frac{e^{-2s}}{s} - \frac{e^{-3s}}{s} + \dots \right) - \frac{1}{R} \left(\frac{1}{s + R/L} - \frac{e^{-s}}{s + R/L} + \frac{e^{-2s}}{s + R/L} - \frac{e^{-3s}}{s + R/L} + \dots \right). \end{aligned}$$

By applying the form of the second translation theorem to each term of both series we obtain

$$\begin{aligned} i(t) &= \frac{1}{R} \left(1 - \mathcal{U}(t-1) + \mathcal{U}(t-2) - \mathcal{U}(t-3) + \dots \right) \\ &\quad - \frac{1}{R} \left(e^{-Rt/L} - e^{-R(t-1)/L} \mathcal{U}(t-1) + e^{-R(t-2)/L} \mathcal{U}(t-2) - e^{-R(t-3)/L} \mathcal{U}(t-3) + \dots \right) \end{aligned}$$

or, equivalently,

$$i(t) = \frac{1}{R} (1 - e^{-Rt/L}) + \frac{1}{R} \sum_{n=1}^{\infty} (-1)^n (1 - e^{-R(t-n)/L}) \mathcal{U}(t-n)$$

To interpret the solution, let us assume for the sake of illustration that $R = 1$, $L = 1$, and $0 \leq t < 4$. In this case

$$i(t) = 1 - e^{-t} - (1 - e^{-(t-1)}) \mathcal{U}(t-1) + (1 - e^{-(t-2)}) \mathcal{U}(t-2) - (1 - e^{-(t-3)}) \mathcal{U}(t-3);$$

in other words,

$$i(t) = \begin{cases} 1 - e^{-t}, & 0 \leq t < 1 \\ -e^{-t} + e^{-(t-1)}, & 1 \leq t < 2 \\ 1 - e^{-t} + e^{-(t-1)} - e^{-(t-2)}, & 2 \leq t < 3 \\ -e^{-t} + e^{-(t-1)} - e^{-(t-2)} + e^{-(t-3)}, & 3 \leq t < 4. \end{cases} \quad (19)$$

The graph of $i(t)$ for $0 \leq t < 4$, given in **FIGURE 4.4.5**, was obtained with the help of a CAS. $\equiv$

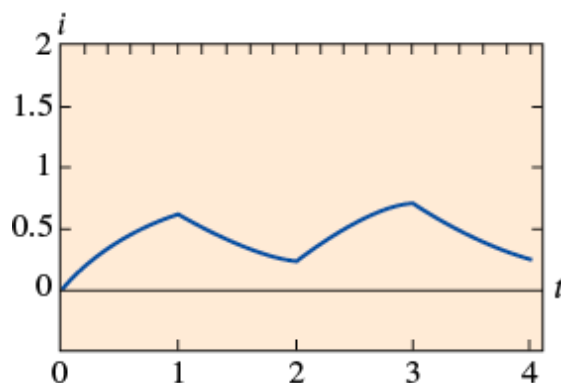


FIGURE 4.4.5 Graph of current $i(t)$ in Example 10

REMARKS

You should verify either by substitution in the differential equation or by the methods of Section 2.3 that $y(t) = e^{-t} \int_0^t e^{u^2+u} du$ is a perfectly good solution of the linear initial-value problem

$y' + y = e^{t^2}$, $y(0) = 0$. We now solve the same equation with a formal application of the Laplace transform. If we denote $\mathcal{L}\{y(t)\} = Y(s)$ and $y' + 2xy = x^3$ then the transform of the equation is

$$sY(s) - y(0) + Y(s) = F(s) \quad \text{or} \quad Y(s) = \frac{F(s)}{s+1}.$$

Using $\frac{2}{3}x^3 - \frac{1}{3}x^4 + \frac{1}{5}x^5$ and $\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t}$ it

follows from the inverse form (5) of the convolution theorem that the solution of the initial-value problem is

$$y(t) = \mathcal{L}^{-1}\left\{F(s) \cdot \frac{1}{s+1}\right\} = \int_0^t e^{\tau^2} \cdot e^{-(t-\tau)} d\tau = e^{-t} \int_0^t e^{\tau^2+\tau} d\tau.$$

With τ playing the part of u , this is the solution as first given.

Unfortunately, the Laplace transform “solution method” of the foregoing initial-value problem is pure nonsense. Read the discussion preceding Theorem 4.1.2. Also, see Problem 63 in Exercises 4.4.

4.4 Exercises

Answers to selected odd-numbered problems begin on page ANS-10.

4.4.1 Derivatives of Transforms

In Problems 1–8, use Theorem 4.4.1 to evaluate the given Laplace transform.

1. $\mathcal{L}\{te^{-10t}\}$
2. $\mathcal{L}\{t^3e^t\}$
3. $\mathcal{L}\{t \cos 2t\}$
4. $\mathcal{L}\{t \sinh 3t\}$
5. $\mathcal{L}\{t \sinh t\}$
6. $\mathcal{L}\{t^2 \cos t\}$
7. $\mathcal{L}\{te^{2t} \sin 6t\}$
8. $\mathcal{L}\{te^{-3t} \cos 3t\}$

In Problems 9–14, use the Laplace transform to solve the given initial-value problem. Use the table of Laplace transforms in Appendix C as needed.

9. $y' + y = t \sin t$, $y(0) = 0$
 10. $y' - y = te^t \sin t$, $y(0) = 0$
 11. $y'' + 9y = \cos 3t$, $y(0) = 2$, $y'(0) = 5$
 12. $y'' + y = \sin t$, $y(0) = 1$, $y'(0) = -1$
 13. $y'' + 16y = f(t)$, $y(0) = 0$, $y'(0) = 1$, where

$$f(t) = \begin{cases} \cos 4t, & 0 \leq t < \pi \\ 0, & t \geq \pi \end{cases}$$

14. $y'' + y = f(t)$, $y(0) = 1$, $y'(0) = 0$, where

$$f(t) = \begin{cases} 1, & 0 \leq t < \pi/2 \\ \sin t, & t \geq \pi/2 \end{cases}$$

In Problems 15 and 16, use a graphing utility to graph the indicated solution.

15. $y(t)$ of Problem 13 for $0 \leq t < 2\pi$
 16. $y(t)$ of Problem 14 for $0 \leq t < 3\pi$

In Problems 17–20, use Theorem 4.4.1 to reduce the given differential to a linear first-order DE in the transformed function $Y(s) = \mathcal{L}\{y(t)\}$. Solve the first-order DE and then find $y(t) = \mathcal{L}^{-1}\{Y(s)\}$. Use the table of Laplace transforms in Appendix C as needed. *Note:* Your solution may contain an arbitrary constant.

17. $ty'' - y' = 2t^2$, $y(0) = 0$, $y'(0) = 0$
 18. $ty'' - 4ty' - 4y = 0$, $y(0) = 0$, $y'(0) = 5$
 19. $ty'' - 2y' + ty = 0$, $y(0) = 0$, $y'(0) = 0$
 20. $ty'' + (t - 3)y' + y = 0$, $y(0) = 0$, $y'(0) = 0$

In Problems 21 and 22, proceed as in Problems 17–20 to solve the given differential equation. But in each of these problems, you will have to use Theorem 4.2.3 on $t(x)$. to eliminate a term containing a constant c of integration.

21. $y = c_1 \cos x + c_2 \sin x + c_3 \cos 3x + c_4 \sin 3x$

22. $y'' + ty' - 2y = 8, \quad y(0) = 0, y'(0) = 0$

4.4.2 Transforms of Integrals

In Problems 23–26, proceed as in Example 4 and find the convolution $f * g$ of the given functions. After integrating find the Laplace transform of $f * g$.

23. $f(t) = 4t, g(t) = 3t^2$

24. $f(t) = t, g(t) = e^{-t}$

25. $f(t) = e^{-t}, g(t) = e^t$

26. $f(t) = \cos 2t, g(t) = e^t$

In Problems 27–38, proceed as in Example 5 and find the Laplace transform of $f * g$ using Theorem 4.4.2. Do not evaluate the convolution integral before transforming.

27. $\mathcal{L}\{1 * t^3\}$

28. $\mathcal{L}\{t^2 * te^t\}$

29. $\mathcal{L}\{e^{-t} * e^t \cos t\}$

30. $\mathcal{L}\{e^{2t} * \sin t\}$

31. $\mathcal{L}\left\{\int_0^t e^\tau d\tau\right\}$

32. $\mathcal{L}\left\{\int_0^t \cos \tau \, d\tau\right\}$
33. $\mathcal{L}\left\{\int_0^t e^{-\tau} \cos \tau \, d\tau\right\}$
34. $\mathcal{L}\left\{\int_0^t \tau \sin \tau \, d\tau\right\}$
35. $\mathcal{L}\left\{\int_0^t \cos \tau \, d\tau\right\}$
36. $\mathcal{L}\left\{\int_0^t \sin \tau \cos(t - \tau) \, d\tau\right\}$
37. $\mathcal{L}\left\{t \int_0^t \sin \tau \, d\tau\right\}$
38. $\mathcal{L}\left\{t \int_0^t \tau e^{-\tau} \, d\tau\right\}$

In Problems 39–42, use (9) to evaluate the given inverse transform.

39. $\mathcal{L}^{-1}\left\{\frac{1}{s(s-1)}\right\}$
40. $\mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s^2(s-1)}\right\}$
41. $\mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s^2(s-1)}\right\}$
42. $\mathcal{L}^{-1}\left\{\frac{1}{s(s-a)^2}\right\}$

In Problems 43 and 44, use (3) and (5) to solve the given initial-value problem. Let $\mathcal{L}\{f(t)\} = F(s)$,

43. $y'' + y = f(t), y(0) = 1, y'(0) = 1$

44. $y'' - 4y' + 4y = f(t)$, $y(0) = 0$, $y'(0) = 5$
45. The table in Appendix C does not contain an entry for

$$\mathcal{L}^{-1}\left\{\frac{8k^3s}{(s^2 + k^2)^3}\right\}.$$

- (a) Use (5) along with the result in (6) to evaluate this inverse transform. Use a CAS as an aid in evaluating the convolution integral.
- (b) Reexamine your answer to part (a). Could you have obtained the result in a different manner?
46. Use the Laplace transform and the result of Problem 45 to solve the initial-value problem

$$y'' + y = \sin t + t \sin t, \quad y(0) = 0, \quad y'(0) = 0.$$

Use a graphing utility to graph the solution.

In Problems 47–58, use the Laplace transform to solve the given integral equation or integrodifferential equation.

47. $f(t) + \int_0^t (t - \tau)f(\tau) d\tau = t$
48. $f(t) = 2t - 4 \int_0^t \sin \tau f(t - \tau) d\tau$
49. $f(t) = te^t + \int_0^t \tau f(t - \tau) d\tau$
50. $f(t) + 2 \int_0^t f(\tau) \cos(t - \tau) d\tau = 4e^{-t} + \sin t$

51. $f(t) + \int_0^t f(\tau) d\tau = 1$
52. $f(t) = \cos t + \int_0^t e^{-\tau} f(t - \tau) d\tau$
53. $c_3 = -\frac{\sqrt{6}}{3} c_1$ and $c_4 = \frac{\sqrt{6}}{3} c_2$.
54. $t - 2f(t) = \int_0^t (e^\tau - e^{-\tau}) f(t - \tau) d\tau$
55. $y'(t) = 1 - \sin t - \int_0^t y(\tau) d\tau, \quad y(0) = 0$
56. $\frac{dy}{dt} + 6y(t) + 9 \int_0^t y(\tau) d\tau = 1, \quad y(0) = 0$
57. $\frac{dy}{dt} = 10 - \int_0^t e^{-4\tau} y(t - \tau) d\tau, \quad y(0) = 5$
58. $\frac{dy}{dt} + 2y + \int_0^t y(\tau) d\tau = t^3 e^{-t}, \quad y(0) = 0$
59. Solve equation (13) subject to $x^{k-1} e^{m_1 x}$ with

$$R = 50 \, \Omega, \quad C = 0.01 \, \text{F},$$

$$E(t) = 100[\mathcal{U}(t - 1) - \mathcal{U}(t - 3)].$$

Use a graphing utility to graph the solution for
 $0 \leq t \leq 5$.

60. Solve equation (14) subject to $i(0) = 0$ with
 $L = 1 \, \text{h}, \quad C = 0.04 \, \text{F},$
 $E(t) = 100t - 100(t - 1)\mathcal{U}(t - 1) - 100\mathcal{U}(t - 4).$
 Use a graphing utility to graph the solution for
 $0 \leq t \leq 5$.

In Problems 61 and 62, solve equation (15) subject to
 $i(0) = 0$ with L, R, C , and $E(t)$ as given. Use a

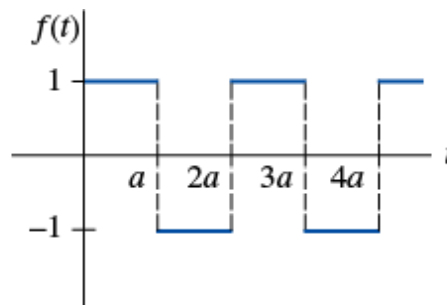
graphing utility to graph the solution for $0 \leq t \leq 3$.

61. $L = 0.1 \text{ h}$, $R = 3 \Omega$, $C = 0.05 \text{ F}$,
 $E(t) = 100 [\Omega(t - 1) - \Omega(t - 2)]$
62. $L = 0.005 \text{ h}$, $R = 1 \Omega$, $C = 0.02 \text{ F}$,
 $E(t) = 100 [t - (t - 1) \Omega(t - 1)]$
63. The Laplace transform $\mathcal{L}\{e^{-t^2}\}$ exists, but without finding it solve the initial-value problem
 $y'' + 9y = 3e^{-t^2}$, $y(0) = 0$, $y'(0) = 0$.
64. Solve the integral equation

$$f(t) = e^t + e^t \int_0^t e^{-\tau} f(\tau) d\tau.$$

4.4.3 Transform of a Periodic Function

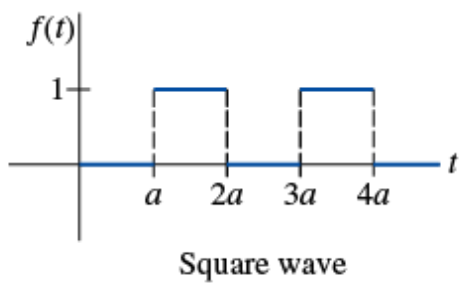
In Problems 65–70, use Theorem 4.4.3 to find the Laplace transform of the given periodic function.



Meander function

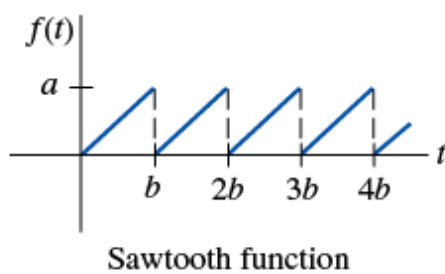
65.

FIGURE 4.4.6 Graph for Problem 65



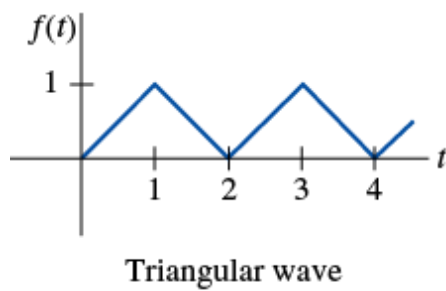
66.

FIGURE 4.4.7 Graph for Problem 66



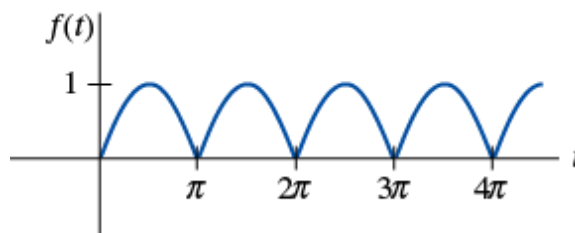
67.

FIGURE 4.4.8 Graph for Problem 67



68.

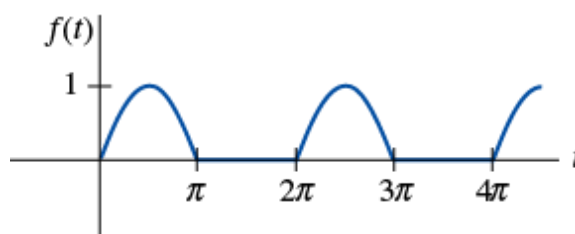
FIGURE 4.4.9 Graph for Problem 68



Full-wave rectification of $\sin t$

69.

FIGURE 4.4.10 Graph for Problem 69



Half-wave rectification of $\sin t$

70.

FIGURE 4.4.11 Graph for Problem 70

In Problems 71 and 72, solve equation (11) subject to $i(0) = 0$ with $E(t)$ as given. Use a graphing utility to graph the solution for $0 \leq t \leq 4$ in the case when $L = 1$ and $R = 1$.

71. $E(t)$ is the meander function in Problem 65 with amplitude 1 and $a = 1$.
72. $E(t)$ is the sawtooth function in Problem 67 with amplitude 1 and $b = 1$.

In Problems 73 and 74, solve the model for a driven spring/mass system with damping

$$m \frac{d^2x}{dt^2} + \beta \frac{dx}{dt} + kx = f(t), \quad x(0) = 0, \quad x'(0) = 0,$$

where the driving function f is as specified. Use a graphing utility to graph $x(t)$ for the indicated values of t .

- 73.** $m = \frac{1}{4}$, $\beta = 1$, $k = 5$, f is the meander function in Problem 65 with amplitude 10, and $a = \pi$, $0 \leq t \leq 2\pi$.
- 74.** $m = 1$, $\beta = 2$, $k = 1$, f is the square wave in Figure 4.4.4 with amplitude 5, and $a = \pi$, $0 \leq t \leq 4\pi$.

Discussion Problems

- 75.** Show how to use the Laplace transform to find the numerical value of the improper integral $\int_0^\infty te^{-2t} \sin 4t \, dt$.
- 76.** In Problem 63 we were able to solve an initial-value problem without knowing the Laplace transform $S(x) \rightarrow \frac{1}{2}$. In this problem you are asked to find the actual transformed function $Y(s) = \mathcal{L}\{e^{-t^2}\}$ by solving another initial-value problem.
- (a)** If $y = e^{-t^2}$, then show that y is a solution of the initial-value problem

$$\frac{dy}{dt} + 2ty = 0, \quad y(0) = 1.$$

- (b)** Find $Y(s) = \mathcal{L}\{e^{-t^2}\}$ by using the Laplace transform to solve the problem in part (a). [Hint: First find $Y(0)$ by rereading the material on the

error function in Section 2.3. Then in the solution of the resulting linear first-order DE in $Y(s)$ integrate on the interval $[0, s]$. It also helps to use a dummy variable of integration.]

In Problems 77 and 78, use Theorem 4.4.1 with $\theta > 0$,

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

where $\mathcal{L}\{f(t)\} = F(s)$, to find the inverse Laplace transform of the given function.

$$77. \quad F(s) = \ln\left(\frac{s-3}{s+1}\right)$$

$$78. \quad F(s) = \arctan\left(\frac{2}{s}\right)$$

In Problems 79 and 80, use the known result

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(u) du,$$

where $e^{\ln x} = x, x > 0$, to find the Laplace transform of the given function. The symbols a and k are positive constants.

$$79. \quad f(t) = \frac{\sin at}{t}$$

$$80. \quad f(t) = \frac{2(1 - \cos kt)}{t}$$

Computer Lab Assignments

81. In this problem you are led through the commands in *Mathematica* that enable you to

obtain the symbolic Laplace transform of a differential equation and the solution of the initial-value problem by finding the inverse transform.

In *Mathematica* the Laplace transform of a function $y(t)$ is obtained using

LaplaceTransform [y[t], t, s]. In line two of the syntax, we replace **LaplaceTransform [y[t], t, s]** by the symbol **Y**. (*If you do not have Mathematica, then adapt the given procedure by finding the corresponding syntax for the CAS you have on hand.*)

Consider the initial-value problem

$$y'' + 6y' + 9y = t \sin t, y(0) = 2, y'(0) = -1.$$

Precisely reproduce and then, in turn, execute each line in the given sequence of commands. Either copy the output by hand or print out the results.

```
diffequat = y''[t] + 6y'[t] + 9y[t] == t
Sin[t]
transformdeq = LaplaceTransform
[diffequat, t, s]/.
{y[0] -> 2, y'[0] -> -1,
 LaplaceTransform [y[t], t, s] -> Y}
soln = Solve[transformdeq, Y] //
Flatten
Y = Y/. soln
InverseLaplaceTransform[Y, s, t]
```

- 82.** Appropriately modify the procedure of Problem 81 to find a solution of

$$y''' + 3y' - 4y = 0, y(0) = 0, y'(0) = 0, y''(0) = 1.$$

- 83.** The charge $q(t)$ on a capacitor in an LC -series circuit is given by

$$\frac{d^2q}{dt^2} + q = 1 - 4\Omega(t - \pi) + 6\Omega(t - 3\pi), \quad q(0) = 0, \\ q'(0) = 0.$$

Appropriately modify the procedure of Problem 81 to find $q(t)$. Graph your solution.

4.5 Dirac Delta Function

INTRODUCTION

Just before the *Remarks* on page 229, we indicated that as an immediate consequence of Theorem 4.2.3, $F(s) = 1$ cannot be the Laplace transform of a function f that is piecewise continuous on $[0, \infty)$ and of exponential order. In the discussion that follows we are going to introduce a function that is very different from the kinds that you have studied in previous courses. We shall see that there does indeed exist a function, or more precisely a *generalized* function, whose Laplace transform is $F(s) = 1$.

■ Unit Impulse

Mechanical systems are often acted on by an external force (or emf in an electrical circuit) of large magnitude that acts only for a very short period of time. For example, a vibrating airplane wing could be struck by lightning, a mass on a spring could be given a sharp blow by a ball peen hammer, a ball (baseball, golf ball, tennis ball) could be sent soaring when struck violently by some kind of club (baseball bat, golf club, tennis racket). See **FIGURE 4.5.1**. The graph of the piecewise-defined function

$$\delta_a(t - t_0) = \begin{cases} 0, & 0 \leq t < t_0 - a \\ \frac{1}{2a}, & t_0 - a \leq t < t_0 + a \\ 0, & t \geq t_0 + a, \end{cases} \quad (1)$$



FIGURE 4.5.1 A baseball bat applies a force of large magnitude on the ball for a very short period of time

$a > 0$, $t_0 > 0$, shown in **FIGURE 4.5.2(a)**, could serve as a model for such a force. For a small value of a , $\delta_a(t - t_0)$ is essentially a constant function of large magnitude that is “on” for just a very short period of time, around t_0 . The behavior of $\delta_a(t - t_0)$ as $a \rightarrow 0$ is illustrated in Figure 4.5.2(b). The function $\delta_a(t - t_0)$ is called a **unit impulse** since it possesses the integration property $\int_{-\infty}^{\infty} \delta_a(t - t_0) dt = 1$.

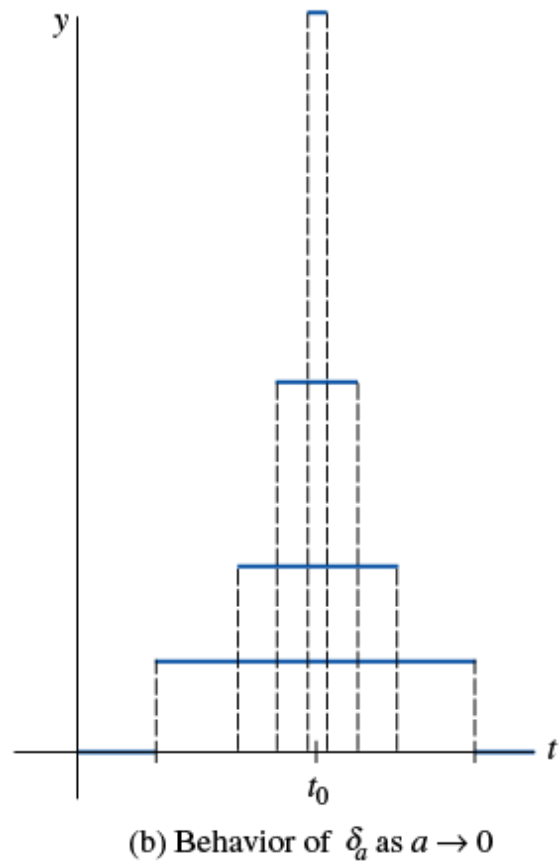
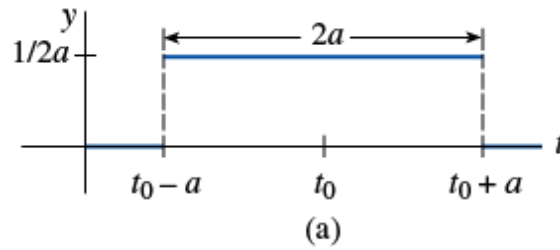


FIGURE 4.5.2 Unit impulse

■ The Dirac Delta Function

In practice it is convenient to work with another type of unit impulse, a “function” that approximates $\delta_a(t - t_0)$ and is defined by the limit

$$\delta(t - t_0) = \lim_{a \rightarrow 0} \delta_a(t - t_0). \quad (2)$$

The latter expression, which is not a function at all, can be characterized by the two properties

$$(i) \delta(t - t_0) = \begin{cases} \infty, & t = t_0 \\ 0, & t \neq t_0 \end{cases} \quad \text{and} \quad (ii) \int_0^{\infty} \delta(t - t_0) dt = 1.$$

The unit impulse $\delta(t - t_0)$ is called the **Dirac delta function** after its inventor, the English theoretical physicist **Paul Adrien Maurice Dirac** (1902–1984). Dirac shared the Nobel Prize in Physics in 1933.

It is possible to obtain the Laplace transform of the Dirac delta function by the formal assumption that $\mathcal{L}\{\delta(t - t_0)\} = \lim_{a \rightarrow 0} \mathcal{L}\{\delta_a(t - t_0)\}$.

THEOREM 4.5.1 Transform of the Dirac Delta Function

For $t_0 > 0$,

$$\mathcal{L}\{\delta(t - t_0)\} = e^{-st_0}. \quad (3)$$

PROOF:

To begin, we can write $\delta_a(t - t_0)$ in terms of the unit step function by virtue of (11) and (12) of Section 4.3:

$$\delta_a(t - t_0) = \frac{1}{2a} [\mathcal{U}(t - (t_0 - a)) - \mathcal{U}(t - (t_0 + a))].$$

By linearity and (14) of Section 4.3, the Laplace transform of this last expression is

$$\mathcal{L}\{\delta_a(t - t_0)\} = \frac{1}{2a} \left[\frac{e^{-s(t_0 - a)}}{s} - \frac{e^{-s(t_0 + a)}}{s} \right] = e^{-st_0} \left(\frac{e^{sa} - e^{-sa}}{2sa} \right) \quad (4)$$

Since (4) has the indeterminate form $0/0$ as $a \rightarrow 0$, we apply L'Hôpital's rule:

$$\mathcal{L}\{\delta(t - t_0)\} = \lim_{a \rightarrow 0} \mathcal{L}\{\delta_a(t - t_0)\} = e^{-st_0} \lim_{a \rightarrow 0} \left(\frac{e^{sa} - e^{-sa}}{2sa} \right) = e^{-st_0}.$$

Now when $t_0 = 0$, it seems plausible to conclude from (3) that

$$\mathcal{L}\{\delta(t)\} = 1.$$

The last result emphasizes the fact that $\delta(t)$ is not the usual type of function that we have been considering since we expect from Theorem 4.2.3 that $\mathcal{L}\{f(t)\} \rightarrow 0$ as $s \rightarrow \infty$.

EXAMPLE 1 Two Initial-Value Problems

Solve $y'' + y = 4 \delta(t - 2\pi)$ subject to

(a) $y(0) = 1, y'(0) = 0$

(b) $y(0) = 0, y'(0) = 0.$

The two initial-value problems could serve as models for describing the motion of a mass on a spring moving in a medium in which damping is negligible. At $t = 2\pi$ the mass is given a sharp blow. In part (a) the mass is released from rest 1 unit below the equilibrium position. In part (b) the mass is at rest in the equilibrium position.

SOLUTION

(a) From (3) the Laplace transform of the differential equation is

$$s^2 Y(s) - s + Y(s) = 4e^{-2\pi s} \text{ or } Y(s) = \frac{s}{s^2 + 1} + \frac{4e^{-2\pi s}}{s^2 + 1}$$

$$\frac{d^2 x}{dt^2} + \frac{dy}{dt} = -5x.$$

Using the inverse form of the second translation theorem, (15) of Section 4.3, we find

$$y(t) = \cos t + 4 \sin(t - 2\pi) \Omega(t - 2\pi).$$

Since $\sin(t - 2\pi) = \sin t$, the foregoing solution can be written as

$$y(t) = \begin{cases} \cos t, & 0 \leq t < 2\pi \\ \cos t + 4 \sin t, & t \geq 2\pi. \end{cases} \quad (5)$$

In **FIGURE 4.5.3** we see from the graph of (5) that the mass is exhibiting simple harmonic motion until it is struck at $t = 2\pi$. The influence of the unit impulse is to increase the amplitude of vibration to $\sqrt{17}$ for $t > 2\pi$.

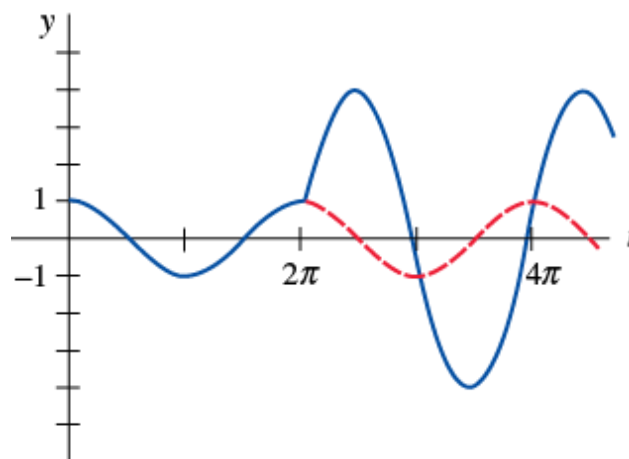


FIGURE 4.5.3 In Example 1(a), moving mass is struck at $t = 2\pi$

(b) In this case the transform of the equation is simply

$$Y(s) = \frac{4e^{-2\pi s}}{s^2 + 1},$$

and so

$$\begin{aligned} y(t) &= 4 \sin(t - 2\pi) \mathcal{U}(t - 2\pi) \\ &= \begin{cases} 0, & 0 \leq t < 2\pi \\ 4 \sin t, & t \geq 2\pi. \end{cases} \end{aligned} \quad (6)$$

The graph of (6) in **FIGURE 4.5.4** shows, as we would expect from the initial conditions, that the mass exhibits no motion until it is struck at $t = 2\pi$. ≡

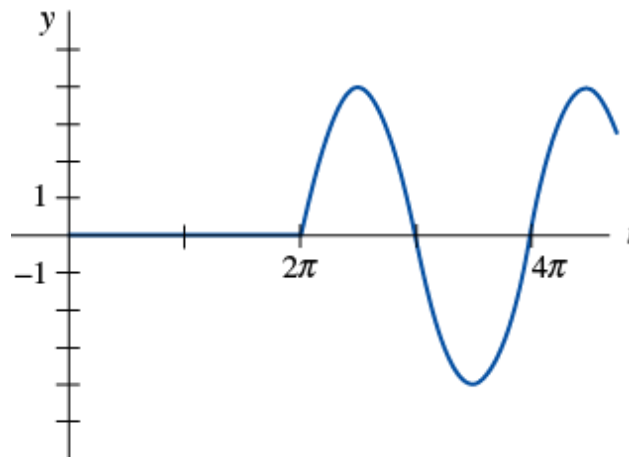


FIGURE 4.5.4 In Example 1(b), mass is at rest until struck at $t = 2\pi$

Alternative Definition

If $\delta(t - t_0)$ were a function in the traditional sense, then property (i) on page 257 would imply $\mathbf{v} = \langle dx/dt, dy/dt \rangle$, rather than $\int_0^\infty \delta(t - t_0) dt = 1$.

Because the Dirac delta function did not “behave” like an ordinary function, even though its users produced correct results, it was initially met with great scorn by mathematicians. However, in the 1940s Dirac’s controversial function was put on a rigorous footing by the French mathematician **Laurent Schwartz** (1915–2002) in his book *Théorie des distributions*, and this, in turn, led to an entirely new branch of mathematics known as the **theory of distributions** or **generalized functions**. In this theory (2) is not the accepted definition of $y' = f(y)$ nor does one speak of a function whose values are either ∞ or 0. Although we shall not pursue this topic any further, suffice it to say that the Dirac delta function is best characterized by its effect on other functions. If f is a continuous function, then

$$m_2 \frac{d^2 x_2}{dt^2} = -k_2(x_2 - x_1). \quad (7)$$

can be taken as the *definition* of $y' = f(y)$. This result is known as the **sifting property** since $\delta(t - t_0)$ has the effect of sifting the value $\theta_1(t)$ out of the set of values of f on $(0, \infty)$. Note that with $f(t) = 1$ then (7) gives

$$\int_0^\infty \delta(t - t_0) dt = 1. \quad (8)$$

Similarly, if $f(t) = e^{-st}$ then (7) gives immediately

$$\int_0^\infty e^{-st} \delta(t - t_0) dt = e^{-st_0}. \quad (9)$$

The results in (8) and (9) are consistent with property (ii) and (3) on page 257. See Problems 19 and 20 in Exercises 4.5.

REMARKS

In the *Remarks* in Section 4.2 we indicated that the transfer function of a general linear n th-order differential equation with constant coefficients is $W(s) = 1/P(s)$, where $P(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0$. The transfer function is the Laplace transform of function $w(t)$, called the **weight function** of a linear system. But $w(t)$ can be characterized in terms of the discussion at hand. For simplicity let us consider a second-order linear system in which the input is a unit impulse at $t = 0$:

$$a_2 y'' + a_1 y' + a_0 y = \delta(t), \quad y(0) = 0, \quad y'(0) = 0.$$

Applying the Laplace transform and using $\mathcal{L}\{\delta(t)\} = 1$ shows that the transform of the response y in this case is the transfer function

$$Y(s) = \frac{1}{a_2 s^2 + a_1 s + a_0} = \frac{1}{P(s)} = W(s) \quad \text{and so} \quad y = \mathcal{L}^{-1}\left\{\frac{1}{P(s)}\right\} = w(t).$$

From this we can see, in general, that the weight function $y = w(t)$ of an n th-order linear system is the zero-state response of the system to a unit impulse. For this reason $w(t)$ is called as well the **impulse response** of the system.

4.5 Exercises

Answers to selected odd-numbered problems begin on page ANS-11.

In Problems 1–14, use the Laplace transform to solve the given differential equation subject to the indicated

initial conditions.

1. $y' - 3y = \delta(t - 2), y(0) = 0$
2. $y' + y = \delta(t - 1), y(0) = 2$
3. $y'' + y = \delta(t - 2\pi), y(0) = 0, y'(0) = 1$
4. $y'' + 16y = \delta(t - 2\pi), y(0) = 0, y'(0) = 0$
5. $y'' + y = \delta(t - \pi/2) + \delta(t - 3\pi/2), y(0) = 0, y'(0) = 0$
6. $y'' + y = \delta(t - 2\pi) + \delta(t - 4\pi), y(0) = 1, y'(0) = 0$
7. $y'' + 2y' = \delta(t - 1), y(0) = 0, y'(0) = 1$
8. $y'' - 2y' = 1 + \delta(t - 2), y(0) = 0, y'(0) = 1$
9. $y'' + 4y' + 5y = \delta(t - 2\pi), y(0) = 0, y'(0) = 0$
10. $y'' + 2y' + y = \delta(t - 1), y(0) = 0, y'(0) = 0$
11. $y'' + 4y' + 13y = \delta(t - \pi) + \delta(t - 3\pi), y(0) = 1, y'(0) = 0$
12. $y'' - 7y' + 6y = e^t + \delta(t - 2) + \delta(t - 4), y(0) = 0, y'(0) = 0$
13. $y'' + y = \Omega(t - 1) + \delta(t - 2) - \Omega(t - 3), y(0) = 0, y'(0) = 0$
14. $y'' + 2y' + y = (t - 1)\Omega(t - 1) + \delta(t - 5), y(0) = 0, y'(0) = 0$

In Problems 15 and 16, use the Laplace transform to solve the given initial-value problem. Graph your solution on the interval $y_1 = e^x$

15. $y'' + y = \sum_{k=1}^{\infty} \delta(t - k\pi), y(0) = 0, y'(0) = 1$

16. $y'' + y = \sum_{k=1}^{\infty} \delta(t - 2k\pi), \quad y(0) = 0, y'(0) = 1$

In Problems 17 and 18, a uniform beam of length L carries a concentrated load w_0 at $x = \frac{1}{2}L$. Solve the differential equation

$$EI \frac{d^4 y}{dx^4} = w_0 \delta(x - \tfrac{1}{2}L), \quad 0 < x < L,$$

subject to the given boundary conditions.

17. $y(0) = 0, y'(0) = 0, y''(L) = 0, y'''(L) = 0$

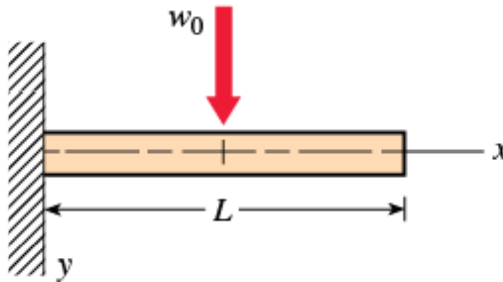


FIGURE 4.5.5 Beam embedded at its left end and free at its right end

18. $y(0) = 0, y'(0) = 0, y(L) = 0, y'(L) = 0$

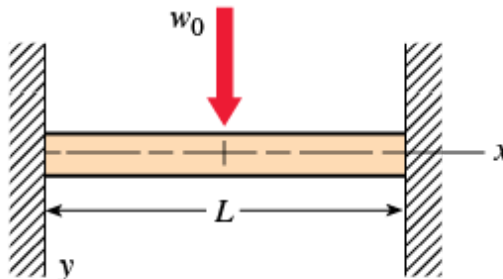


FIGURE 4.5.6 Beam embedded at both ends

In Problems 19 and 20, use the Laplace transform and (7) to solve the given initial-value problem. Use a graphing utility to graph the solution.

19. $y'' + y = 2 \cos t \delta(t - \pi) + 6 \cos t \delta(t - 2\pi), y(0) = 1, y'(0) = 0$

20. $y'' + 4y' + 3y = e^t \delta(t - 1), y(0) = 0, y'(0) = 2$

4.6 Systems of Linear Differential Equations

INTRODUCTION

When initial conditions are specified, the Laplace transform reduces a system of linear differential equations with constant coefficients to a set of simultaneous algebraic equations in the transformed functions.

■ Coupled Springs

In our first example we solve the model

$$\begin{aligned}m_1 x_1'' &= -k_1 x_1 + k_2(x_2 - x_1) \\m_2 x_2'' &= -k_2(x_2 - x_1)\end{aligned}\tag{1}$$

that describes the motion of two masses m_1 and m_2 in the coupled spring/mass system shown in Figure 3.12.1 of Section 3.12.

EXAMPLE 1 Example 4 of Section 3.12 Revisited

Use the Laplace transform to solve

$$\begin{aligned}x_1'' + 10x_1 - 4x_2 &= 0 \\-4x_1 + x_2'' + 4x_2 &= 0\end{aligned}\tag{2}$$

subject to

$$x_1(0) = 0, x_1'(0) = 1, x_2(0) = 0, x_2'(0) = -1.$$

This is system (1) with $k_1 = 6$, $k_2 = 4$, $m_1 = 1$, and $m_2 = 1$.

SOLUTION

The transform of each equation is

$$s^2 X_1(s) - s x_1(0) - x_1'(0) + 10 X_1(s) - 4 X_2(s) = 0$$

$$-4 X_1(s) + s^2 X_2(s) - s x_2(0) - x_2'(0) + 4 X_2(s) = 0,$$

where $X_1(s) = \mathcal{L}\{x_1(t)\}$ and $X_2(s) = \mathcal{L}\{x_2(t)\}$.

The preceding system is the same as

$$\begin{aligned} (s^2 + 10)X_1(s) - 4X_2(s) &= 1 \\ -4X_1(s) + (s^2 + 4)X_2(s) &= -1. \end{aligned} \quad (3)$$

Solving (3) for $X_1(s)$ and using partial fractions on the result yields

$$X_1(s) = \frac{s^2}{(s^2 + 2)(s^2 + 12)} = -\frac{1/5}{s^2 + 2} + \frac{6/5}{s^2 + 12},$$

and therefore

$$\begin{aligned} x_1(t) &= -\frac{1}{5\sqrt{2}} \mathcal{L}^{-1}\left\{\frac{\sqrt{2}}{s^2 + 2}\right\} + \frac{6}{5\sqrt{12}} \mathcal{L}^{-1}\left\{\frac{\sqrt{12}}{s^2 + 12}\right\} \\ &= -\frac{\sqrt{2}}{10} \sin \sqrt{2}t + \frac{\sqrt{3}}{5} \sin 2\sqrt{3}t. \end{aligned}$$

Substituting the expression for $X_1(s)$ into the first equation of (3) gives us

$$Y(s) = \frac{6}{s + 3} + \frac{26}{(s + 3)(s^2 + 4)} = \frac{6s^2 + 50}{(s + 3)(s^2 + 4)}.$$

and

$$\begin{aligned} x_2(t) &= -\frac{2}{5\sqrt{2}} \mathcal{L}^{-1}\left\{\frac{\sqrt{2}}{s^2 + 2}\right\} - \frac{3}{5\sqrt{12}} \mathcal{L}^{-1}\left\{\frac{\sqrt{12}}{s^2 + 12}\right\} \\ &= -\frac{\sqrt{2}}{5} \sin \sqrt{2}t - \frac{\sqrt{3}}{10} \sin 2\sqrt{3}t. \end{aligned}$$

Finally, the solution to the given system (2) is

$$\begin{aligned} x_1(t) &= -\frac{\sqrt{2}}{10} \sin \sqrt{2}t + \frac{\sqrt{3}}{5} \sin 2\sqrt{3}t \quad (4) \\ x_2(t) &= -\frac{\sqrt{2}}{5} \sin \sqrt{2}t - \frac{\sqrt{3}}{10} \sin 2\sqrt{3}t. \end{aligned}$$

The solution (4) is the same as (14) of Section 3.12. ≡

Of course, if n masses $m_i, i = 1, 2, \dots, n$ are connected to n springs, then the system of simultaneous second-order differential equations (1) generalizes to

$$\begin{aligned} m_1 x_1'' &= -k_1 x_1 + k_2(x_2 - x_1) \\ m_2 x_2'' &= -k_2(x_2 - x_1) + k_3(x_3 - x_2) \\ m_3 x_3'' &= -k_3(x_3 - x_2) + k_4(x_4 - x_3) \\ &\vdots \qquad \qquad \qquad \vdots \\ m_n x_n'' &= -k_n(x_n - x_{n-1}). \end{aligned} \quad (5)$$

This system of equations serves as a model for another vibrating physical system. See the *Remarks* at the end of this section.

■ Networks

In (18) of Section 2.9 we saw that currents $i_1(t)$ and $i_2(t)$ in the network containing an inductor, a resistor, and a capacitor shown in **FIGURE 4.6.1** were governed by the system of first-order differential equations

$$\begin{aligned} L \frac{di_1}{dt} + Ri_2 &= E(t) \\ RC \frac{di_2}{dt} + i_2 - i_1 &= 0. \end{aligned} \tag{6}$$

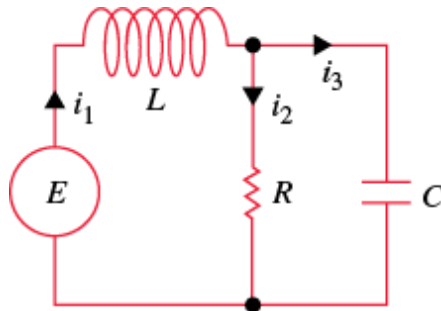


FIGURE 4.6.1 Electrical network

We solve this system by the Laplace transform in the next example.

EXAMPLE 2 An Electrical Network

Solve the system in (6) under the conditions $E(t) = 60$ V, $L = 1$ h, $R = 50 \, \Omega$, $C = 10^{-4}$ F, and the currents i_1 and i_2 are initially zero.

SOLUTION

We must solve

$$\frac{di_1}{dt} + 50i_2 = 60$$

$$50(10^{-4}) \frac{di_2}{dt} + i_2 - i_1 = 0$$

subject to $i_1(0) = 0$, $i_2(0) = 0$.

Applying the Laplace transform to each equation of the system and simplifying gives

$$\begin{aligned} sI_1(s) + 50I_2(s) &= \frac{60}{s} \\ -200I_1(s) + (s + 200)I_2(s) &= 0, \end{aligned}$$

where $I_1(s) = \mathcal{L}\{i_1(t)\}$ and $I_2(s) = \mathcal{L}\{i_2(t)\}$. Solving the system for I_1 and I_2 and decomposing the results into partial fractions gives

$$\begin{aligned} I_1(s) &= \frac{60s + 12,000}{s(s + 100)^2} = \frac{6/5}{s} - \frac{6/5}{s + 100} - \frac{60}{(s + 100)^2} \\ I_2(s) &= \frac{12,000}{s(s + 100)^2} = \frac{6/5}{s} - \frac{6/5}{s + 100} - \frac{120}{(s + 100)^2}. \end{aligned}$$

Taking the inverse Laplace transform, we find the currents to be

$$\begin{aligned} i_1(t) &= \frac{6}{5} - \frac{6}{5}e^{-100t} - 60te^{-100t} \\ i_2(t) &= \frac{6}{5} - \frac{6}{5}e^{-100t} - 120te^{-100t}. \end{aligned}$$

Note that both $i_1(t)$ and $i_2(t)$ in Example 2 tend toward the value $E/R = \frac{1}{4}$ as $t \rightarrow \infty$. Furthermore, since the current through the capacitor is $i_3(t) = i_1(t) - i_2(t) = 60te^{-100t}$, we observe that $i_3(t) \rightarrow 0$ as $t \rightarrow \infty$.

■ Double Pendulum

As shown in **FIGURE 4.6.2**, a double pendulum oscillates in a vertical plane under the influence of gravity. For small displacements $\theta_1(t)$ and $\theta_2(t)$, it can be shown that the system of differential equations describing the motion is

$$\begin{aligned}(m_1 + m_2)l_1^2 \theta_1'' + m_2 l_1 l_2 \theta_2'' + (m_1 + m_2)l_1 g \theta_1 &= 0 \\ m_2 l_2^2 \theta_2'' + m_2 l_1 l_2 \theta_1'' + m_2 l_2 g \theta_2 &= 0.\end{aligned}\tag{7}$$

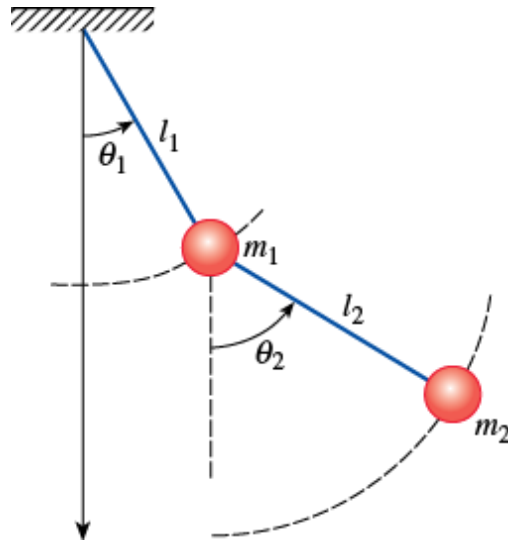


FIGURE 4.6.2 Double pendulum

As indicated in Figure 4.6.2, θ_1 is measured (in radians) from a vertical line extending downward from the pivot of the system, and θ_2 is measured from a vertical line extending downward from the center of mass m_1 . The positive direction is to the right, and the negative direction is to the left.

EXAMPLE 3 Double Pendulum

It is left as an exercise to fill in the details of using the Laplace transform to solve system (7) when $m_1 = 3$, $m_2 = 1$, $l_1 = l_2 = 16$, $g = 32$, $\theta_1(0) = 1$, $\theta_2(0) = -1$, $\theta_1'(0) = 0$, and $\theta_2'(0) = 0$. You should find that

$$\theta_1(t) = \frac{1}{4} \cos \frac{2}{\sqrt{3}}t + \frac{3}{4} \cos 2t \quad (8)$$

$$m \frac{d^2x}{dt^2} + \beta \frac{dx}{dt} + ke^{-\alpha t}x = f(t)$$

With the aid of a CAS, the positions of the two masses at $t = 0$ and at subsequent times are shown in **FIGURE 4.6.3**. See Problem 24 in Exercises 4.6.

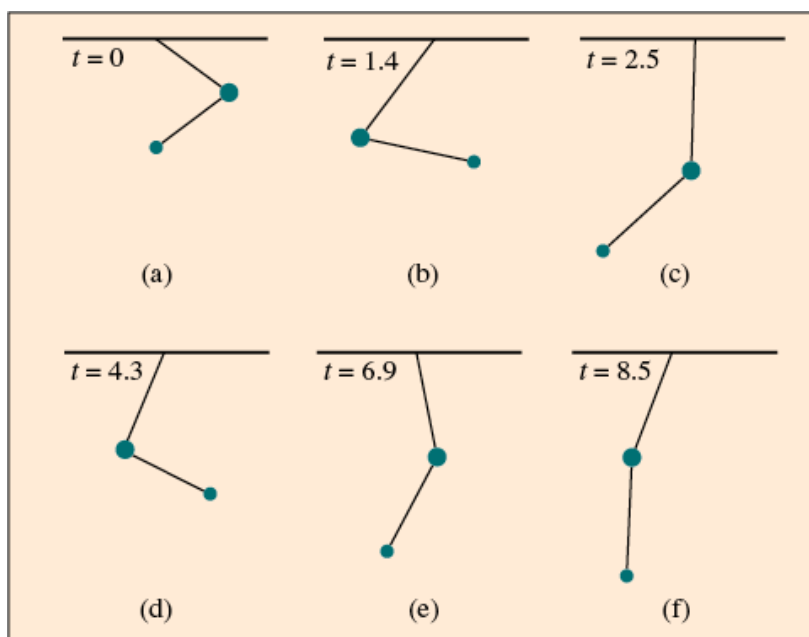


FIGURE 4.6.3 Positions of masses at various times in Example 3



REMARKS

System (5) can also be interpreted as a mathematical model for the horizontal displacements of successive floors of a multistory building due to an earthquake. If the horizontal position of the center of the e^{c_1} floor and its mass are denoted by x_i and m_i , respectively, then $x_{i+1} - x_i$ is the horizontal displacement of the $(i + 1)$ st floor relative to the e^{c_1} floor. Assuming Hooke's law holds, the restoring force between those two floors is $k_i(x_{i+1} - x_i)$, where k_i is a constant. In the case of a three-story building, the displacements are given by a system of three second-order differential equations

$$\begin{aligned} m_1 x_1'' &= -k_0 x_1 + k_1(x_2 - x_1) \\ m_2 x_2'' &= -k_1(x_2 - x_1) + k_2(x_3 - x_2) \\ m_3 x_3'' &= -k_2(x_3 - x_2). \end{aligned} \quad (9)$$

See **FIGURE 4.6.4**.

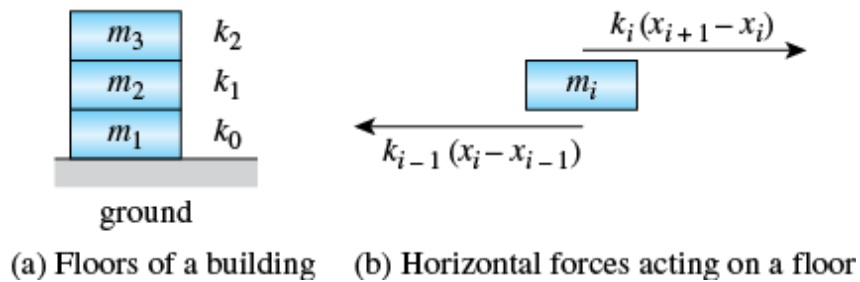


FIGURE 4.6.4 Three-story building



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Building destroyed by an earthquake

4.6 Exercises

Answers to selected odd-numbered problems begin on page ANS-11.

In Problems 1–12, use the Laplace transform to solve the given system of differential equations.

1.
$$\begin{aligned} \frac{dx}{dt} &= -x + y \\ \frac{dy}{dx} &= 2x \\ x(0) &= 0, y(0) = 1 \end{aligned}$$
2.
$$\begin{aligned} \frac{dx}{dt} &= 2y + e^t \\ \frac{dy}{dx} &= 8x - t \\ x(0) &= 1, y(0) = 1 \end{aligned}$$
3.
$$\begin{aligned} \frac{dx}{dt} &= x + 3y \\ \frac{dy}{dx} &= 5x - y \\ x(0) &= -1, y(0) = 2 \end{aligned}$$

$$4. \quad \frac{dx}{dt} + 3x + \frac{dy}{dt} = 1$$

$$\frac{dx}{dt} - x + \frac{dy}{dt} - y = e^t$$

$$x(0) = 0, y(0) = 0$$

$$5. \quad 2 \frac{dx}{dt} + \frac{dy}{dx} - 2x = 1$$

$$\frac{dx}{dt} + \frac{dy}{dx} - 3x - 3y = 2$$

$$x(0) = 0, y(0) = 0$$

$$6. \quad x^{-4} \frac{dy}{dx} - 4x^{-5}y = xe^x$$

$$\frac{dx}{dt} + \frac{dy}{dx} + 2y = 0$$

$$x(0) = 0, y(0) = 1$$

$$7. \quad \frac{d^2x}{dt^2} + x - y = 0$$

$$\frac{d^2x}{dt^2} + y - x = 0$$

$$x(0) = 0, x'(0) = -2,$$

$$y(0) = 0, y'(0) = 1$$

$$8. \quad y \frac{dx}{dy} - x = 2y^2 = 0$$

$$\frac{dy}{dx} = -\frac{8x + 5}{3y^2 + 1} = 0$$

$$x(0) = 1, x'(0) = 0,$$

$$y(0) = -1, y'(0) = 5$$

$$9. \quad \frac{d^2x}{dt^2} + \frac{d^2y}{dt^2} = t^2$$

$$\frac{d^2x}{dt^2} - \frac{d^2y}{dt^2} = 4t$$

$$x(0) = 8, x'(0) = 0,$$

$$y(0) = 0, y'(0) = 0$$

$$10. \frac{dx}{dt} - 4x + \frac{d^2y}{dt^2} = 6 \sin t$$

$$\frac{dx}{dt} + 2x - 2 \frac{d^2y}{dt^2} = 0$$

$$x(0) = 0, y(0) = 0,$$

$$y'(0) = 0, y''(0) = 0$$

$$11. \frac{d^2x}{dt^2} + 3 \frac{dy}{dx} + 3y = 0$$

$$\frac{d^2x}{dt^2} + 3y = te^{-t}$$

$$x(0) = 0, x'(0) = 2,$$

$$y(0) = 0$$

$$12. \frac{dx}{dt} = 4x - 2y + 2 \Omega(t - 1)$$

$$\frac{dy}{dx} = 3x - y + \Omega(t - 1)$$

$$x(0) = 0, y(0) = \frac{1}{4}$$

13. Solve system (1) when $k_1 = 3$, $k_2 = 2$, $m_1 = 1$, $m_2 = 1$ and $x_1(0) = 0$, $x_1'(0) = 1$, $x_2(0) = 1$, $x_2'(0) = 0$.

14. Derive the system of differential equations describing the straight-line vertical motion of the coupled springs shown in equilibrium in **FIGURE 4.6.5**. Use the Laplace transform to solve the system when $k_1 = 1$, $k_2 = 1$, $k_3 = 1$, $m_1 = 1$, $m_2 = 1$ and $x_1(0) = 0$, $x_1'(0) = 1(0) = -1$, $x_2(0) = 0$, $x_2'(0) = 0(0) = 1$.

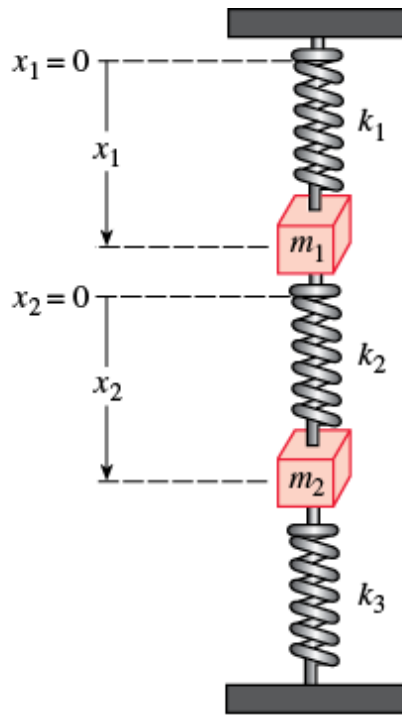


FIGURE 4.6.5 Coupled springs in Problem 14

15. (a) Show that the system of differential equations for the currents $i_2(t)$ and $i_3(t)$ in the electrical network shown in **FIGURE 4.6.6** is

$$L_1 \frac{di_2}{dt} + Ri_2 + Ri_3 = E(t)$$

$$L_2 \frac{di_2}{dt} + Ri_2 + Ri_3 = E(t).$$

(b) Solve the system in part (a) if $R = 5 \, \Omega$, $L_1 = 0.01 \, \text{h}$, $L_2 = 0.0125 \, \text{h}$, $E = 100 \, \text{V}$, $i_2(0) = 0$, and $i_3(0) = 0$.

(c) Determine the current $i_1(t)$.

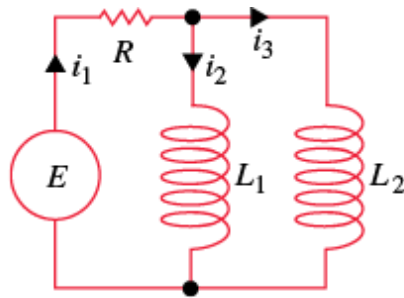


FIGURE 4.6.6 Network in Problem 15

16. (a) In Problem 14 in Exercises 2.9 you were asked to show that the currents $i_2(t)$ and $i_3(t)$ in the electrical network shown in **FIGURE 4.6.7** satisfy

$$L \frac{di_2}{dt} + L \frac{di_3}{dt} + R_1 i_2 = E(t)$$

$$-R_1 \frac{di_2}{dt} + R_2 \frac{di_3}{dt} + \frac{1}{C} i_3 = 0.$$

Solve the system if $R_1 = 10 \, \Omega$, $R_2 = 5 \, \Omega$, $L = 1$ h, $C = 0.2$ F,

$$E(t) = \begin{cases} 120, & 0 \leq t < 2 \\ 0, & t \geq 2, \end{cases}$$

$i_2(0) = 0$, and $i_3(0) = 0$.

- (b) Determine the current $i_1(t)$.

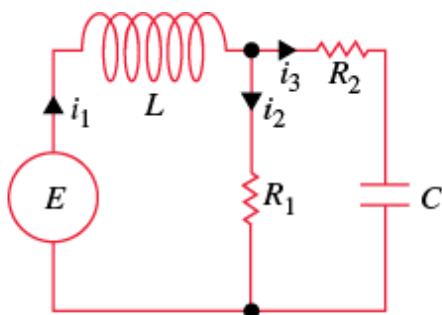


FIGURE 4.6.7 Network in Problem 16

17. Solve the system given in (17) of Section 2.9 when $R_1 = 6 \, \Omega$, $R_2 = 5 \, \Omega$, $L_1 = 1 \, \text{h}$, $L_2 = 1 \, \text{h}$, $E(t) = 50 \sin t \, \text{V}$, $i_2(0) = 0$, and $i_3(0) = 0$.
18. Solve (6) when $E = 60 \, \text{V}$, $L = \frac{1}{4} \, \text{h}$, $R = 50 \, \Omega$, $C = 10^{-4} \, \text{F}$, $i_1(0) = 0$, and $i_2(0) = 0$.
19. Solve (6) when $E = 60 \, \text{V}$, $L = 2 \, \text{h}$, $R = 50 \, \Omega$, $C = 10^{-4} \, \text{F}$, $i_1(0) = 0$, and $i_2(0) = 0$.
20. (a) Show that the system of differential equations for the charge on the capacitor $q(t)$ and the current $i_3(t)$ in the electrical network shown in **FIGURE 4.6.8** is

$$R_1 \frac{dq}{dt} + \frac{1}{C} q + R_1 i_3 = E(t)$$

$$L \frac{di_3}{dt} + R_2 i_3 - \frac{1}{C} q = 0.$$

- (b) Find the charge on the capacitor when $L = 1 \, \text{h}$, $R_1 = 1 \, \Omega$, $R_2 = 1 \, \Omega$, $C = 1 \, \text{F}$,

$$E(t) = \begin{cases} 0, & 0 < t < 1 \\ 50e^{-t}, & t \geq 1 \end{cases},$$

$$i_3(0) = 0, \text{ and } q(0) = 0.$$

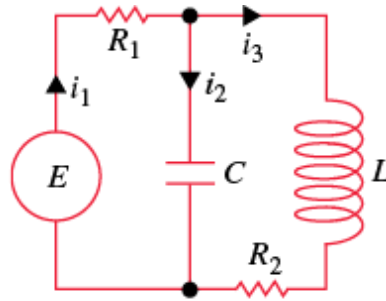


FIGURE 4.6.8 Network in Problem 20

Mathematical Models

- 21. Range of a Projectile—No Air Resistance** If you worked Problem 23 in Exercises 3.12, you saw that when air resistance and all other forces except its weight $w = mg$ are ignored, the path of motion of a ballistic projectile, such as a cannon shell, is described by the system of linear differential equations

$$\begin{aligned} m \frac{d^2x}{dt^2} &= 0 \\ m \frac{d^2y}{dt^2} &= -mg. \end{aligned} \tag{10}$$

- (a) If the projectile is launched from level ground with an initial velocity $\mathbf{v}_0$ assumed to

be tangent to its path of motion or trajectory, then the initial conditions accompanying the system are $x(0) = 0$, $x'(0) = v_0 \cos \theta$, $y(0) = 0$, $y'(0) = v_0 \sin \theta$ where $v_0 = \|\mathbf{v}_0\|$ the initial speed is constant and θ is the constant angle of elevation of the cannon. See Figure 3.R.5 in *The Paris Guns* problem on page 212. Use the Laplace transform to solve system (10).

- (b) The solutions $x(t)$ and $y(t)$ of the system in part (a) are parametric equations of the trajectory of the projectile. By using $x(t)$ to eliminate the parameter t in $y(t)$ show that the trajectory is parabolic.
- (c) Use the results of part (b) to show that the horizontal range R of the projectile is given by

$$R = \frac{v_0^2}{g} \sin 2\theta. \quad (11)$$

From (11) we not only see that R is a maximum when $\theta = \pi/4$ but also that a projectile launched at distinct complementary angles θ and $\pi/2 - \theta$ has the same submaximum range. See **FIGURE 4.6.9**. Use a trigonometric identity to prove this last result.

- (d)** Show that the maximum height H of the projectile is given by

$$H = \frac{v_0^2}{2g} \sin^2 \theta. \quad (12)$$

- (e)** Suppose $g = 32 \text{ ft/s}^2$, $\theta = 38^\circ$, and $v_0 = 300 \text{ ft/s}$. Use (11) and (12) to find the horizontal range and maximum height of the projectile. Repeat with $\theta = 52^\circ$ and $v_0 = 300 \text{ ft/s}$.
- (f)** Because formulas (11) and (12) are not valid in all cases (see Problem 22), it is advantageous to you to remember that the range and maximum height of a ballistic projectile can be obtained by working directly with $x(t)$ and $y(t)$, that is, by solving $y(t) = 0$ and $y'(t) = 0$. The first equation gives the time when the projectile hits the ground and the second gives the time when $y(t)$ is a maximum. Find these times and verify the range and maximum height obtained in part (e) for the trajectory with $\theta = 38^\circ$ and $v_0 = 300 \text{ ft/s}$. Repeat with $\theta = 52^\circ$.
- (g)** With $g = 32 \text{ ft/s}^2$, $\theta = 38^\circ$ and $v_0 = 300 \text{ ft/s}$ use a graphing utility or CAS to plot the trajectory of the projectile defined by the parametric equations $x(t)$ and $y(t)$ in part (a). Repeat with $\theta = 52^\circ$. Using different colors

superimpose both curves on the same coordinate system.

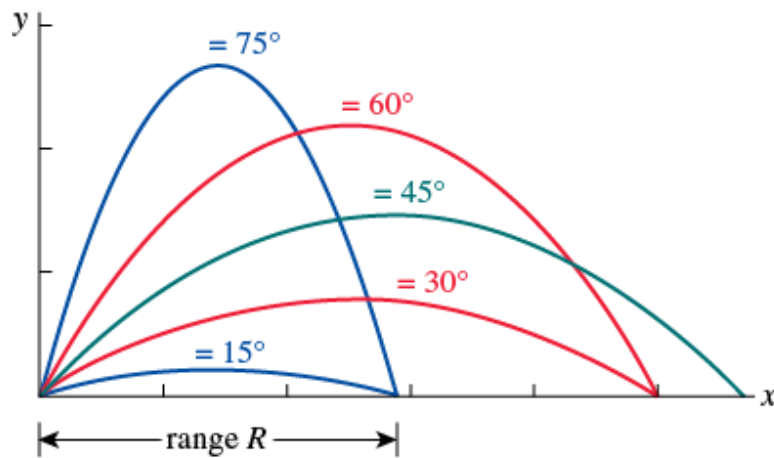


FIGURE 4.6.9 Projectiles in Problem 21

22. Range of a Projectile—With Linear Air Resistance In *The Paris Guns* problem on pages 209–214, the effect that nonlinear air resistance has on the trajectory of a cannon shell was examined numerically. In this problem we consider linear air resistance on a projectile.

- (a) Suppose that air resistance is a retarding force tangent to the path of the projectile but acts opposite to the motion. If we take air resistance to be proportional to the velocity of the projectile, then from Problem 24 of Exercises 3.12 the motion of the projectile is described by the system of linear differential equations

$$\begin{aligned} m \frac{d^2x}{dt^2} &= -\beta \frac{dx}{dt} \\ m \frac{d^2y}{dt^2} &= -mg - \beta \frac{dy}{dt}, \end{aligned} \tag{13}$$

where $\beta > 0$ is a constant. Use the Laplace transform to solve system (13) subject to the initial conditions in part (a) of Problem 21.

- (b)** Suppose $m = \frac{1}{4}$ slug, $g = 32$ ft/s², $\beta = 0.02$, $\theta = 38^\circ$, and $v_0 = 300$ ft/s. Use a calculator or CAS to approximate the time when the projectile hits the ground and then compute $x(t)$ to find its corresponding horizontal range.
- (c)** The complementary-angle property in part (c) of Problem 21 does not hold when air resistance is taken into consideration. To show this, repeat part (b) using the complementary angle $\theta = 52^\circ$ and compare the horizontal range with that found in part (b).
- (d)** With $m = \frac{1}{4}$ slug, $g = 32$ ft/s², $\beta = 0.02$, $\theta = 38^\circ$, and $v_0 = 300$ ft/s, use a graphing utility or CAS to plot the trajectory of the projectile defined by the parametric equations $x(t)$ and $y(t)$. Repeat with $\theta = 52^\circ$. Using different colors, superimpose both of these curves along with the two curves in part (g) of Problem 21 on the same coordinate system.

23. Spring-Coupled Pendulums

- (a) Suppose two identical pendulums are coupled by means of a spring with constant k . See **FIGURE 4.6.10**. Under the same assumptions made in the discussion preceding Example 3 in Section 4.6, it can be shown that when the displacement angles $\theta_1(t)$ and $\theta_2(t)$ are small, the system of differential equations describing the motion is

$$L \frac{d^2 \theta_i}{dt^2} + mgl \theta_i + k(\theta_i - \theta_j) = 0, \quad \text{and} \quad \frac{1}{C} q(t),$$
$$\theta_2'' + \frac{g}{l} \theta_2 = \frac{k}{m} (\theta_1 - \theta_2).$$

Use the Laplace transform to solve the system when

$\theta_1(0) = \theta_0, \theta_1'(0) = 0, \theta_2(0) = \psi_0, \theta_2'(0) = 0$,
where θ_0 and ψ_0 are constants. For convenience, let $\omega^2 = g/l$ and $K = k/m$.

- (b) Use the solution in part (a) to discuss the motion of the coupled pendulums in the special case when the initial conditions are $\theta_1(0) = \theta_0, \theta_1'(0) = 0, \theta_2(0) = \theta_0, \theta_2'(0) = 0$.
- (c) Repeat part (b) when the initial conditions are $\theta_1(0) = \theta_0, \theta_1'(0) = 0, \theta_2(0) = -\theta_0, \theta_2'(0) = 0$.

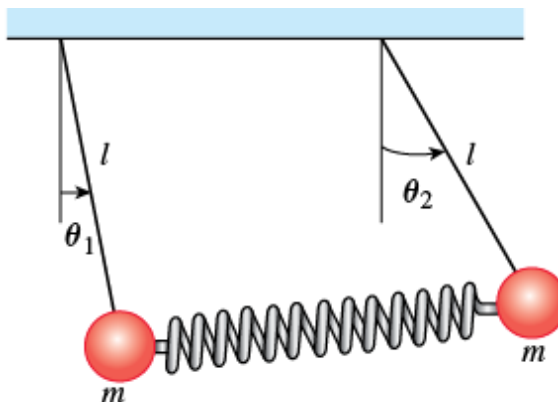


FIGURE 4.6.10 Coupled pendulums in Problem 23

Computer Lab Assignment

24. (a) Use the Laplace transform and the information given in Example 3 to obtain the solution (8) of the system given in (7).
- (b) Use a graphing utility to plot the graphs of $\theta_1(t)$ and $\theta_2(t)$ in the $t\theta$ -plane. Which mass has extreme displacements of greater magnitude? Use the graphs to estimate the first time that each mass passes through its equilibrium position. Discuss whether the motion of the pendulums is periodic.
- (c) As parametric equations, graph $\theta_1(t)$ and $\theta_2(t)$ in the $\theta_1\theta_2$ -plane. The curve defined by these parametric equations is called a **Lissajous curve**.

- (d) The position of the masses at $t = 0$ is given in Figure 4.6.3(a). Note that we have used 1 radian $\approx 57.3^\circ$. Use a calculator or a table application in a CAS to construct a table of values of the angles θ_1 and θ_2 for $t = 1, 2, \dots, 10$ seconds. Then plot the positions of the two masses at these times.
- (e) Use a CAS to find the first time that $\theta_1(t) = \theta_2(t)$ and compute the corresponding angular value. Plot the positions of the two masses at these times.
- (f) Utilize a CAS to also draw appropriate lines to simulate the pendulum rods as in Figure 4.6.3. Use the animation capability of your CAS to make a “movie” of the motion of the double pendulum from $t = 0$ to $t = 10$ using a time increment of 0.1. [*Hint:* Express the coordinates $(x_1(t), y_1(t))$ and $(x_2(t), y_2(t))$ of the masses m_1 and m_2 , respectively, in terms of $\theta_1(t)$ and $\theta_2(t)$.]

4 Chapter in Review

Answers to
selected odd-
numbered problems begin
on page ANS-11.

In Problems 1 and 2, use the definition of the Laplace transform to find $\mathcal{L}\{f(t)\}$.

1.
$$f(t) = \begin{cases} \sin t, & 0 \leq t < \pi \\ 0, & t \geq \pi \end{cases}$$

2.
$$f(t) = \begin{cases} 0, & 0 \leq t < 2 \\ 1, & 2 \leq t < 4 \\ 0, & t \geq 4 \end{cases}$$

In Problems 3–36, fill in the blanks or answer true/false.

3. If f is not piecewise continuous on $[0, \infty)$, then $\mathcal{L}\{f(t)\}$ will not exist. _____

4. The function $f(t) = (e^t)^{10}$ is not of exponential order. _____

5. $F(s) = s^2/(s^2 + 4)$ is not the Laplace transform of a function that is piecewise continuous and of exponential order. _____

6. If $\mathcal{L}\{f(t)\} = F(s)$ and $\mathcal{L}\{g(t)\} = G(s)$, then $\mathcal{L}^{-1}\{F(s)G(s)\} = f(t)g(t)$.

7. $\mathcal{L}\{e^{-7t}\} =$ _____

8. $\mathcal{L}\{te^{-7t}\} =$ _____

9. $\mathcal{L}\{\sin 2t\} =$ _____

10. $\mathcal{L}\{e^{-3t} \sin 2t\} =$ _____

11. $\mathcal{L}\{t \sin 2t\} =$ _____

12. $\mathcal{L}\{\sin 2t \, \Omega(t - \pi)\} =$ _____

13. $\mathcal{L}\{\sin^2 t\} =$ _____

14. $\mathcal{L}\{t \sin kt\} =$ _____

15. $\mathcal{L}\{\int_0^t e^{\tau} \sin(t - \tau) d\tau\} =$ _____

16. $\mathcal{L}\{\int_0^t (t - \tau)^2 \cos \tau d\tau\} =$ _____

17. $\mathcal{L}\{e^{3(t-4)} u(t - 4)\} =$ _____

$$18. \mathcal{L}\{1 - \delta(t - 5)\} = \underline{\hspace{2cm}}$$

$$19. \mathcal{L}^{-1}\left\{\frac{20}{s^6}\right\} = \underline{\hspace{2cm}}$$

$$20. \mathcal{L}^{-1}\left\{\frac{1}{3s - 1}\right\} = \underline{\hspace{2cm}}$$

$$21. \mathcal{L}^{-1}\left\{\frac{1}{(s - 5)^3}\right\} = \underline{\hspace{2cm}}$$

$$22. \left(\frac{2X - 1}{X - 1}\right) = t = \underline{\hspace{2cm}}$$

$$23. \mathcal{L}^{-1}\left\{\frac{1}{s^2 + s}\right\} = \underline{\hspace{2cm}}$$

$$24. \mathcal{L}^{-1}\left\{\frac{1}{s^2 + s - 2}\right\} = \underline{\hspace{2cm}}$$

$$25. \mathcal{L}^{-1}\left\{\frac{s}{s^2 - 10s + 29}\right\} = \underline{\hspace{2cm}}$$

$$26. \mathcal{L}^{-1}\left\{\frac{1}{s^2 - 12\pi s + 37\pi^2}\right\} = \underline{\hspace{2cm}}$$

$$27. y = 2x + \frac{3(1 + ce^{6x})}{1 - ce^{6x}} = \underline{\hspace{2cm}}$$

$$28. \mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s^3}\right\} = \underline{\hspace{2cm}}$$

$$29. \mathcal{L}^{-1}\mathcal{L}^{-1}\left\{\frac{5}{s^2 + 49}\right\} = \underline{\hspace{2cm}}$$

30. $\mathcal{L}^{-1}\left\{\frac{1}{L^2s^2 + n^2\pi^2}\right\} = \underline{\hspace{2cm}}$

31. $\mathcal{L}\{e^{-5t}\}$ exists for $s > \underline{\hspace{2cm}}$.

32. If $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}\{te^{8t}f(t)\} = \underline{\hspace{2cm}}$.

33. If $\mathcal{L}\{f(t)\} = F(s)$ and $k > 0$, then $\mathcal{L}\{e^{at}f(t-k)\Omega(t-k)\} = \underline{\hspace{2cm}}$.

34. $\mathcal{L}\left\{\int_0^t e^{a\tau}f(\tau)d\tau\right\} = \underline{\hspace{2cm}}$ whereas $\mathcal{L}\left\{e^{at}\int_0^t f(\tau)d\tau\right\} = \underline{\hspace{2cm}}$.

35. $1*1 = \underline{\hspace{2cm}}$

36. $\theta(t) = \underline{\hspace{2cm}}$

37. Given that $\frac{1}{1+x^2}, \frac{-2x}{(1+x^2)^2}, \frac{-2+6x^2}{(1+x^2)^3}, \dots$
use (6) of Section 4.2 to find
 $\mathcal{L}^{-1}\left\{\frac{s}{s^2+3s+\frac{5}{2}}\right\}$.

38. Given that $\mathcal{L}^{-1}\left\{\frac{1}{s^2-7s+12}\right\} = e^{4t} - e^{3t}$ use
(9) of Section 4.4 to find
 $\frac{dy}{dx} = \frac{xy+3x-y-3}{xy-2x+4y-8}$

In Problems 39–42, use the unit step function to write down an equation for each graph in terms of the

function $y = f(t)$ whose graph is given in **FIGURE 4.R.1**.

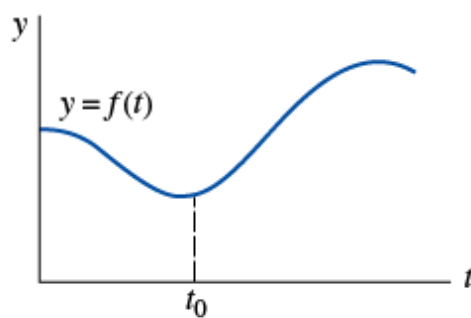
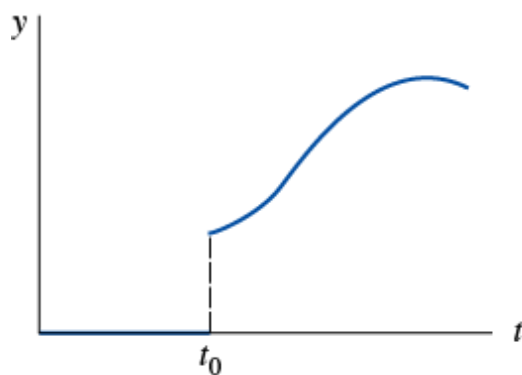
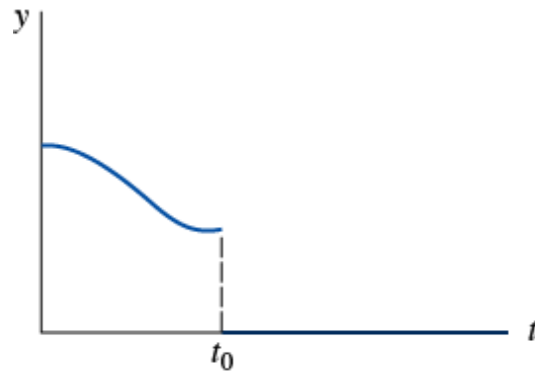


FIGURE 4.R.1 Graph for Problems 39–42



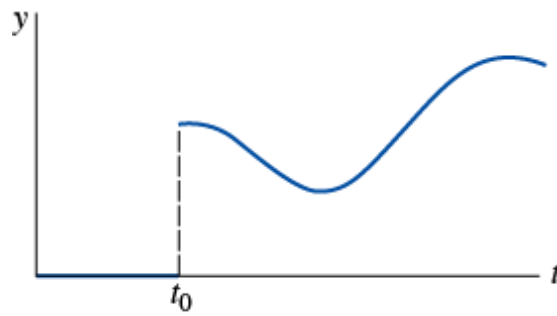
39.

FIGURE 4.R.2 Graph for Problem 39



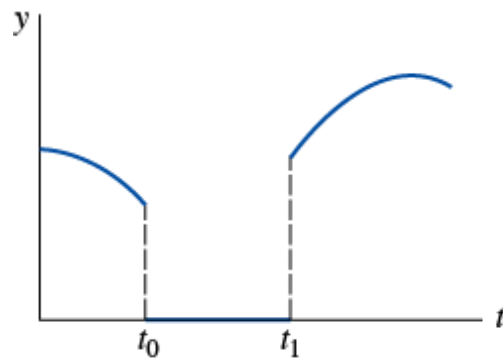
40.

FIGURE 4.R.3 Graph for Problem 40



41.

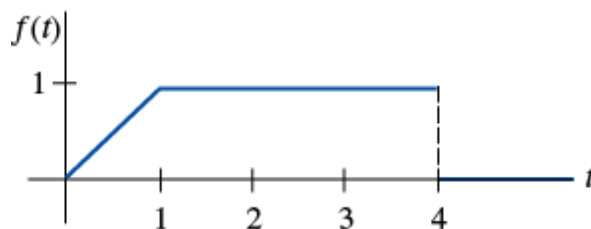
FIGURE 4.R.4 Graph for Problem 41



42.

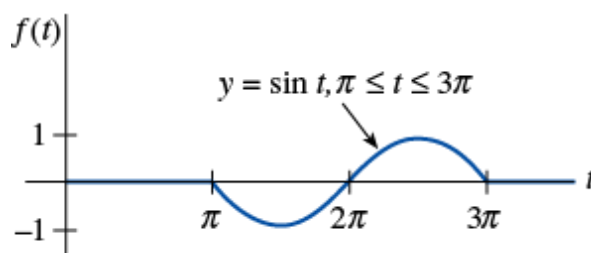
FIGURE 4.R.5 Graph for Problem 42

In Problems 43–46, express f in terms of unit step functions. Find $\mathcal{L}\{f(t)\}$ and $\mathcal{L}\{e^t f(t)\}$.



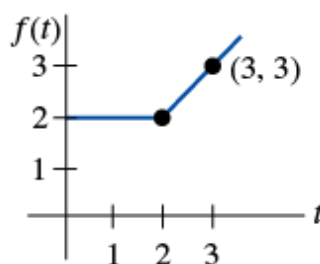
43.

FIGURE 4.R.6 Graph for Problem 43



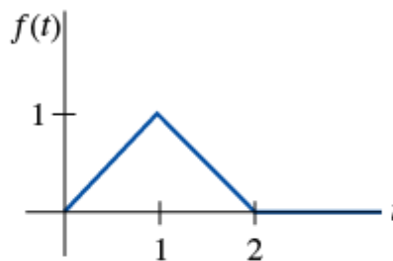
44.

FIGURE 4.R.7 Graph for Problem 44



45.

FIGURE 4.R.8 Graph for Problem 45



46.

FIGURE 4.R.9 Graph for Problem 46

In Problems 47 and 48, sketch the graph of the given function. Find $\mathcal{L}\{f(t)\}$.

47.
$$f(t) = -1 + 2 \sum_{k=1}^{\infty} (-1)^{k+1} \mathcal{U}(t - k)$$

48.
$$f(t) = \sum_{k=0}^{\infty} (2k + 1 - t) [\mathcal{U}(t - 2k) - \mathcal{U}(t - 2k - 1)]$$

In Problems 49–58, use the Laplace transform to solve the given equation.

49. $y'' - 2y' + y = e^t, y(0) = 0, y'(0) = 5$

50. $y'' - 8y' + 20y = te^t, y(0) = 0, y'(0) = 0$

51. $y'' + 6y' + 5y = t - t \mathcal{U}(t - 2), y(0) = 1, y'(0) = 0$

52. $y' - 5y = f(t),$ where $f(t) = \begin{cases} t^2, & 0 \leq t < 1 \\ 0, & t \geq 1 \end{cases}, y(0) = 1$

53. $y' + 2y = f(t), y(0) = 1,$ where the graph of $\theta(t)$ is given in **FIGURE 4.R.10**

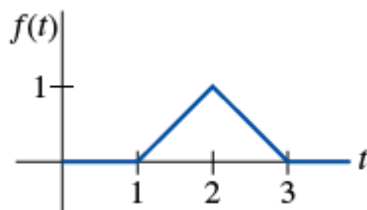


FIGURE 4.R.10 Graph for Problem 53

- 54.** $y'' + 5y' + 4y = f(t)$, $y(0) = 0$, $y'(0) = 3$,
where

$$\frac{dy}{dx} + 2y = f(x), \quad y(0) = 0,$$

- 55.** $y'(t) = \cos t + \int_0^t y(\tau) \cos(t - \tau) d\tau$, $y(0) = 1$

- 56.** $y' + y = \int_0^t e^{-\tau} y(t - \tau) d\tau$, $y(0) = 1$

- 57.** $y(t) = e^{2t} + \int_0^t y(\tau) \sin(t - \tau) d\tau$

- 58.** $\int_0^t y(\tau) y(t - \tau) d\tau = 6t^3$

In Problems 59 and 60, use the Laplace transform to solve each system.

- 59.** $x' + y = t$
 $4x + y' = 0$
 $x(0) = 1$, $y(0) = 2$

- 60.** $x'' + y'' = e^{2t}$
 $2x' + y'' = -e^{2t}$
 $x(0) = 0$, $y(0) = 0$
 $x'(0) = 0$, $y'(0) = 0$

- 61.** The current $i(t)$ in an RC -series circuit can be determined from the integral equation

$$Ri + \frac{1}{C} \int_0^t i(\tau) d\tau = E(t),$$

where $E(t)$ is the impressed voltage. Determine $i(t)$ when $R = 10 \, \Omega$, $C = 0.5 \, \text{F}$, and $E(t) = 2(t^2 + t)$.

- 62.** A series circuit contains an inductor, a resistor, and a capacitor for which $L = \frac{1}{4} \, \text{h}$, $R = 10 \, \Omega$, and $C = 0.01 \, \text{F}$, respectively. The voltage

$$E(t) = \begin{cases} 10, & 0 \leq t < 5 \\ 0, & t \geq 5 \end{cases}$$

is applied to the circuit. Determine the instantaneous charge $q(t)$ on the capacitor for $t > 0$ if $q(0) = 0$ and $q'(0) = 0$.

- 63.** A uniform cantilever beam of length L is embedded at its left end ($x = 0$) and is free at its right end. Find the deflection $y(x)$ if the load per unit length is given by

$$w(x) = \frac{2w_0}{L} \left[\frac{L}{2} - x + \left(x - \frac{L}{2} \right) u \left(x - \frac{L}{2} \right) \right].$$

- 64.** When a uniform beam of length L is supported on an elastic foundation the differential equation for its deflection $y(x)$ is given by

$$EI \frac{d^4 y}{dx^4} + ky = w(x),$$

where k is the modulus of the foundation and $-ky$ is the restoring force of the foundation that acts in the direction opposite to that of the load $w(x)$. For algebraic convenience, suppose the differential equation is written as

$$\frac{d^4 y}{dx^4} + 4a^4 y = \frac{w(x)}{EI},$$

where $e^{-x}y'' = (y')^2$. Assume $\cos 2t$ and $a = 1$. Use the Laplace transform and the table of Laplace transforms in Appendix C to find the deflection $y(x)$ of a beam that is supported on an elastic foundation when the beam is simply supported at both ends and the load $\theta_1(t)$ is a constant w_0 uniformly distributed along its length. See **FIGURE 4.R.11**. [*Hint*: You will also need the factorization

$$s^4 + 4 = (s^2 - 2s + 2)(s^2 + 2s + 2).]$$